

SIASUN 新松

Operating Manual for
Arc Welding
Application of
SIASUN Robot

Industrial Robot Technical Information



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1 Standard Configuration for Arc Welding Robot

1.1 Configuration for Arc Welding Robot

1.1.1 Cabinet configuration

Click [System Setup] - [Cabinet Configuration], enter the password: sysiasun, and enter the cabinet configuration interface. Select according to the information on the nameplate of the robot control cabinet, click [Save], and restart the robot after receiving a success prompt.

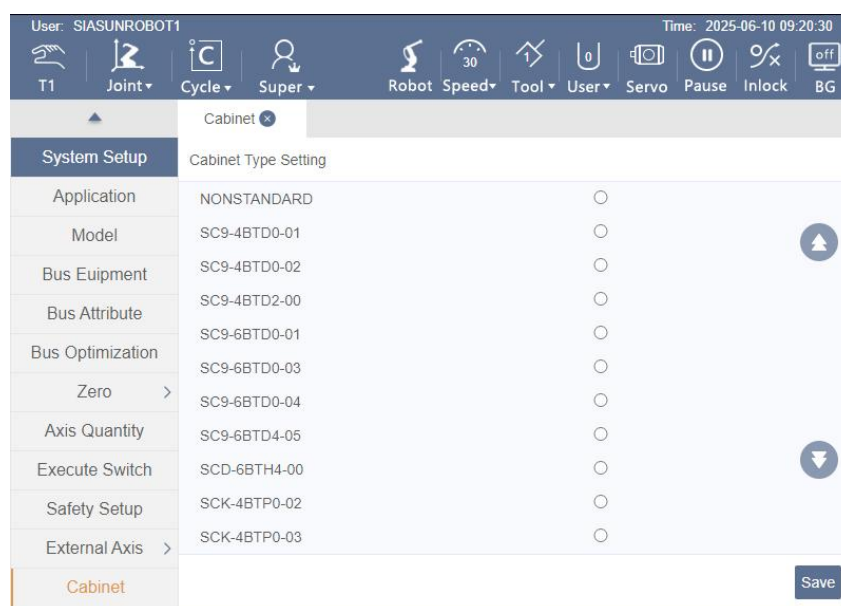


Figure 1- 1 Cabinet Configuration Interface

1.1.2 Robot Model Setting

Click [System Setup] - [Model], enter the password: sysiasun, enter the robot model setting interface, and check the model of the machine. If it does not match the model of the main body, manually select the correct model, click [Save], and restart the robot after receiving a success prompt.

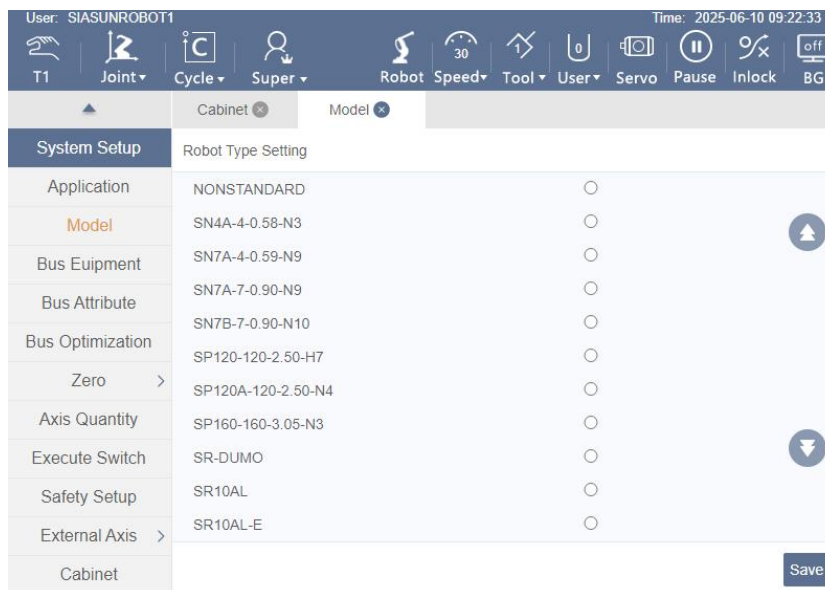


Figure 1-2 Robot Model Configuration Interface

1.1.3 Axis Quantity Setting

When the workstation has external axes, go to [System Setup] - [Axis Quantity], enter the password: sysiasun, enter the axis quantity interface, fill in the actual number of external axes, click [Save], and restart the robot after receiving a success prompt. For specific external axis information, please refer to Operating Manual for SIASUN Robot.

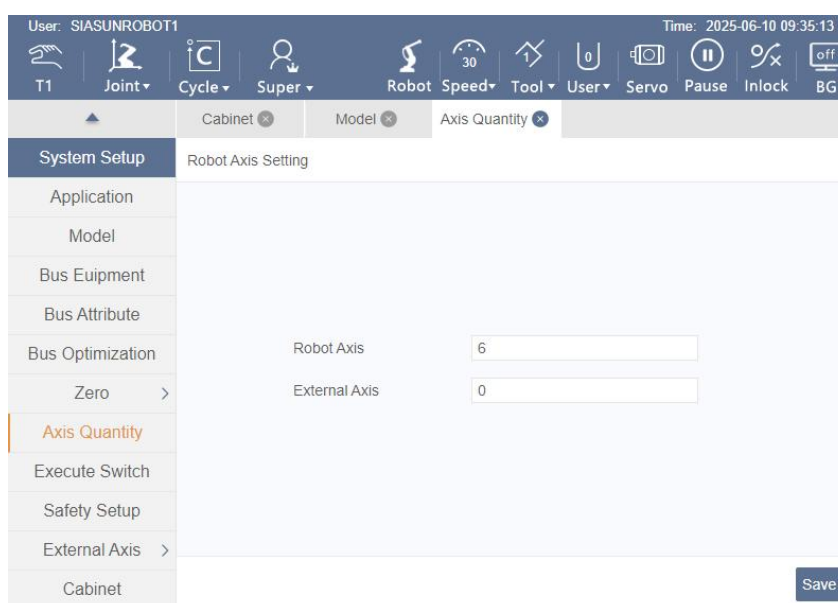


Figure 1-3 Axis Quantity Setting Interface

1.1.4 Bus Device Configuration

When the workstation has an external axis, go to [System Setup] - [Bus Equipment], enter the password: sysiasun, and enter the bus device configuration interface. Click Previous Page or Next Page to select the type of each communication node. If the current communication node is IO board, click IO Indicator, and then click [Save]. After receiving a success prompt, restart the robot.

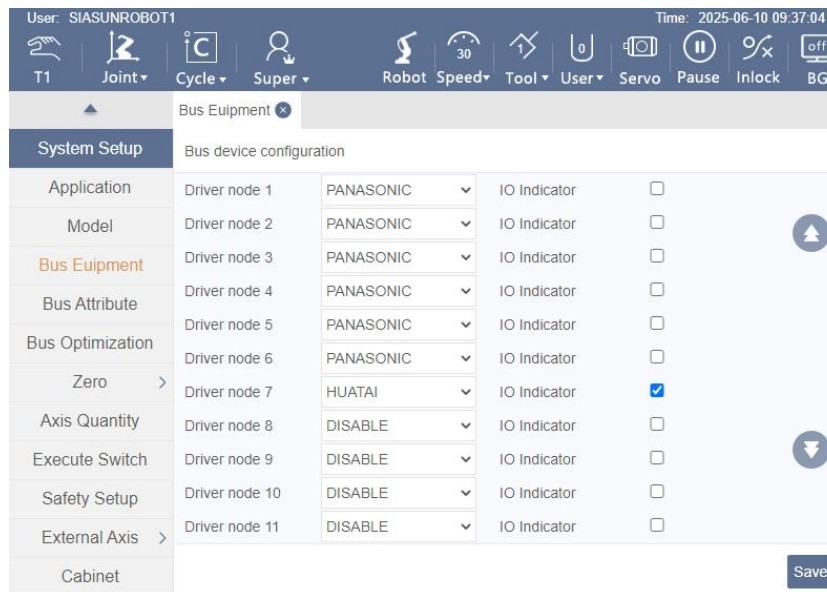


Figure 1-4 Bus Device Configuration Interface

1.1.5 Remote IO Configuration

Click [IO] - [Remote IO Map] - [Input], and configure correctly according to the on-site electrical conditions. By default, External Stop, External Pause, and External Servo Off are set to normally closed under shielding. If necessary, release shielding and perform IO setting. After the configuration is completed, click [Save].

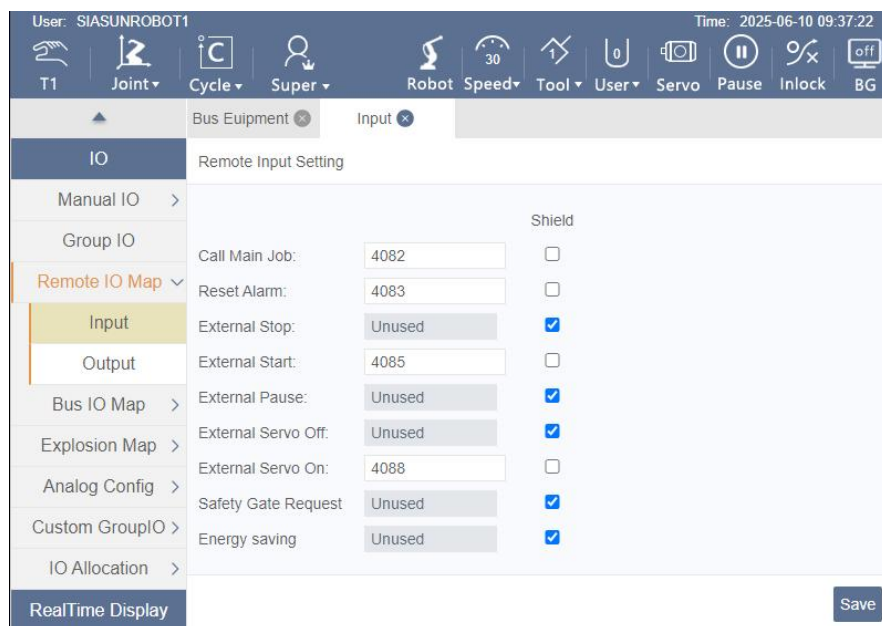


Figure 1-5 Remote Input Configuration Interface

1.1.6 Robot Application Setting

Click [System Setup] - [Application], enter the password: sysiasun, and enter the application setting interface, where you can view the applications currently configured for the robot. The application can only be configured under the system user. If any modification is needed, please contact after-sales service.

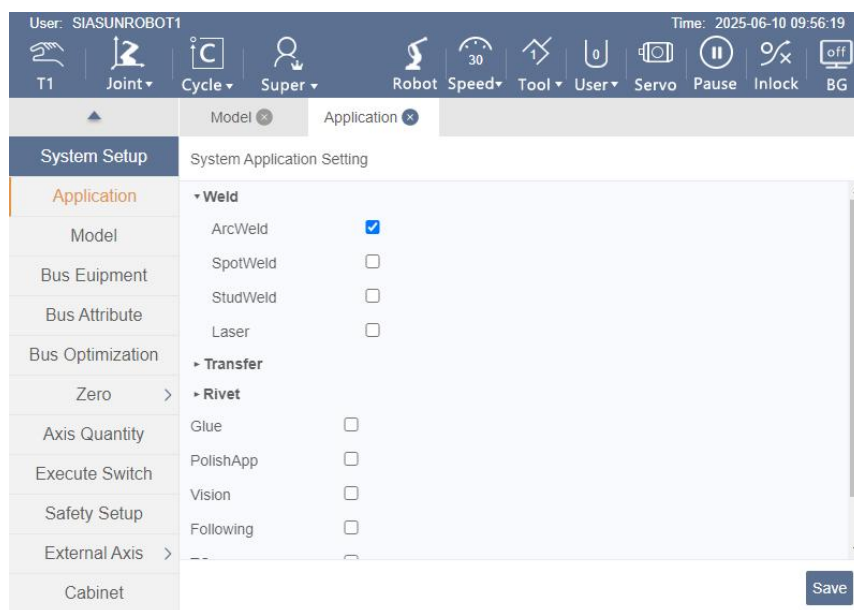


Figure 1-6 Application Configuration Interface

1.1.7 Welder Configuration

Click [Arc Weld] - [Basic Config] - [Welder Config], select the corresponding brand and model of the welder, click [Save], and restart the robot and the welder. For all supported brands of welders, please refer to Table 1-1 List of System-supported Welders.

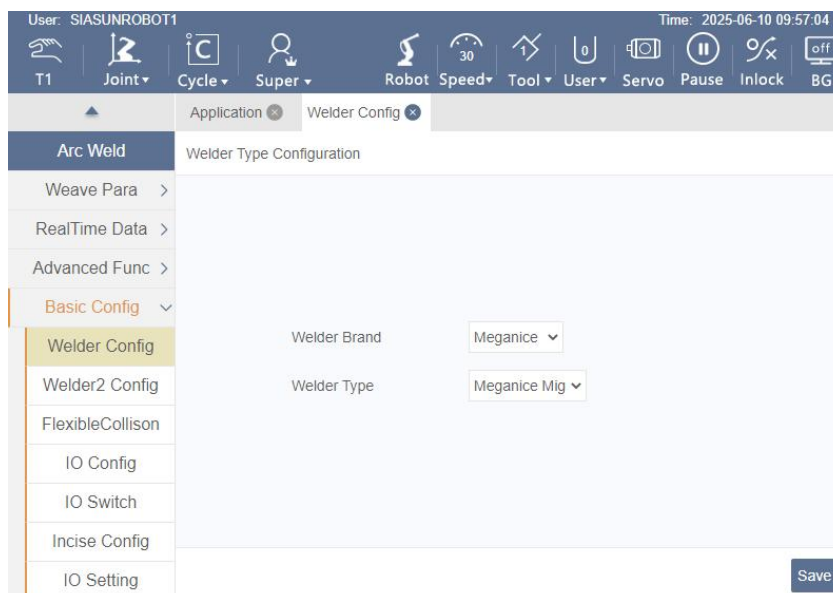


Figure 1- 7 Welder Setting Interface

Table 1- 1 List of System-supported Welders

No.	Welder type	Communication mode	Notes
1	General	DeviceNet/hard-wired	
2	Megmeet MIG	DeviceNet	In the internal menu of Megmeet welder, the communication part is set as follows: Artsen: N00 SiA, N01 2, N02 1, N05 OFF, N06 ON, N07 ON, P05 ON. DEX2: FA0 ON, FA1 ON, FA3 2, FA4 OFF, FA5 OFF, FA6 OFF, FA9 SiA.
3	Aotai	CAN/DeviceNet	In the parameters of Aotai MIG welding machine teaching box, the ID is set according to the settings on

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			the welder communication box, and the baud rate is set to 125. Restart the welder after setting.
4	Aotai TIG	CAN	
5	Aotai cutter	CAN	
6	Fronius MIG/CMT/TPSi	DeviceNet	
7	Lincoln MIG	Arclink	It is recommended to give priority to Arclink communication
8	Panasonic	DeviceNet	Panasonic 500GS: CN8 connector of the welder motherboard should be connected to 8B robot interface; the first position of the dip switch of the communication interface box should be set to ON.
9	Kemppi MIG	DeviceNet	
10	OTC MIG	DeviceNet	
11	ESAB MIG	DeviceNet	
12	Migatron TIG	DeviceNet	
13	LORCH MIG	DeviceNet	
14	EWM MIG	DeviceNet	
15	Kaierda MIG	DeviceNet	
16	Meganic MIG	CAN	
17	Hypertherm cutter	CAN/DeviceNet	

1.1.8 Welder Kit Configuration

The welding machine type kit configuration currently includes three models: Aotai Thin Plate, Megmeet Thin Plate, and Megmeet Medium-Thick Plate. After kit configuration is completed, users can start using it directly without any additional setup (default configurations include I/O, communication setup, bus mapping, etc.).

Figure 1-8 shows the configuration of Aotai kit.

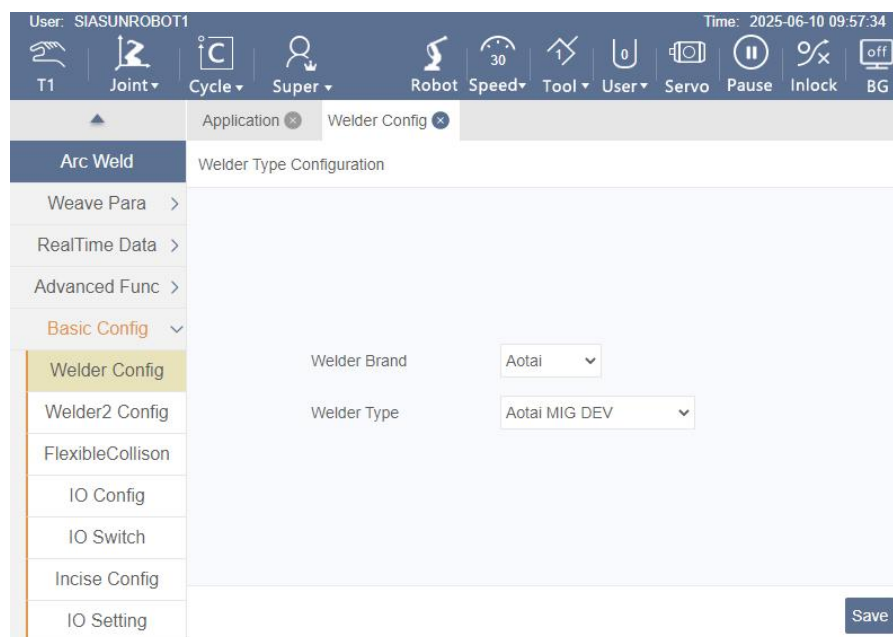


Figure 1-8 Welder Kit Configuration Interface

1.1.9 DeviceNet Communication Setup of Universal Welder

1.1.9.1 Communication Setup

- 1) Click [User Configuration] - [Peripheral Setup] - [Switch Setup], turn on the dnet switch and click [Save].

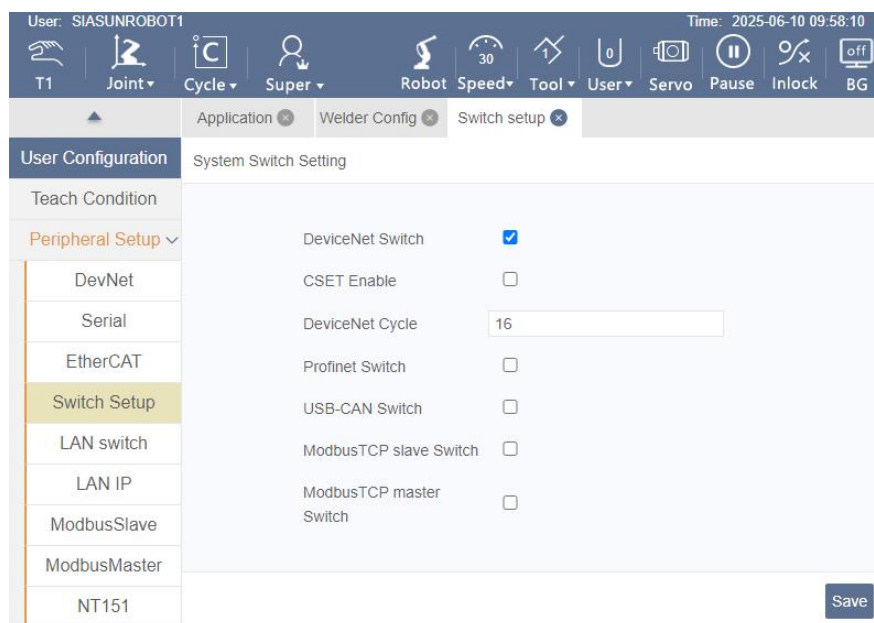


Figure 1-9 System Switch Setup Interface

- 2) Click [User Configuration] - [Peripheral Setup] - [DevNet], enter the DeviceNet slave

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setting interface, and configure the corresponding welder as shown in the following figure. Set the welder according to the model in Table 1-2, click [Save], and restart the robot after receiving a success prompt.

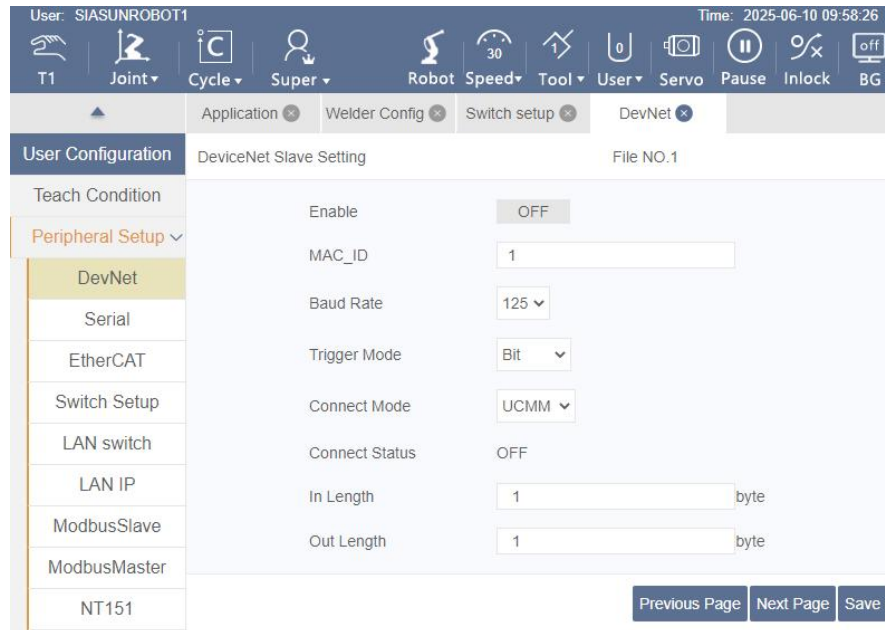


Figure 1- 10 DeviceNet Slave Setting

Table 1-2 DeviceNet Slave Setting of Welders

DeviceNet Slave Setting	Megmeet PMCM/PLUS/PMCM Tandem	Aotai MIG DEV	Fronius MIG	Fronius TWIN	Fronius TPSi	Lorch MIG
Enable state	ON	ON	ON	ON	ON	ON
MAC ID	2	8	11	11	6	2
Baud rate	125	125	500	125	500	500
Trigger mode	Polling	Polling	Polling	Polling	Polling	Polling
Connection mode	Group 2	Group 2	Group 2	UCMM	UCMM	UCMM
Input length	13	13	13	22	40	16
Output length	12	12	12	22	40	16

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DeviceNet Slave Setting	ESAB MIG	OTC MIG	Kemppi MIG	Migatroni c TIG	Panasonic DeviceNet	Kaierda Devicenet	EWM Devicenet
Enable state	ON	ON	ON	ON	ON	ON	ON
MAC ID	3	62	3	11	3	Keep	Keep
Baud rate	500	500	500	500	250	consistent on the communicat ion module of the welder	consistent on the communica tion module of the welder
Trigger mode	Polling	Polling	Polling	Polling	Polling	Polling	Polling
Connection mode	UMCC	UCMM	UCMM	UCMM	UCMM	Group 2	Group 2
Input length	16	32	8	17	16	1	12
Output length	16	32	8	13	18	6	10

1.1.9.2 IO Mapping Configuration

Click [IO] - [Bus IO Map] - [Input IO Map] to configure the IO mapping. According to the communication mode, select the corresponding IO mapping configuration interface, set the bus ID (MAC ID), IO start character and length (input and output lengths of the welder protocol). (Note that there should be no overlapped IO.) After the configuration is completed, start from the initial IO number and correspond to the input and output bits in the welder protocol. As shown in the following figure, the starting bit starts from IO number 1, totaling 13 bytes.

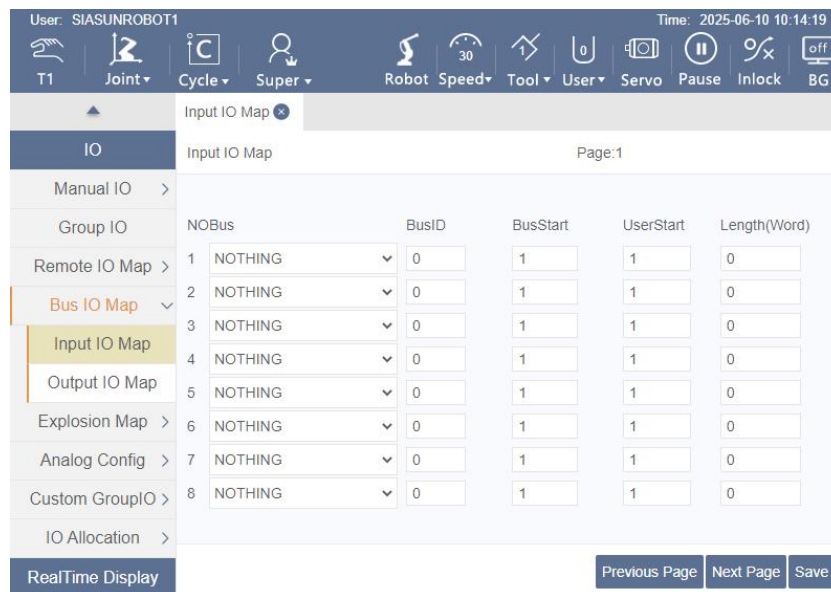


Figure 1-11 Input IO Mapping

Table 1-3 Recommended Configuration Table of Common Welders

Welder brand	Bus ID	Bus IO	IO start character	Length (character)
Fronius TPSi	Consistent with the welder ID.	The default is 0.	Set as the case may be.	20
SKS			If set to 2, the robot end starts from 17-bit IO.	6

1.1.9.3 IO Input Configuration of Welder

Click [Arc Weld] - [Welder Para] - [Welder IO Input], enter the IO input configuration interface of the welder, displaying the data received by the robot, as shown in Figure 1-12.

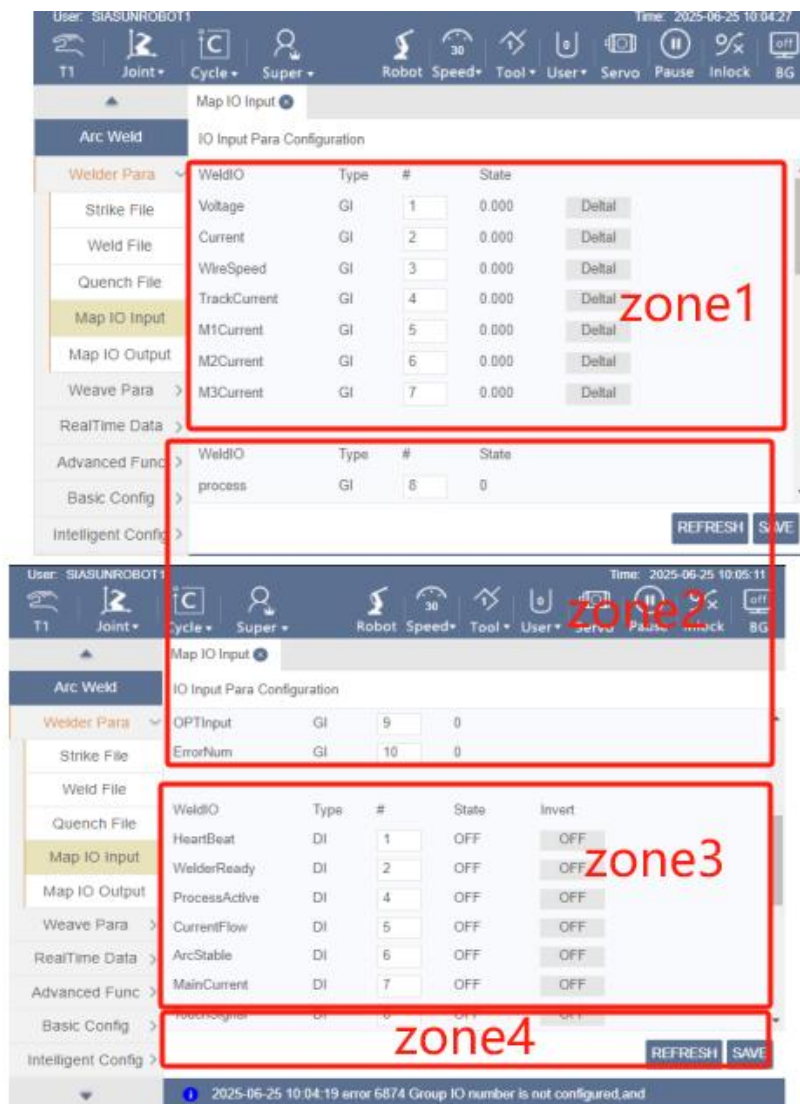


Figure 1-12 Welder IO Input Interface

1) Zone 1:

Welding signal (Column 1): The signal content input by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "GI" represents the custom group IO input, and "#" corresponds to the group number (corresponding to the custom group IO configuration interface); "DI" represents IO input, and "#" corresponds to IO number.

Status (Column 4): Displays the values calculated according to the data transformation relationship.

Detail (Column 5): Press Detail, and the data conversion input interface corresponding to the file number will pop up, as shown in Figure 1-13. The signal type and the corresponding group IO number can be displayed.

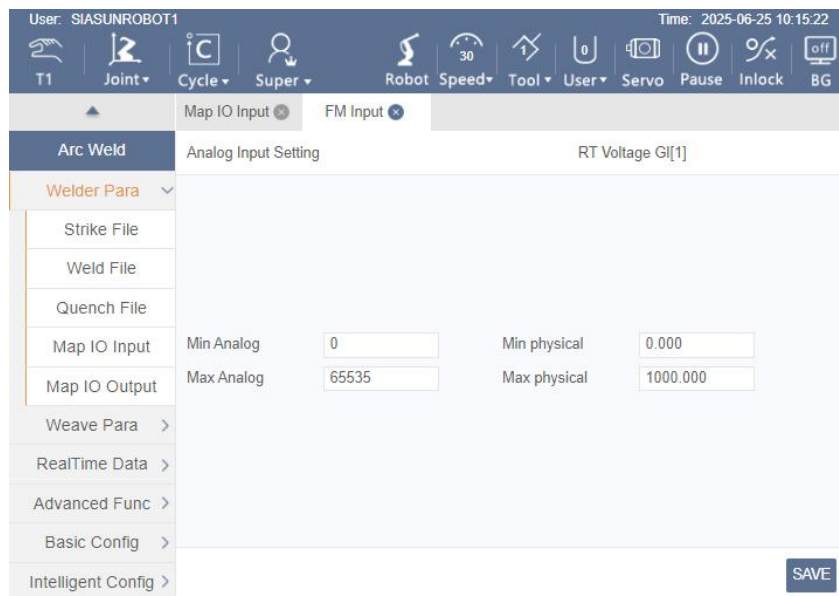


Figure 1-13 Real-time Current Data Conversion Interface

Table 1-4 Recommended Configuration Table of IO Input

IO Input Configuration	Fronius TPSi	SKS
Real-time current	1	1
Real-time voltage	2	2
Wire feeding speed	3	3
Tracking current	4	4

Table 1-5 Recommended Configuration Table of Data Conversion Input

Data Conversion Input		Fronius TPSi	SKS
Real-time current	Minimum analog quantity	0	0
	Maximum analog quantity	65535	255
	Minimum physical quantity	0	0
	Maximum physical quantity	6553.5	510
Real-time voltage	Minimum analog quantity	0	0

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	Maximum analog quantity	65535	255
	Minimum physical quantity	0	0
	Maximum physical quantity	655.35	51
Wire feeding speed	Minimum analog quantity	0	0
	Maximum analog quantity	32767	255
	Minimum physical quantity	0	0
	Maximum physical quantity	327.67	25.5
Tracking current	Minimum analog quantity	0	0
	Maximum analog quantity	65536	255
	Minimum physical quantity	0	0
	Maximum physical quantity	65536	510

2) Zone 2:

Welding signal (Column 1): The signal content input by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "GI" represents the custom group IO input, and "#" corresponds to the group number (corresponding to the custom group IO configuration interface);

Status (Column 4): Displays the values of the configured group I/O.

Table 1-6 Recommended Configuration Table of Welding Signal (Error Code)

Welding Signal (Error Code)	Fronius TPSi	SKS
		5

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3) Zone 3:

Welding signal (Column 1): The signal content input by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "DI" represents IO input, and "#" represents the corresponding IO number.

Status (Column 4): Displays the corresponding IO status, ON or OFF.

Whether it is reverse (Column 5): ON means reverse, and the signal judgement input by the welder is reversed. It is OFF by default, 0->0 and 1->1. If the reverse switch is set to ON, it will be 0->1 and 1->0. For example, by default, it is normal 0 when the welding torch collision signal is 0; it is collision 1 when the welding torch collision signal is 1. After ON is reversed, when the welding torch collision signal is 0, it is collision 1; when the welding torch collision signal is 1, it is normal 0.

Table 1-7 Recommended Configuration Table of Welding IO Input

Welding IO Input	Fronius TPSi (IO start character is 2)	SKS
Successful arc striking	37	35
Welding torch collision	41	0
During welding	37	35
Uphill end		0
Touch-sensing	40	36
Welder ready	34	40
Wire feeder ready		0
Welder error		37
Wire feeder error		

4) Zone 4:

Refresh: Press Refresh to update the content of the current configuration, including the display of values.

Save: Save the modified configuration.

1.1.9.4 IO Output Configuration of Welder

Click [Arc Weld] - [Welder Para] - [Welder IO Output], enter the IO output configuration

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interface of the welder, displaying the data received by the robot, as shown in Figure 1-14.

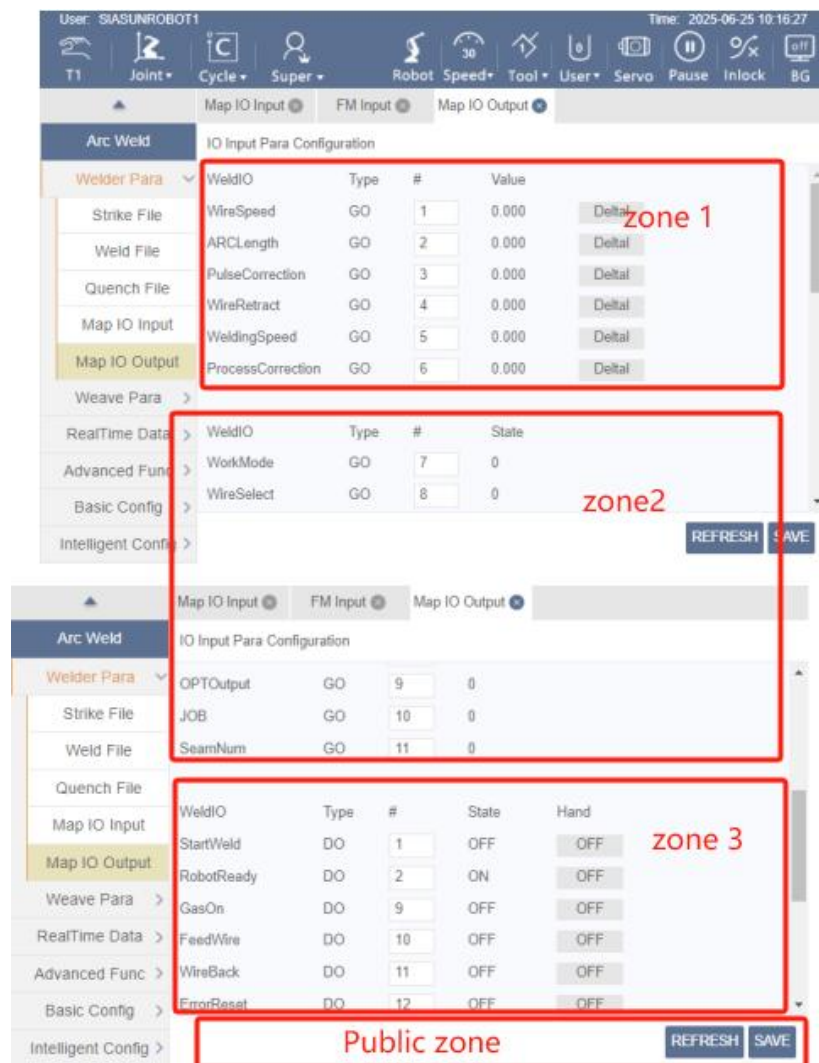


Figure 1-14 IO Output Interface of Welder

1) Zone 1:

Welding signal (Column 1): The signal content output by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "GO" represents the group IO output, and "#" represents the corresponding group number (corresponding to the custom group IO configuration interface);

Status (Column 4): Displays the values calculated according to the data transformation relationship.

Detail (Column 5): The corresponding data conversion output interface pops up. The data conversion output interface is used with the IO input and output interfaces of the welder. It is used to adjust the linear relationship of parameters such as output welding current and voltage, as shown in Figure 1-15.

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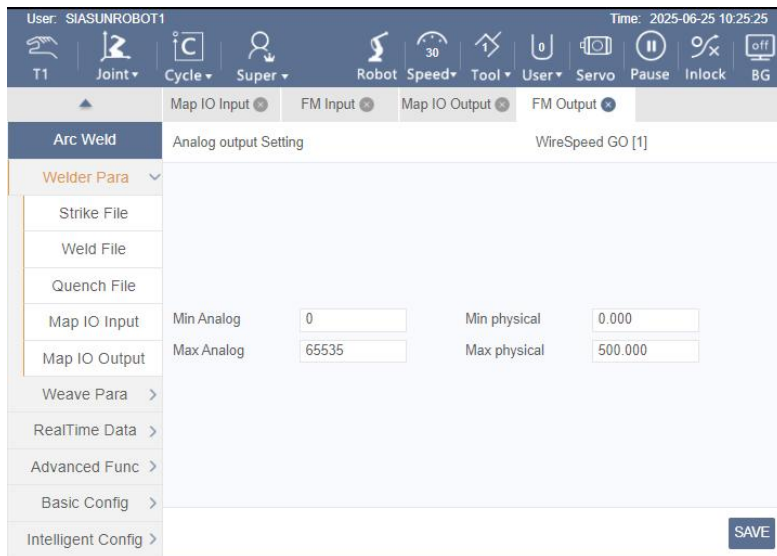


Figure 1-15 Welding Voltage Data Conversion Interface

Table 1-8 Recommended Configuration Table of Welding IO Output

IO Output Configuration	Fronius TPSi JOB mode	SKS JOB mode
Welding current		
Welding voltage		
Wire feeding speed		

2) Zone 2:

Welding signal (Column 1): The signal content output by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "GO" represents the group IO output, and "#" represents the corresponding group number (corresponding to the custom group IO configuration interface);

Status (Column 4): Displays the values of the configured group I/O.

Table 1-9 Recommended Configuration Table of IO Output

IO Output Setting	Fronius TPSi JOB mode	SKS
Operating mode	1	0
Storage channel	2	1

Note: For SKS welder, the groups G and P need to be issued through the group IO.

3) Zone 3:

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Welding signal (Column 1): The signal content output by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "DO" represents IO output, and "#" represents the corresponding IO number.

Status (Column 4): Displays the corresponding IO status, ON or OFF.

Table 1-10 Recommended Configuration Table of Welding IO Output

Welding IO Output	Fronius TPSi JOB mode	SKS
Welding start	33	33
Robot ready	34	0
Air supply	41	34
Wire feeding	42	35
Wire back drawing	43	36
Fault resetting	44	0
Sensing enable	45	38
Burnback Enable		0

4) Public zone:

Refresh: Press Refresh to update the content of the current configuration, including the display of values.

Save: Save the modified configuration.

5) Notes:

- a) According to the IO output configuration of the welder, if the corresponding welding parameter positions are configured, the corresponding parameters will be displayed in the welding parameters (General File, Strike File, Weld File, Quench File, and Burnback file), as shown in the following figure. The IO output in the welding parameters is configured with the welding mode, welding current, and welding voltage. Then the welding parameters in the striking file display the corresponding items.

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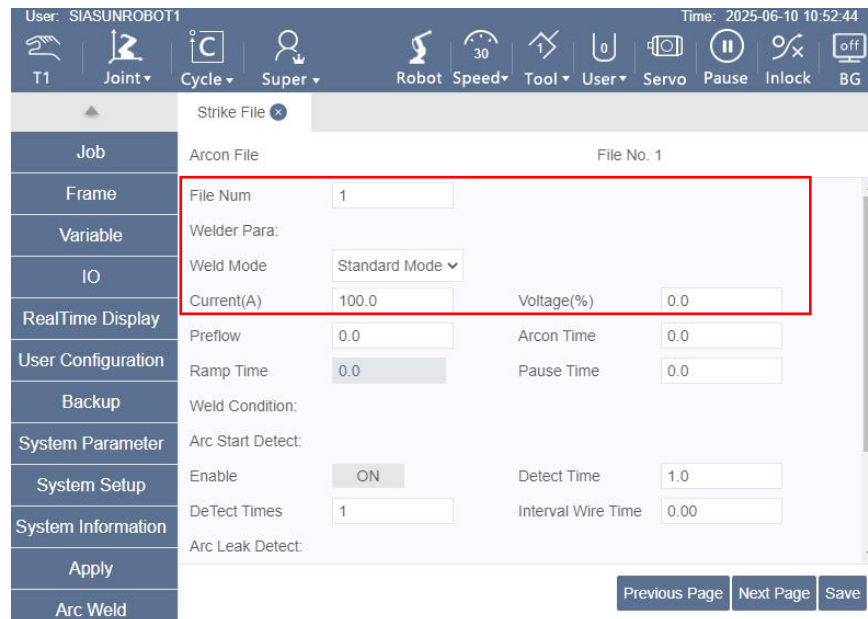


Figure 1-16 Strike File Configuration Interface

- b) In the general parameters, the welding torch collision switch requires the configuration of the welding torch collision signal in the input signal of the welder.
- c) At present, only Lincoln arclink welders have burn-back files, while burnback files of other welders are not displayed.

Table 1-11 Welding Mode Configuration Table

	Fronius TPSi	SKS
	JOB mode	JOB mode
Welding Mode	2	

1.1.9.5 Custom Group IO Configuration

Click [IO] - [Custom Group IO] - [Input/Output], and enter the custom group IO configuration interface.

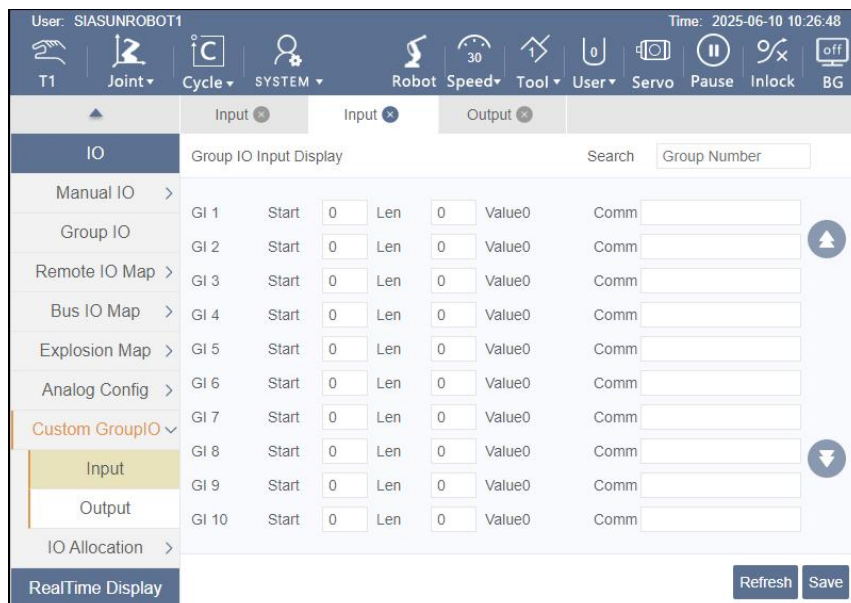


Figure 1-17 Custom Group IO Input Configuration Interface

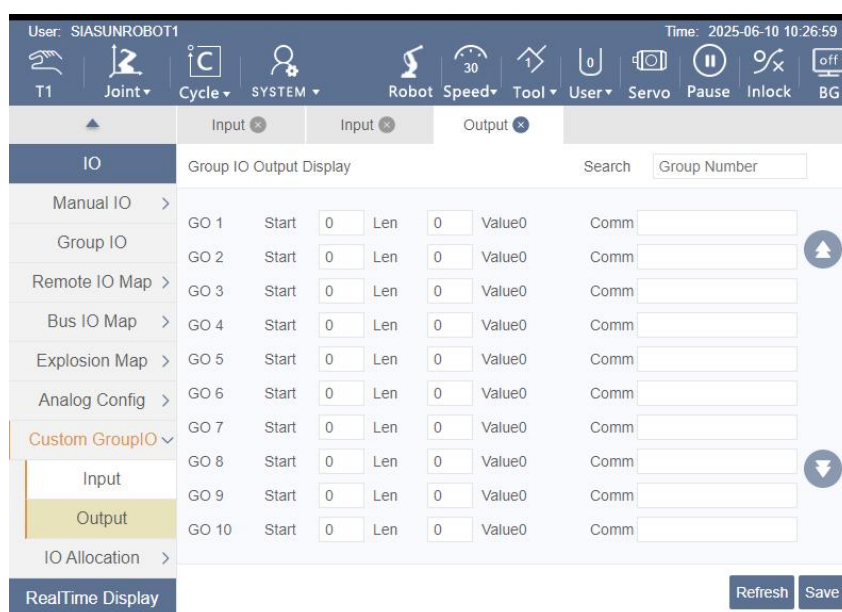


Figure 1-18 Custom Group IO Output Configuration Interface

Each horizontal row represents a set of IO configurations, with GI/GO1 as the group IO number, totaling 100 groups. The starting IO is the starting IO number corresponding to the current group IO (range: 1-1024), the length corresponds to the current group IO (range: 1-32), the value is the status value of the current group IO, and the comment can be added by the operator; press "Refresh" to refresh the value of the current group IO. Press "Save" to save the current configuration.

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Table 1-12 Recommended Configuration Table of Custom Group IO Input

Custom Group IO Input		Fronius TPSi	SKS
GI 1	Starting IO	113	81
	Length	16	16
	Notes	Real-time current	Real-time current
GI 2	Starting IO	97	65
	Length	16	16
	Notes	Real-time voltage	Real-time voltage
GI 3	Starting IO	129	97
	Length	16	16
	Notes	Wire feeding speed	Wire feeding speed
GI 4	Starting IO	145	81
	Length	16	16
	Notes	Tracking coefficient	Tracking current
GI 5	Starting IO	161	41
	Length	16	7
	Notes	Alarm code	Alarm code
GI 6	Starting IO		
	Length		
	Notes		

Table 1-10 Recommended Configuration Table of Custom Group IO

Custom group IO		Fronius TPSi JOB mode	SKS JOB mode
GO 1	Starting IO	35	49
	Length	5	3
	Notes	Operating mode	Group P

GO 2	Starting IO	97	
	Length	16	
	Notes	Storage channel	
GO 3	Starting IO		
	Length		
	Notes		

1.1.10 Analog Communication Configuration of Universal Welder

The welder and the robot are hard-wired, and the current, voltage, etc. are transmitted through analog input and output modules (Beckhoff or Huatai).

1.1.10.1 Basic Configuration

Check hard-wiring between the welder and the robot, and sort the signal lines of the user's IO board corresponding to the welder.

Configure the bus device according to the wiring situation. The nodes corresponding to the analog input and output modules are shown in Figure 1-19.

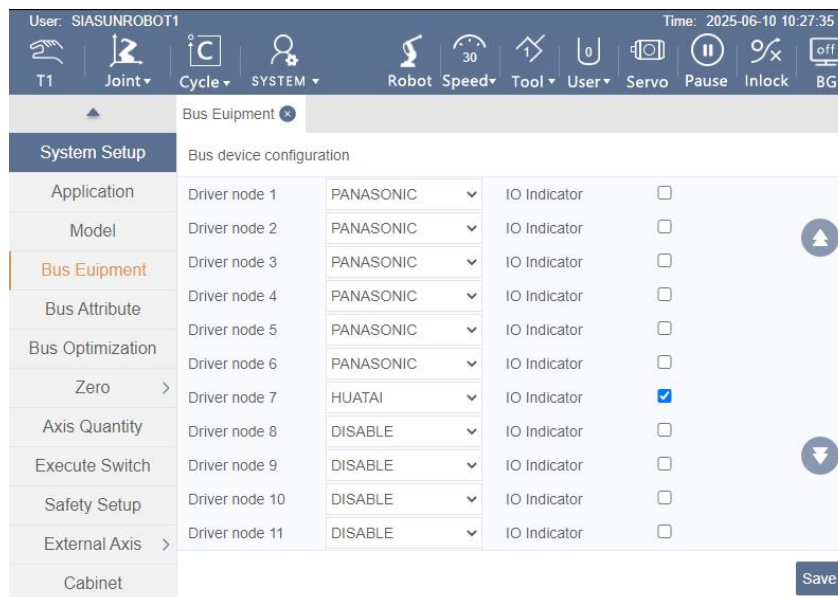


Figure 1-19 Bus Device Configuration Interface

1.1.10.2 IO Input Configuration of Welder

Click [Arc Welding] - [Welder Para] - [Welder IO Input], click Previous Page and Next Page to configure each function. After the configuration is completed, click Save.

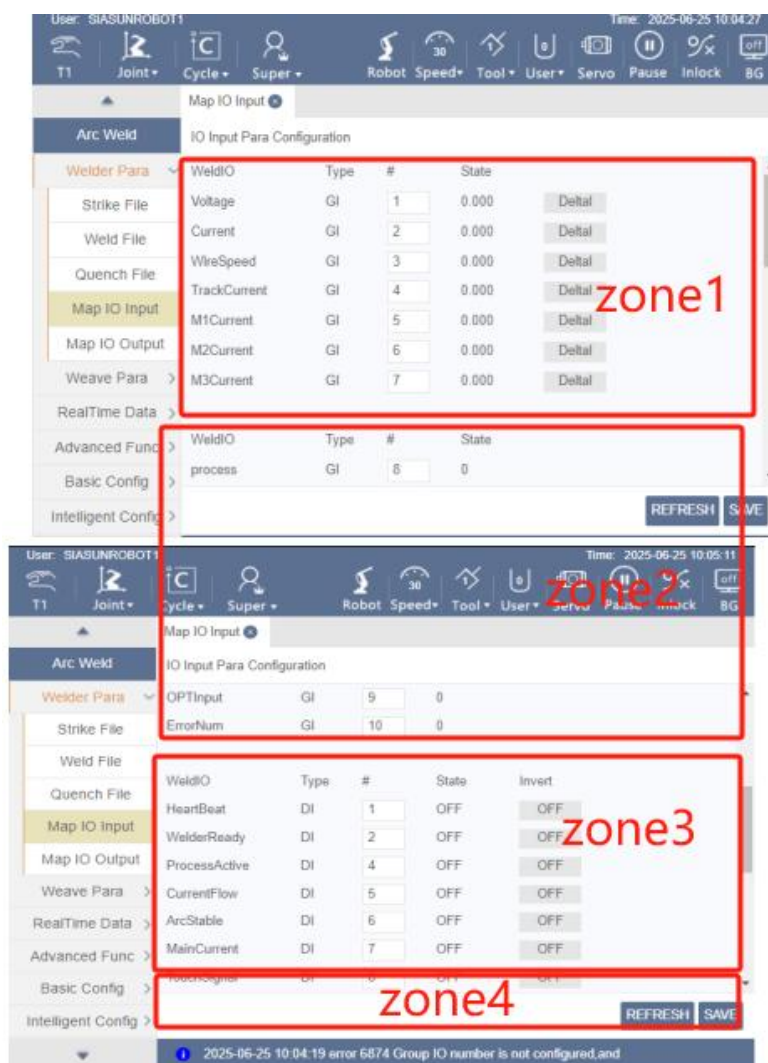


Figure 1-20 IO Input Configuration Interface of Welder

1) Zone 1:

Welding signal (Column 1): The signal content input by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Status (Column 2): Displays the values calculated according to the analog quantity IO output transformation relationship.

Detail: Press Detail, and the analog IO input interface pops up. The analog IO input interface is the path [IO] - [Analog Config] - [Input], which is used with the IO input interface of the welder.

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It is used to adjust the linear relationship of input parameters such as current and voltage, as shown in Figure 1-21. (The default file numbers 1-4 represent real-time current - tracking current)

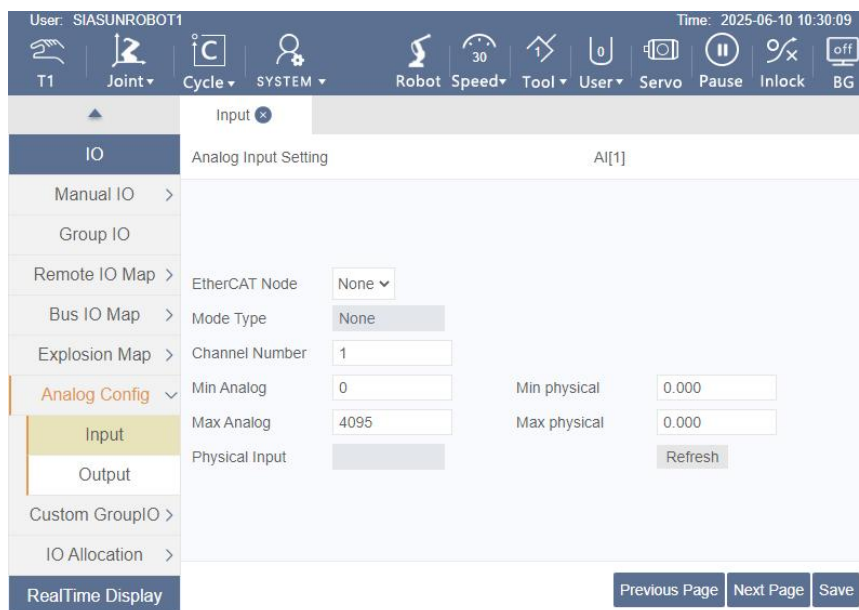


Figure 1-21 Analog Input Parameter Configuration Interface

2) Zone 2:

Welding signal (Column 1): The signal content input by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "GI" represents the custom group IO input, and "#" corresponds to the group number (corresponding to the custom group IO configuration interface);

Status (Column 4): Displays the values of the configured group I/O.

Table 1-14 Recommended Configuration Table of Welding Signal (Error Code)

Welding Signal (Error Code)	Fronius TPSi	SKS
		5

3) Zone 3:

Welding signal (Column 1): The signal content input by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "DI" represents IO input, and "#" represents the corresponding IO number.

Status (Column 4): Displays the values of the configured I/O.

Whether it is reverse (Column 5): ON means reverse, and the signal judgement input by the welder is reversed. It is OFF by default, 0->0 and 1->1. If the reverse switch is set to ON, it will

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be 0->1 and 1->0. For example, by default, it is normal 0 when the welding torch collision signal is 0; it is collision 1 when the welding torch collision signal is 1. After ON is reversed, when the welding torch collision signal is 0, it is collision 1; when the welding torch collision signal is 1, it is normal 0.

Table 1-15 Recommended Configuration Table of Welding IO Input

Welding IO Input	Fronius TPSi (IO start character is 2)	SKS
Successful arc striking	37	35
Welding torch collision	41	0
During welding	37	35
Uphill end		0
Touch-sensing	40	36
Welder ready	34	40
Wire feeder ready		0
Welder error		37
Wire feeder error		

4) Public zone:

A: Refresh: Press Refresh to update the content of the current configuration, including the display of values.

B: Save: Save the modified configuration.

1.1.10.3 IO Output Configuration of Welder

Click [Arc Weld] - [Welder Para] - [Welder IO Output], enter the IO output configuration interface of the welder, displaying the data received by the robot, as shown in Figure 1-22.

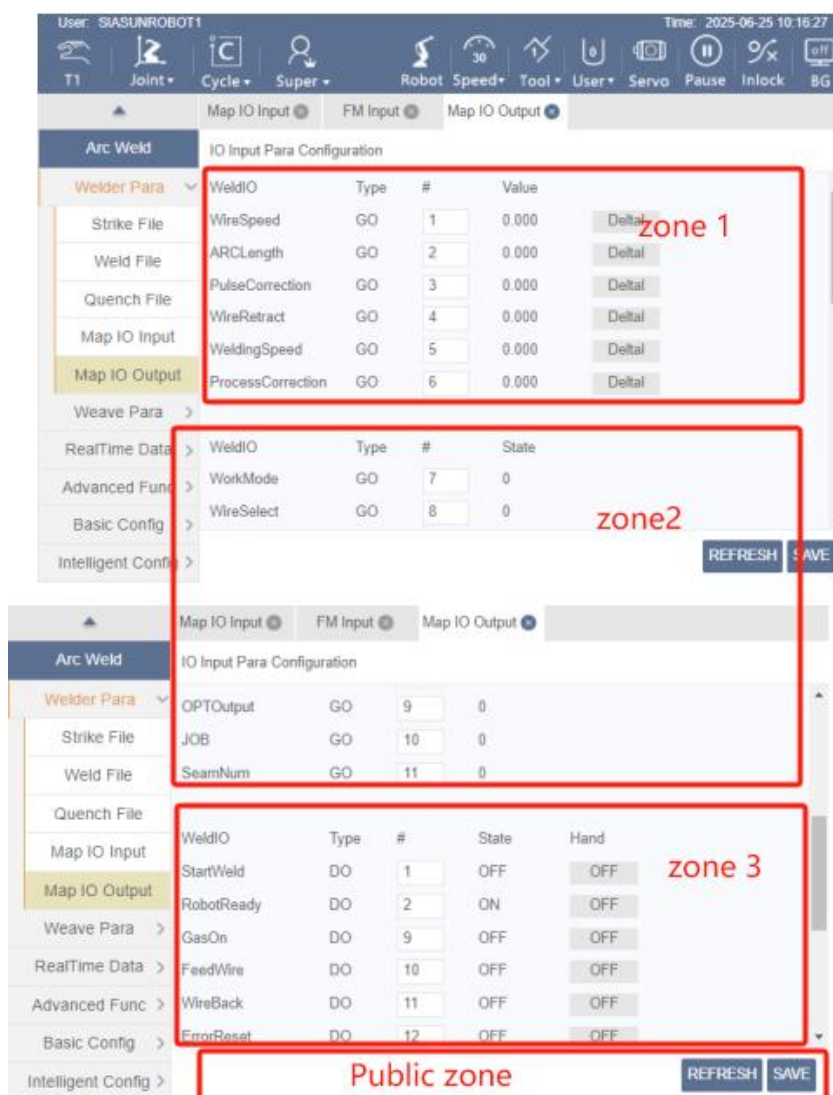


Figure 1-22 IO Output Configuration Interface of Welder

1) Zone 1:

Welding signal (Column 1): The signal content output by the universal welder, with the corresponding configuration of the signal in the horizontal direction.

Detail (Column 2): The corresponding analog IO output interface pops up. Analog IO output interface is used with the IO output interface of the welder. It is used to adjust the linear relationship of parameters such as output welding current and voltage, as shown in Figure 1-23. (The default analog IO output files 1-9 correspond to the welding current - parameter 5).

Table 1-6 Recommended Configuration Table of Data Conversion Output

Data Conversion Output		Real-point Analog Module
Welding	Minimum analog quantity	0

current	Maximum analog quantity	32767
	Minimum physical quantity	0
	Maximum physical quantity	2000
Welding voltage	Minimum analog quantity	0
	Maximum analog quantity	32767
	Minimum physical quantity	0
	Maximum physical quantity	2000
Wire feeding speed	Minimum analog quantity	0
	Maximum analog quantity	10
	Minimum physical quantity	0
	Maximum physical quantity	Fill in according to the parameters of the wire feeder

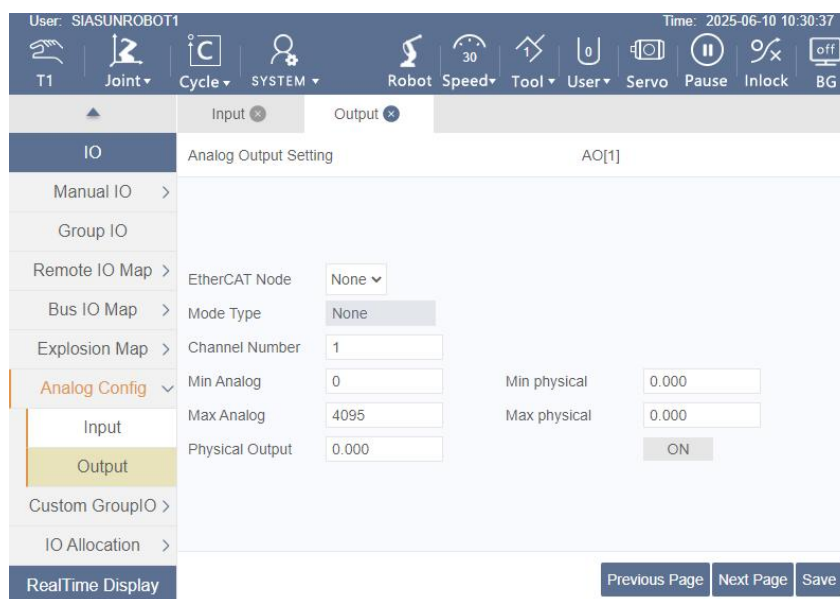


Figure 1-23 Analog Output Parameter Configuration Interface

2) Zone 2:

Welding signal (Column 1): The signal content output by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "GO" represents the group IO output, and "#" represents the corresponding group number (corresponding to the custom group IO configuration interface);

Status (Column 4): Displays the values of the configured group I/O output.

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3) Zone 3:

Welding signal (Column 1): The signal content output by the universal welder, with the corresponding configuration of the signal in the horizontal direction;

Type # (Columns 2 and 3): "DO" represents IO output, and "#" represents the corresponding IO number.

Status (Column 4): Displays the corresponding IO status, ON or OFF.

Table 1-17 Recommended Configuration Table of Welding IO Output

Welding IO Output	Fill in the port number according to the IO module wiring and the functions used
Welding start	For example, OUT#1
Robot ready	OUT#2
Air supply	OUT#3
Wire feeding	OUT#4
Wire back drawing	
Fault resetting	
Sensing enable	
Burnback Enable	

4) Public zone:

Refresh: Press Refresh to update the content of the current configuration, including the display of values.

Save: Save the modified configuration.

5) Notes:

- a) According to the IO output configuration of the welder, if the corresponding welding parameter positions are configured, the corresponding parameters will be displayed in the welding parameters (General File, Strike File, Weld File, Quench File, and Burnback file), as shown in the following figure. The IO output in the welding parameters is configured with the welding mode, welding current, and welding voltage. Then the welding parameters in the striking file display the corresponding items.

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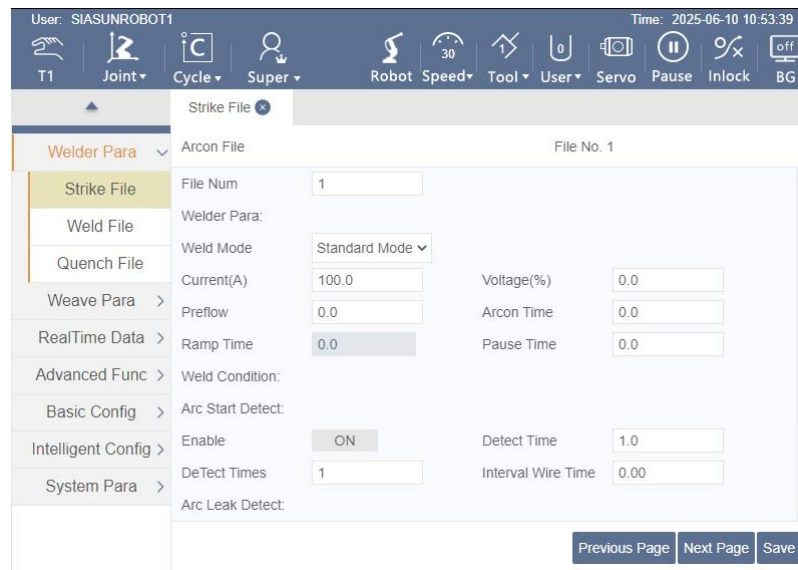


Figure 1-24 Strike File Configuration Interface

- b) In the general parameters, the welding torch collision switch requires the configuration of the welding torch collision signal in the IO input signals of the welder.
- c) At present, only Lincoln arclink welders have burn-back files, while burnback files of other welders are not displayed.

1.1.11 Execute Switch Configuration

Click [System Setup] - [Execute Switch], set the appropriate switch according to on-site environment, and click [Save]. The application configuration of conventional arc welding is shown in Figure 1-25.

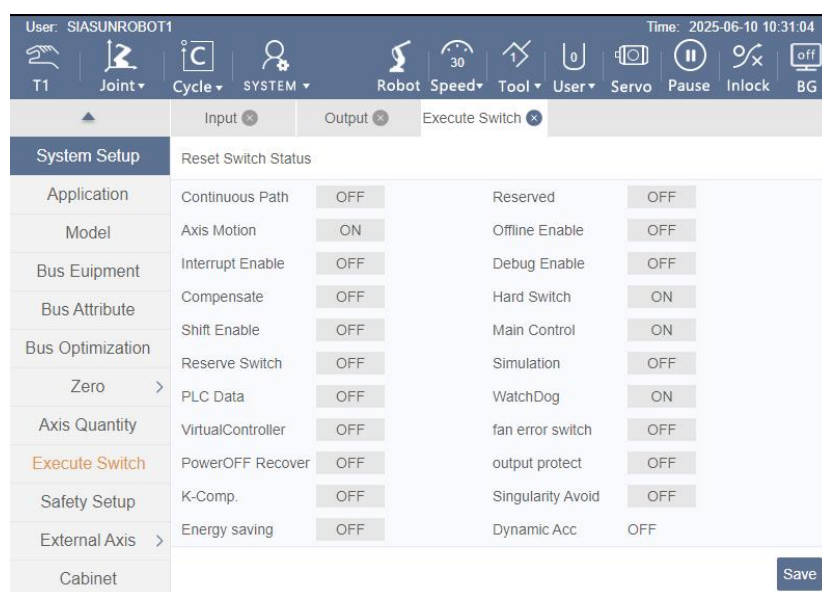


Figure 1-25 Execute Switch Configuration Interface

1.2 Zero-digit Check of Arc Welding Robot

1.2.1 Zero position setting

Click [System Parameter] - [Zero-digit] - [Zero Setup], and check whether the zero-digit setting value of the robot is consistent with the zero-digit code disk label.

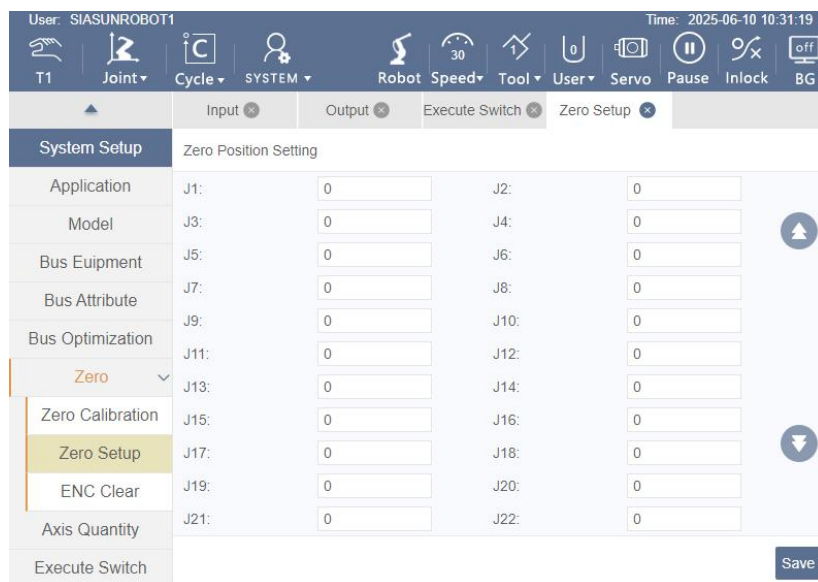


Figure 1-26 Zero-digit Setting Interface

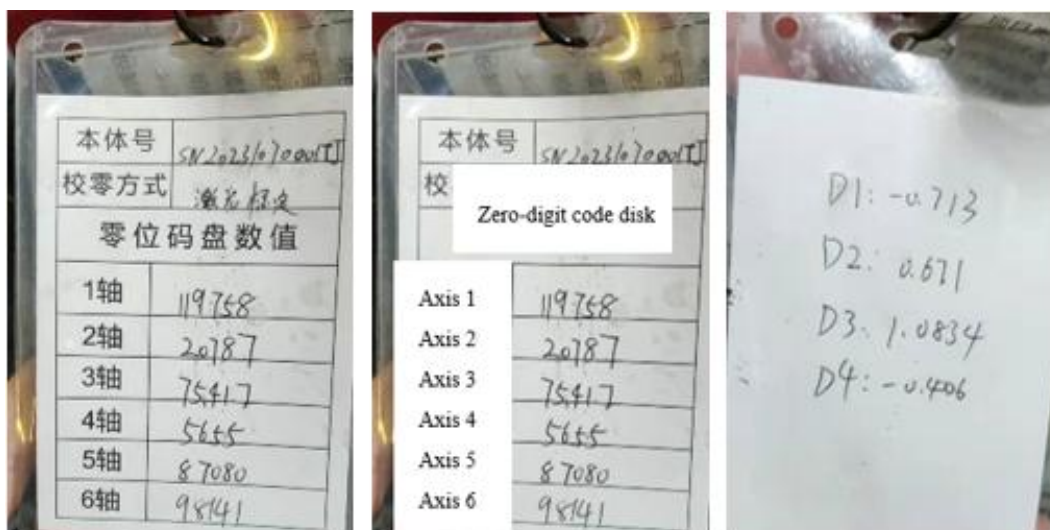


Figure 1-27 Zero-digit code disk Label

1.2.2 Check Zero-digit Label

Click [RealTime Display] - [RealTime DISP] - [Joint Value], operate the robot, move each

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axis to the zero-digit label of the robot, and check the current joint value. Taking SR10C robot as an example, the real-time display of joint values is shown in Figure 1-28. Check whether the factory zero-digit is correct.

Note: For different types of robots, the real-time joint values of the five axes corresponding to the zero-digit tag position are different.

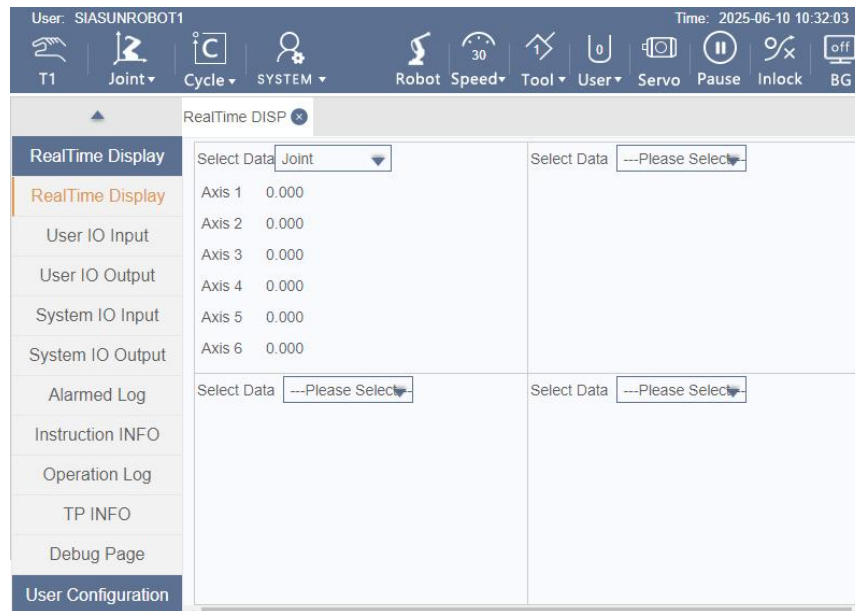


Figure 1-28 Joint Value Real-time Display Interface

1.3 TCP Calibration of Arc Welding Robot

1.3.1 5-point Calibration

The 5-point method can determine the TCP, that is, the x, y and z values of the origin of tool frame relative to the flange center point. Use the manual motion robot to make the robot TCP point to the same point in five different poses. The system will automatically calculate the TCP position value. During teaching, try to approach the target point with five different poses in different directions.

Note: The tip of the external calibration needle should be higher than the origin of the rectangular frame (the point where the first axis intersects the second axis), as shown in Figure 1-29.

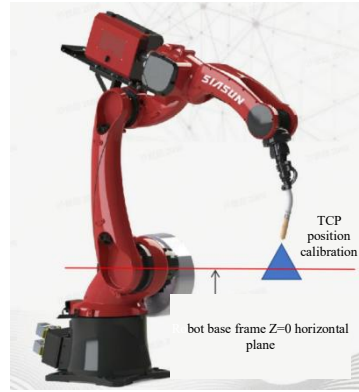


Figure 1-29 Schematic Diagram of External Calibration Needle Requirement

After calibration by the 5-point method, only the TCP points (x, y, z) of the tool can be determined. At this time, the pose values (Rx, Ry, Rz) of the tool are the default (0,0,0).

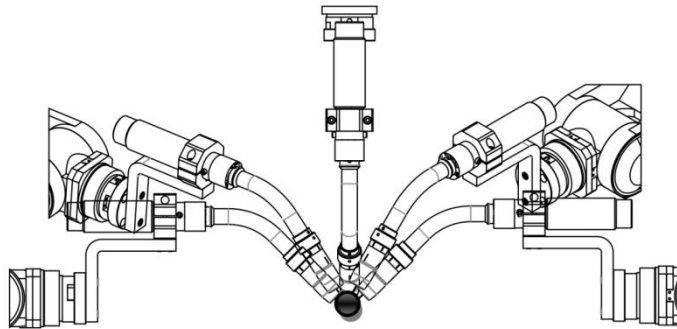


Figure 1-30 5-point Calibration Diagram

1.3.2 Tool Pose Calibration

Use the 5-point method to calibrate the tool TCP, and then calibrate the pose. Calibrate the tool pose at any point in space, then calibrate the x-direction of the tool along the parallel direction of the X-axis, and finally calibrate the Z positive direction of the tool at a point on the XZ plane in the positive direction of the tool z-axis.

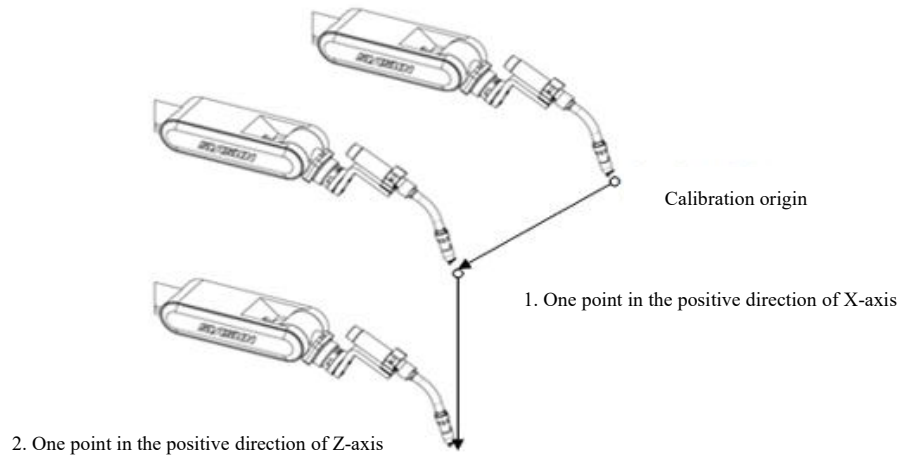


Figure 1-31 Pose Calibration Diagram

1.3.3 Calibration of Tool TCP

- 1) Click [Frame] - [Tool Frame] - [Calibration] to enter the tool frame calibration interface.

The tool number to be calibrated can be selected through the page up and down buttons.

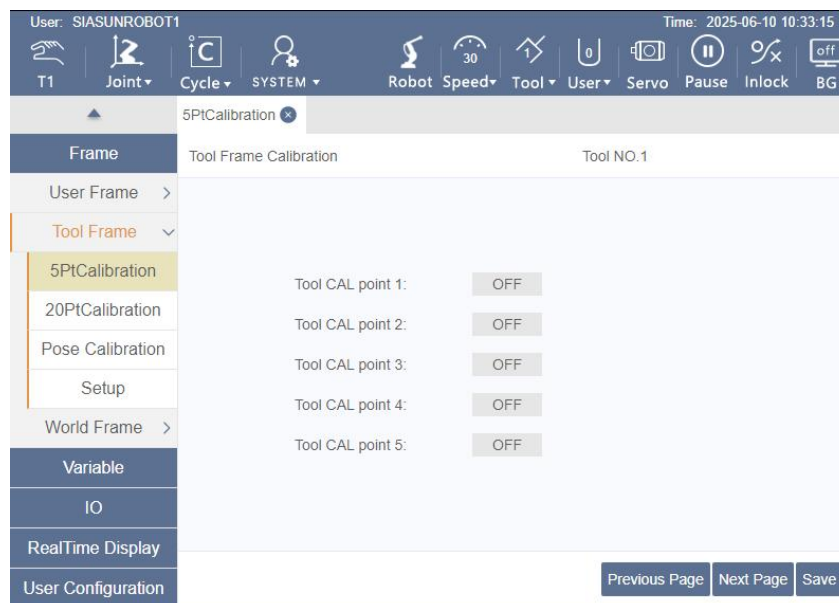


Figure 1-32 Tool Frame Calibration Interface

- 2) Approach the same target point in five different poses in sequence and click "OFF".

When "OFF" changes to "ON", this point is successfully calibrated. Click [Save] to save and end the calibration. Press "Next Page" to calibrate another tool frame. There are a total of 8 tool frames. The number is displayed on the tool number in the upper right corner of the interface.

1.3.4 Pose Calibration

- 1) Click [Frame] - [Tool Frame] - [Pose Calibrate], and click Previous Page or Next Page to select the desired tool number.

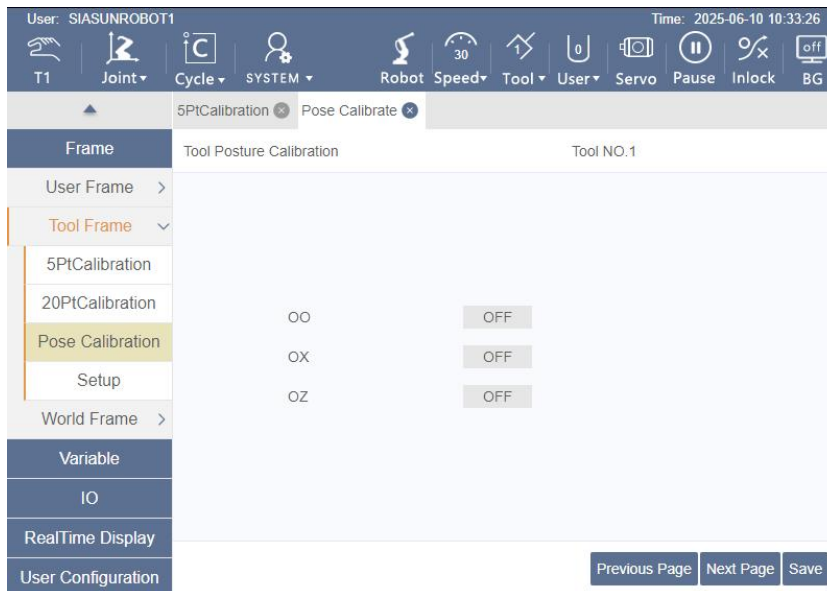


Figure 1-33 Tool Frame Pose Calibration Interface

- 2) First, calibrate the selected points of TCP in space, and then calibrate a point in the X direction of the tool frame and a point on the XZ plane in the positive direction of the z-axis. At the three points, click OFF/ON to calibrate.
- 3) After calibration is completed, press [Save].

Notes:

- 1) If the tool frame has been calibrated for the calibrated coordinate number, the pose value of this tool frame may not be (0,0,0). Please go to the tool setting and change it to (0,0,0) again, and then recalibrate the pose.
- 2) TCP and pose in the same frame need to be calibrated under the same tool number.
- 3) If the system has external axis configuration, for the calibration of the external axes, please refer to Operating Manual for SIASUN Robot.

1.4 Welder Parameter Configuration

1.4.1 Welding Instructions

Welding instructions mainly include strike instructions, quench instructions, and welding

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parameter modification instructions. Strike processing is not supported in Step mode. Descriptions are as follows:

ARCON	Function	Welding start.	
	Format	ARCON #<Parameter 1> #<Parameter 2>	
	Description	Parameter 1	Strike file number, range: 1-32.
		Parameter 2	Weld file number, range: 1-32.
	Examples	ARCON #1 #3 // Strike using the strike file number 1 and begin welding with the welding parameters in the weld file number 3.	
ARCOFF	Function	Welding end.	
	Format	ARCOFF #<Parameter 1>	
	Description	Parameter 1	Quench file number, range: 1-32.
	Examples	ARCOFF #1 // Quench using the quench file number 1.	
ARCON2	Function	Quick-change function of welding torch: When connecting the two welders, the welder 2 begins welding.	
	Format	ARCON2 #<Parameter 1> #<Parameter 2>	
	Description	Parameter 1	Strike file number, range: 1-32.
		Parameter 2	Weld file number, range: 1-32.
	Examples	ARCON2 #1 #1 // Strike using the strike file number 1 and begin welding with the welding parameters in the weld file number 1.	
ARCOFF2	Function	Quick-change function of welding torch: When connecting the two welders, the welder 2 ends welding.	
	Format	ARCOFF #<Parameter 1>	
	Description	Parameter 1	Quench file number, range: 1-32.
	Examples	ARCOFF2 #1 // Quench using the quench file number 1.	
PARA_S	Function	It is used when modifying welding parameters.	
	Format	PARA SWITCH #<Parameter 1>	
	Description	Parameter 1	Weld file number, range: 1-32.
	Examples	PARA SWITCH #1 // Perform welding using the parameters of weld file number 1.	

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ARCON_U I	Function	The strike instruction can set the strike current and voltage through the instruction parameters.	
	Format	ARCON_UI #Parameter 1 #Parameter 2 I=Parameter 3 U=Parameter 4	
	Description	Parameter 1	Strike file number, range: 1-32.
		Parameter 2	Weld file number, range: 1-32.
		Parameter 3	Strike current value.
		Parameter 4	Strike voltage value.
	Examples	ARCON_UI #1 #1 I=200 U=20	
ARCOFF_ UI	Function	The quench instruction can set the quench current and voltage as well as the quench time through the instruction parameters.	
	Format	ARCOFF_UI #Parameter 1 I=Parameter 2 U=Parameter 3 T=Parameter 4	
	Description	Parameter 1	Quench file number, range: 1-32.
		Parameter 2	Quench current.
		Parameter 3	Quench voltage.
		Parameter 4	Quench time.
	Examples	ARCOFF_UI #1 I=150 U=15 T=1.0	
PARA_S_U I	Function	The welding parameter instruction can set the welding current and voltage through the instruction parameters.	
	Format	PARA_S_UI #Parameter 1 I=Parameter 2 U=Parameter 3	
	Description	Parameter 1	Weld file number, range: 1-32.
		Parameter 2	Switch the welding current.
		Parameter 3	Switch the welding voltage.
	Examples	PARA_S_UI #1 I=250 U=24	
ARCON_U I2	Function	The strike instruction of the welder 2 can set the strike current and voltage through the instruction parameters.	
	Format	ARCON_UI2 #Parameter 1 #Parameter 2 I=Parameter 3 U=Parameter 4	
	Description	Parameter 1	Strike file number, range: 1-32.

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		Parameter 2	Weld file number, range: 1-32.
		Parameter 3	Strike current value.
		Parameter 4	Strike voltage value.
	Examples	ARCON_UI2 #1 #1 I=200 U=20	
ARCOFF_ UI2	Function	The quench instruction of the welder 2 can set the quench current and voltage as well as the quench time through the instruction parameters.	
	Format	ARCOFF_UI2 #Parameter 1 I=Parameter 2 U=Parameter 3 T=Parameter 4	
	Description	Parameter 1	Quench file number, range: 1-32.
		Parameter 2	Quench current.
		Parameter 3	Quench voltage.
		Parameter 4	Quench time.
	Examples	ARCOFF_UI2 #1 I=150 U=20 T=1.0	
PARA_S_U I2	Function	The welding parameter instruction of the welder 2 can set the welding current and voltage through the instruction parameters.	
	Format	PARA_S_UI2 #Parameter 1 I=Parameter 2 U=Parameter 3	
	Description	Parameter 1	Weld file number, range: 1-32.
		Parameter 2	Switch the welding current.
		Parameter 3	Switch the welding voltage.
	Examples	PARA_S_UI2 #1 I=250 U=24	

During the welding process, the strike, weld and quench parameters can be modified by calling different files through instructions. However, there is only one file for the burnback parameters, and all the burnback processes of welding within the system use the same parameters.

The welding parameter modification instruction is used to modify the welding parameters when the multi-pass welding function is not in use. If the welding parameters are required to be modified for the weld line within the multi-pass welding cycle interval needs, use the multi-pass welding database instruction (for details, see 5 Multi-pass Welding Instructions).

1.4.2 Welding Process Flow

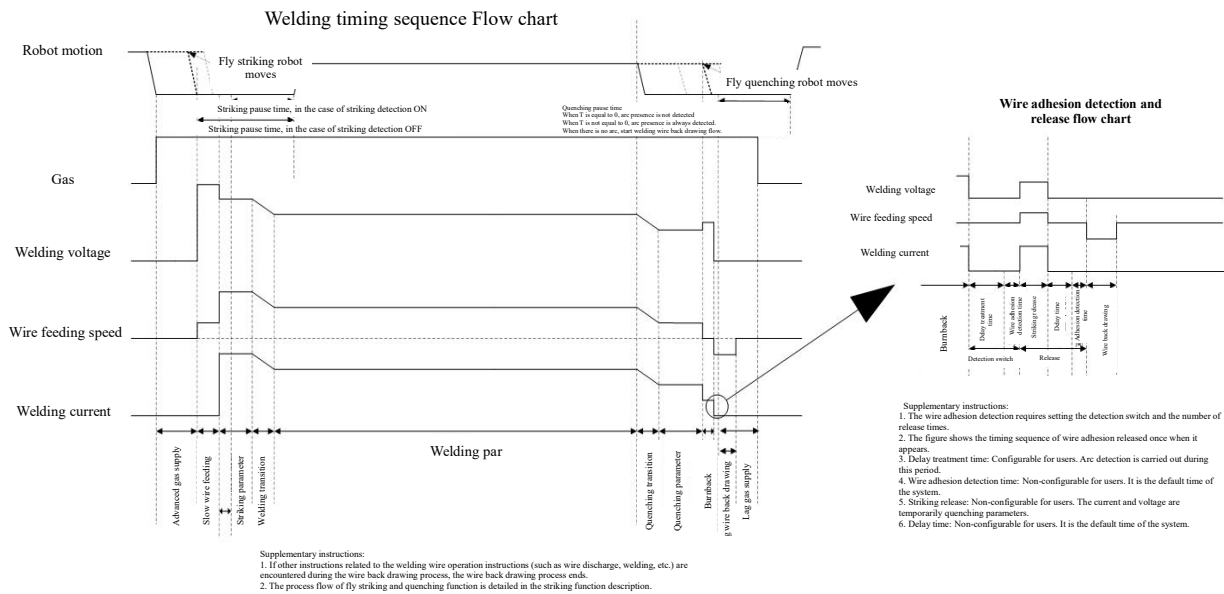


Figure 1-34 Welding Process Sequence Diagram

1.4.2.1 Striking Flow

When the ARCON instruction and the breakpoint recover striking, the striking process is executed. After striking is successful, the timing of the striking time begins. After the striking time is reached, the welding parameters are sent. During this process, the timing is carried out in parallel with the striking pause time. That is, after the pause time, the robot starts to move and weld. There is no need to wait for the welding parameters to be sent. The welding parameters can be sent during the robot welding process.

1.4.2.2 Quenching Flow

The quenching flow is executed by ARCOFF instruction and during the quenching pause. When the wire adhesion detection is enabled, there is a certain delay before the detection (which can be set in the quench file) to ensure that the detection is carried out after the arc disappears.

After the wire adhesion detection is completed, the timing for wire back drawing, post-gas supply and quenching pause time starts simultaneously. Once the pause time is reached, the robot can leave without waiting for the end of wire drawing and post-gas supply. When the welding process is paused, wire adhesion detection and wire back drawing are not carried out.

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If the quenching pause time is greater than 0, a detection of whether there is an arc will be conducted at the end of the pause. If there is an arc, an alarm will be triggered. No detection will be conducted when the pause time is 0.

When an emergency stop occurs during the welding process, the quenching process is no longer executed, and the arc is directly and quickly extinguished.

1.4.3 Welding Parameter Setting

When using the welding function, it is recommended to set all the striking time, quenching time, and pre-gas and post-gas supply times inside the welder (including when calling the JOB) to 0. All these process flows should be set by the robot.

1.4.3.1 Strike File

Click [Arc Weld] - [Welder Para] - [Strike File], and set the corresponding parameter.

Figure 1-35 Strike File Setting Interface

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Parameter description:

- 1) File number: You can choose files 1-32, or switch files by click Previous Page and Next Page.
- 2) Welder Parameter:
 - a) Welder mode: JOB mode, unified mode, pulse mode, etc. can be selected. Note: In DC mode, when the unified mode is set to ON, the voltage increases or decreases in the form of percentage. When the unified mode is set to OFF, the voltage is set in the form of a specific value.
 - b) Welder related process parameters: welding current, voltage, JOB number, etc.
 - c) Pre-gas supply time: If this parameter is used, set the pre-gas supply time of the welder to 0.
 - d) Striking time: The time for welding using the striking parameter after successful striking. If this parameter is used, the striking time of the welder is set to 0.
 - e) Transition time: The time for transition to the welding parameters after the striking time ends. Currently, it is temporarily invalid.
 - f) Pause time: The duration during which the robot remains in place at the striking point after striking.

Note: If the parameters in the teaching box file are used to control the welding process, the corresponding parameters on the welder panel need to be set to 0. For example, if the strike file of the teaching box sets the pre-gas supply time to 0.5, the pre-gas supply time in the welder should be set to 0. Otherwise, normal operation will be affected.

3) Welding condition (Arc Start Detect):

Quench File	Arc Start Detect:	
Weave Para >	Enable	☒ ON
RealTime Data >	Detect Time	1.0
Advanced Func >	DeTect Times	1
Basic Config >	Arc Leak Detect:	
Intelligent Config >	Enable	☒ ON
System Para >	Detect Time	1.0
	Restart/Offset:	
	Restart Enable	☐ OFF
	Offset Enable	☐ OFF
	Assist Function:	☐ OFF

Figure 1-36 Arc Start Detect Setting

- a) Enable: Arc start detection switch.
- b) Detect Time: If the arc is not struck each time the detection time is exceeded, it is

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considered that striking failed; this time of Aotai welder needs to be superimposed with the pre-gas supply time set inside the welder.

- c) **Detect Times:** Set the number of re-striking times. After one striking failure detection, it can automatically re-strike at the original position before reaching the detection times and then conduct the striking detection again. If striking is not successful after reaching the detection times, the alarm will be triggered.
 - d) **Interval time for wire discharge:** A certain amount of wire is discharged between one striking failure detection and the next striking. It can also be set to a negative value for wire drawing.
- 4) **Welding condition (Arc Leak Detect):**
- a) **Enable:** Arc leak detection switch.
 - b) **Detect Time:** If there is no arc after each detection time during the welding process, it is considered arc leak.
- 5) **Assist Function (Fly Weld):** In the striking parameters, set Assist Function switch to ON, and the interface will refresh to display configuration information such as Fly Weld.

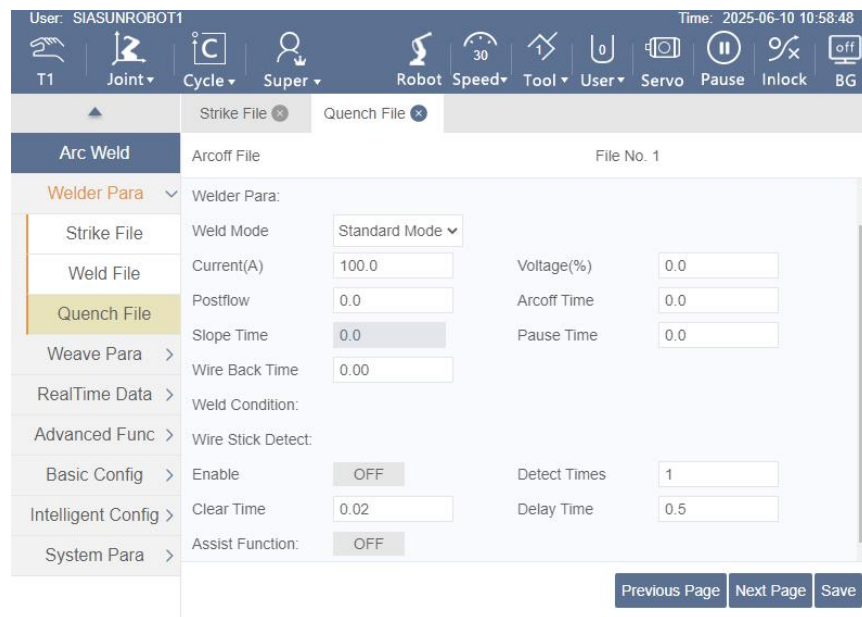


Figure 1-37 Fly Weld Setting

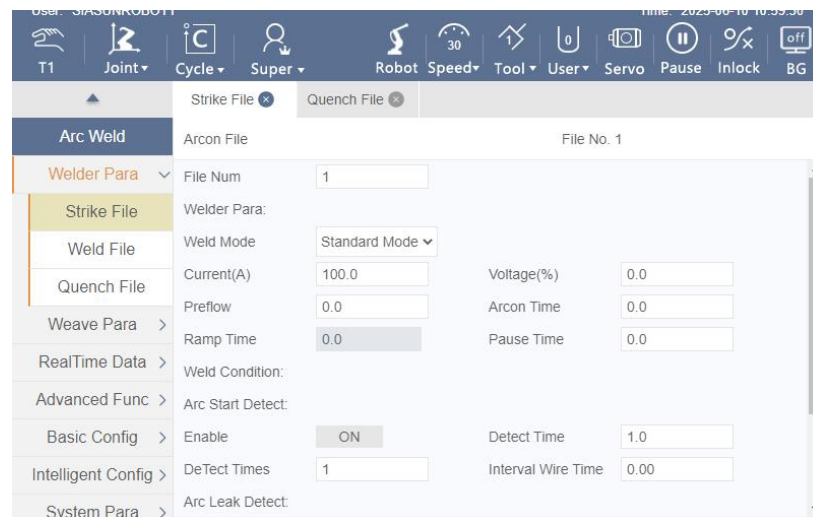
- a) **Enable:** Fly Weld function switch.
- b) **Pre-striking time:** Set the pre-striking time for fly weld (for details, see 2.12 Fly Weld Instructions).

1.4.3.2 Restart/Offset (Strike File)

It includes the offset striking function and the arc restart function.

The offset striking function is mainly applied when arc cannot be directly struck at the starting position due to the influence of gaps or insulating objects such as welding slag at the starting position, and it is necessary to offset the striking point position, strike arc and then return to the starting position for welding. Both single-pass and multi-pass welding may require offset striking, and it needs to be applied to each pass of multi-layer and multi-pass welding.

The arc restart function is automatically performed by the robot due to factors such as the formation of oxide scale at the striking point on the workpiece surface, which leads to a decrease in conductivity and makes it impossible to start the arc. The robot lifts a certain height from the striking point and offset a certain amount to other positions. After striking the arc, the robot will return to the previous striking point for welding.



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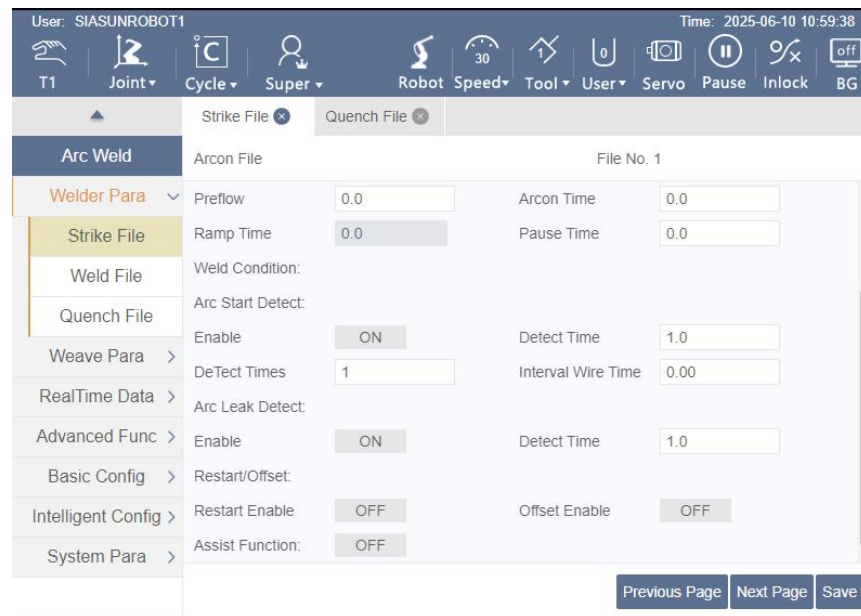


Figure 1-38 Strike File Configuration Interface

Parameter description:

- 1) Common Parameters: Common parameter items for arc restart and striking point offset.
 - a) Offset Speed: The speed at which the starting point moves towards the offset point.
 - b) Return Speed: The speed at which the offset point returns to the starting point.
 - c) Offset Weld File: The weld file number invoked during arc restart and offset striking.
- 2) Arc Restart:
 - a) Enable switch: When the switch is ON, the arc restart function is enabled and valid.
When the switch is OFF, the arc restart function is disabled and invalid.
 - b) Restart times: The number of times the arc restarts.
 - c) Wire discharge time: The wire discharge time of the welding torch at the beginning of arc striking.
 - d) Wire back drawing time: The time it takes for the welding torch to draw back the welding wire after striking fails.
 - e) Welding torch lifting capacity: The distance that the welding torch is lifted upwards after striking fails.
 - f) Straight-line X-direction offset: The distance that the welding torch offsets along the welding direction after the straight-line path striking fails, and then strike the arc again after the offset.
 - g) Arc Y-direction offset: The distance that the welding torch offsets in the left and right

directions after the arc path striking fails, and then strike the arc again after offset.

- h) Pause time: Before arc restart and re-striking, ensure that the robot pauses at the arc striking point during the pre-gas supply time.
- 3) Striking offset:
 - a) Enable switch: Offset the arc striking switch.
 - b) X/Y/Z direction offset: The offset in three directions relative to the starting point in the weld coordinate direction.
- 4) Use steps for offset/restart function: In Strike File, after the parameters are determining, when the welding enable switch is turned on and the striking instruction is executed, the striking point offset can be executed. The robot first moves to the offset point, with the motion speed being the offset speed. After reaching the offset point, the robot strikes the arc. If striking is successful, the robot returns to the starting point. At this time, the motion is linear ARCL motion, with the motion speed being the return speed. If striking fails, the robot restarts the arc at this position. The offset direction of the arc restart remains in the direction of the weld frame until the striking is successful, and then the robot returns to the starting point.
- 5) Precautions for use of Offset/Restart function:
 - a) Define the arc restart file number to be called and modify the parameters of the corresponding file.
 - b) Welding Enable does not perform offset striking in OFF state.
 - c) Pay attention to the offset weld file called, corresponding to the striking parameters from the offset point to the starting point. After reaching the starting point, reissue the corresponding striking parameters.
 - d) When an abnormality occurs during the arc restart process the robot pauses, and it is restarted again, the arc restart will be executed once more. The robot will directly move to the position where the welding torch is lifted to perform the arc restart. After ensuring the successful arc restart, it will return to the starting point. If it still fails, the arc restart offset parameter can be considered for modification.
 - e) Arc restart pause time setting: If the pause time setting is greater than the striking detection time, the arc restart striking detection time is the striking detection time; otherwise, it is the striking detection time minus the pause time.

1.4.3.3 Weld File

Click [Arc Weld] - [Welder Para] - [Weld File], and set the corresponding parameter.

User: SIASUNROBOT1 Time: 2025-06-10 11:00:25

T1 Joint Cycle Super Robot Speed 30 Tool User Servo Pause Inlock BG

Strike File Quench File Weld File

Arc Weld Weld File File No. 1

Welder Para

Strike File

Weld File

Quench File

Weave Para

RealTime Data

Advanced Func

Basic Config

Intelligent Config

System Para

File Num 1

Welder Para:

Weld Mode JOB Mode JOB 1

Previous Page Next Page Save

User: SIASUNROBOT1 Time: 2025-06-10 11:00:43

T1 Joint Cycle Super Robot Speed 30 Tool User Servo Pause Inlock BG

Strike File Quench File Weld File

Arc Weld Weld File File No. 1

Welder Para

Strike File

Weld File

Quench File

Weave Para

RealTime Data

Advanced Func

Basic Config

Intelligent Config

System Para

File Num 1

Welder Para:

Weld Mode Standard Mode

Current(A) 100.0 Voltage(%) 0.0

Previous Page Next Page Save

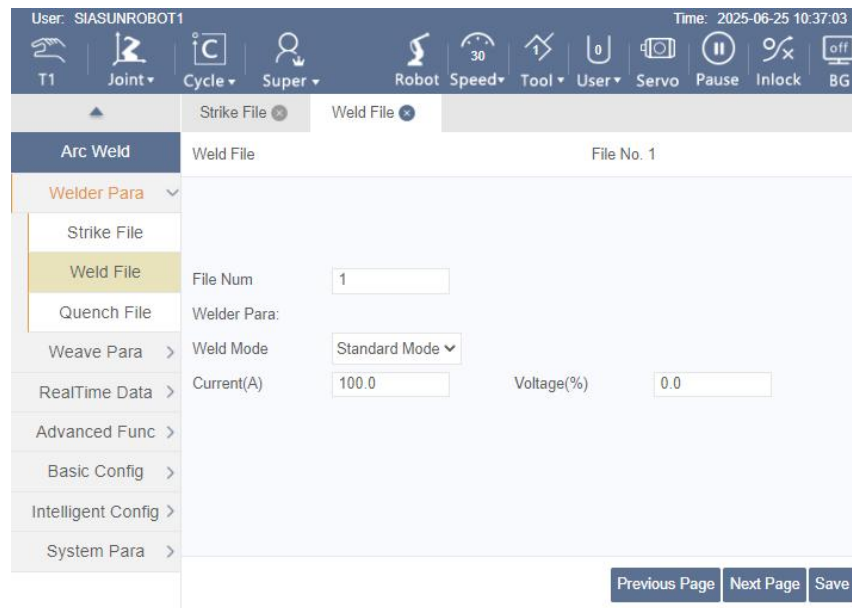


Figure 1-39 Weld File Configuration Interface

Parameter description:

- 1) File number: You can choose files 1-32, or switch files by click Previous Page and Next Page.
- 2) Welding mode: JOB mode, unified mode, pulse mode, etc. can be selected. Note: In DC mode, when the unified mode is set to ON, the voltage increases or decreases in the form of percentage. When the unified mode is set to OFF, the voltage is set in the form of a specific value.

1.4.3.4 Quench File

Click [Arc Weld] - [Welder Para] - [Weld File], and set the corresponding parameter.

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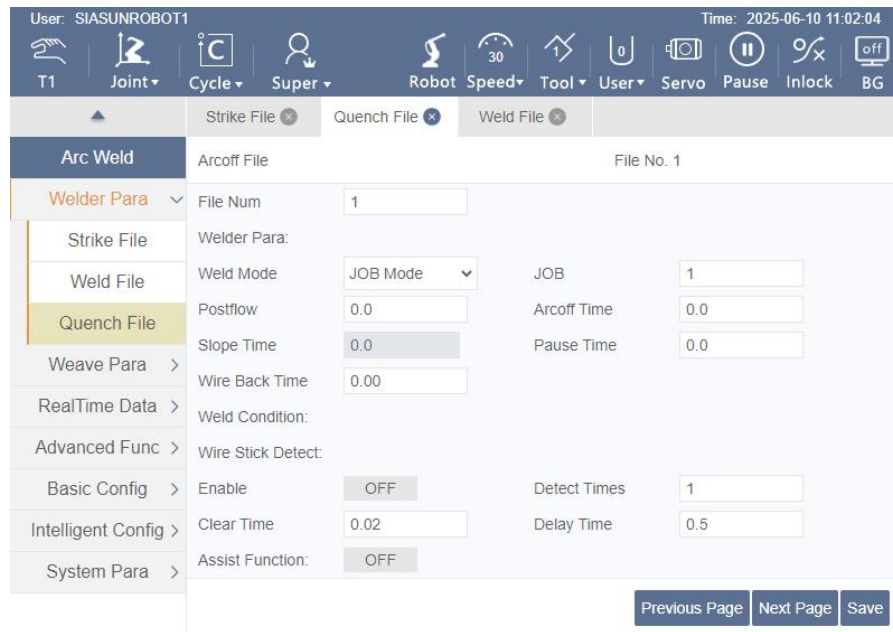


Figure 1-40 Quench File Configuration Interface

Parameter description:

- 1) File number: You can choose files 1-32, or switch files by click Previous Page and Next Page.
- 2) Welder Parameter:
 - a) Welder mode: welding current, voltage, JOB number, etc.
 - b) Post-gas supply time: Delay gas supply time. If this parameter is used, set the post-gas supply time of the welder to 0.
 - c) Quenching time: The time for welding with quenching parameters after starting quenching. If this parameter is used, the quenching time of the welder is set to 0.
 - d) Pause time: The time that the robot stays in place at the quenching point after quenching.
 - e) Wire drawing time: The time for wire back drawing after arc quenching. If this parameter is used, the time for wire back drawing after arc quenching of the welder is set to 0.

Note: If the parameters in the teaching box file are used to control the welding process, the corresponding parameters on the welder panel need to be set to 0. For example, if the strike file of the teaching box sets the post-gas supply time to 0.5, the post-gas supply time in the welder should be set to 0. Otherwise, normal operation will be affected.

- 3) Welding condition (Wire Adhesion Detect):

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- a) Enable: Wire adhesion detection switch.
 - b) DeTect Times: After the occurrence of wire adhesion, the wire adhesion can be released before reaching the detection times. If the wire adhesion still occurs after reaching the detection times, an alarm for wire adhesion will be triggered.
 - c) Wire adhesion release time: The duration of sparking each time the wire adhesion is released.
 - d) Delay detection time: Before the start of the wire adhesion detection, a certain delay time is required to ensure that the arc disappears. If the arc does not disappear within this time, an alarm will be triggered.
- 4) Assist Function (Weave Arcoff): In the quenching parameters, set Assist Function switch to ON, and the interface will refresh to display configuration information such as Weave Arcoff and Fly Weld.

RealTime Data >	Assist Function:	ON
Advanced Func >	Wave Arcoff:	
Basic Config >	Enable	OFF
Intelligent Config >	Fly Weld:	
System Para >	Pre End Time	0.00

Previous Page Next Page Save

Figure 1-41 Assist Function Configuration

Enable: Weave arc off function switch.

5) Assist Function (Fly Weld):

Pre End Time: Set the pre-arc off time for fly weld. (for details, see 2.12 Fly Weld Instructions).

1.4.3.5 Burnback File (Lincoln welder)

Click [Arc Weld] - [Welder Para] - [Burnback File], and set the corresponding parameter. The burnback parameter is valid for all welding processes.

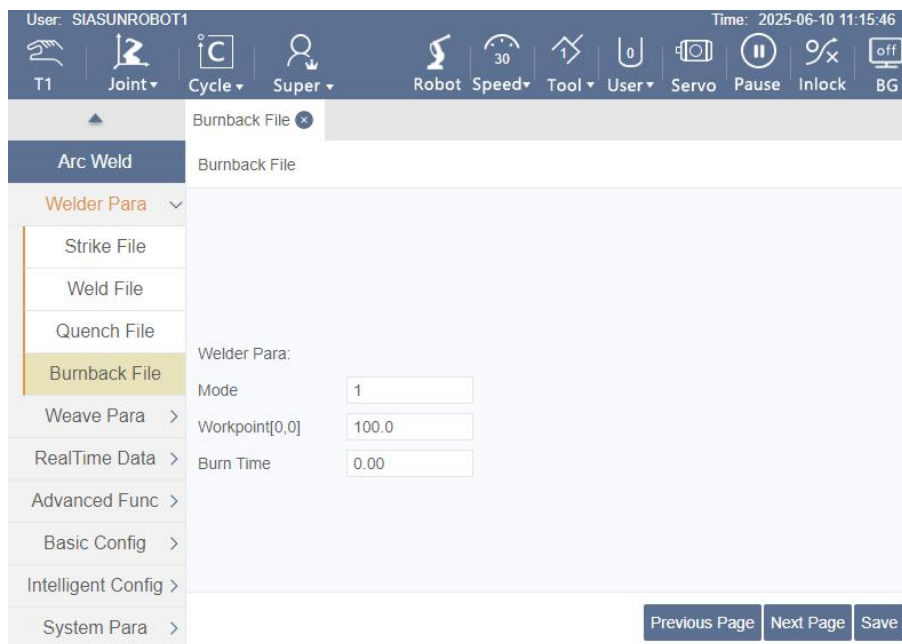


Figure 1-42 Burnback File Configuration Interface

Parameter description:

- 1) Welder related process parameters: welding current, voltage, JOB number, etc.
- 2) Burnback time: Burnback parameter, burnback time.

1.4.4 Shortcut

Op1 shortcut is used for switching applications, as shown in Figure 1-43:

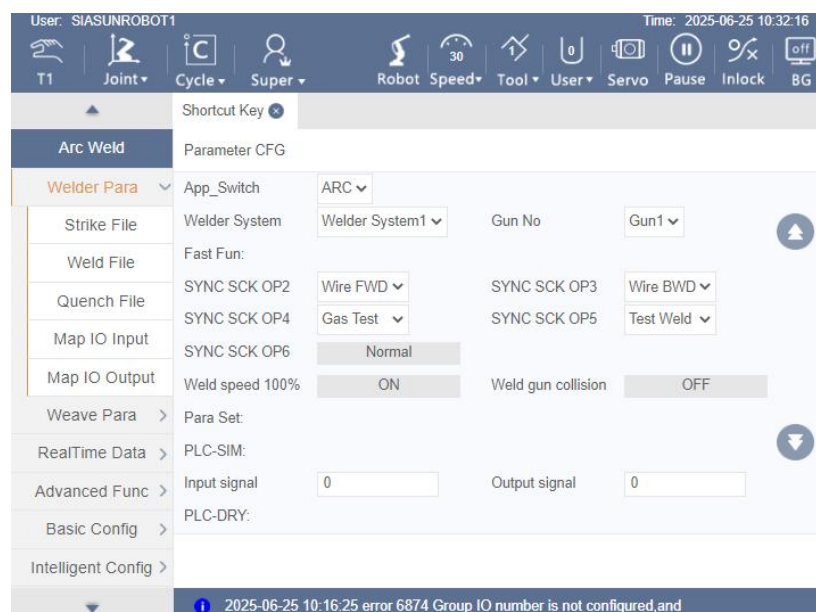


Figure 1-43 OP Shortcut Configuration Interface

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- 1) Welder system and welding torch number: Configure the welder 1 as welder system 1.

Welder system 1 is equipped with a multi-torch welder type, and the welding torch number displays multiple torches. Configure welding machine 2 as welder system 2.

Currently, the robot supports up to 2 welder systems at most.

- 2) Shortcut function:

- a) op2-op5 can be used to switch the types of shortcut functions. Press the corresponding OP key to achieve the corresponding function.

The shortcut op6 can achieve mode switching, corresponding to the welding enable switch;

- b) Welding torch collision: This is the internal welding torch collision switch. For LORCH, Fronius and general welders (to display the welding torch collision switch, the welding collision signal in the input IO of the welding machine should be set to non-0), the internal welding torch collision switch can be set here. For other welders, the external welding torch collision switch can be set in [Arc Weld] - [Basic Configuration] - [Status IO Configuration].

- c) Welding speed 100%: When this switch is set to ON, the welding speed does not change with the speed ratio in the execution mode.

- d) Manual wire feeding speed: For some welders, the manual wire feeding and drawing speeds can be set.

2 Auxiliary Welding Function

2.1 Description and Configuration of Status IO Monitoring Function

Basic welding states mainly refer to the configuration of states such as welding torch collision, water flow, gas pressure, wire shortage monitoring, compressed air, and shielding gas into the user's IO to monitor the IO status in real time. Under normal circumstances, these states have default values. In special cases, the IO sequence number values can be changed. The robot is responsible for real-time monitoring of the corresponding IO table status and will give an alarm in case of abnormal status. The IO sequence number and enable status of the monitoring parameters can be configured through the IO configuration interface of the welding monitoring status. After successful configuration, the system will detect the IO status and give an alarm when the IO status is abnormal.

2.1.1 Parameter Configuration Description

2.1.1.1 [Status IO Configuration] Interface Configuration

Click [Arc Weld] - [Basic Config] - [Status IO Config] to configure the relevant parameters. The configuration of the welding monitoring status IO interface is as follows. Customers can configure the relevant parameters corresponding to IO and enable status through the configuration interface. If Enable is ON, the abnormal status of the IO signal of this monitoring item will be detected and an alarm will be triggered. If Enable is OFF, this monitoring item will not be detected.

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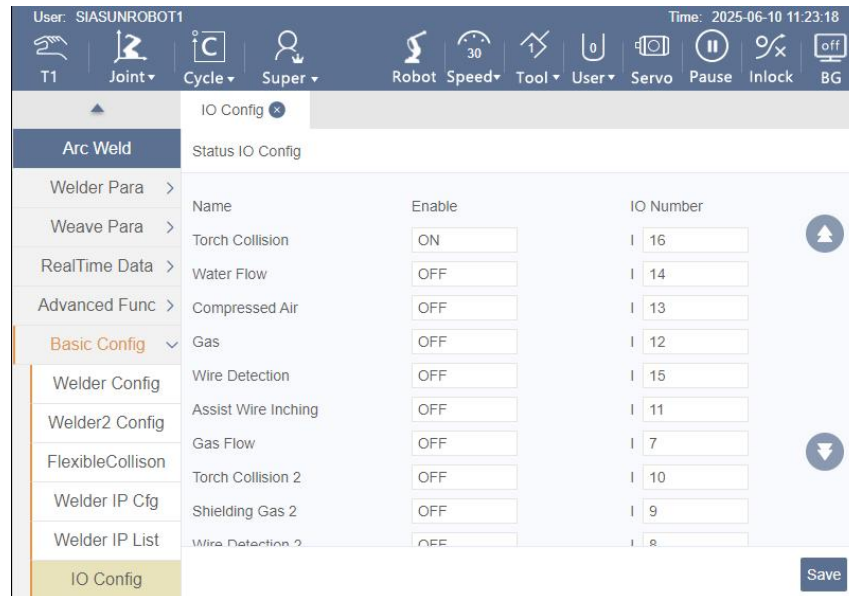


Figure 2-1 Status IO Configuration Interface

For the welding torch collision wiring, taking WELLLINK IO module as an example, observe whether the common input terminal of the module (within the red box) is connected to 24V or 0V. 24V corresponds to the NPN mode input, and 0V corresponds to the PNP mode input.

For the two-core wire for welding torch anti-collision detection, one core is connected to the IO interface set on the status IO configuration page, and the other core is selected to be connected to 0V (NPN mode) or 24V (PNP mode) in the green box.

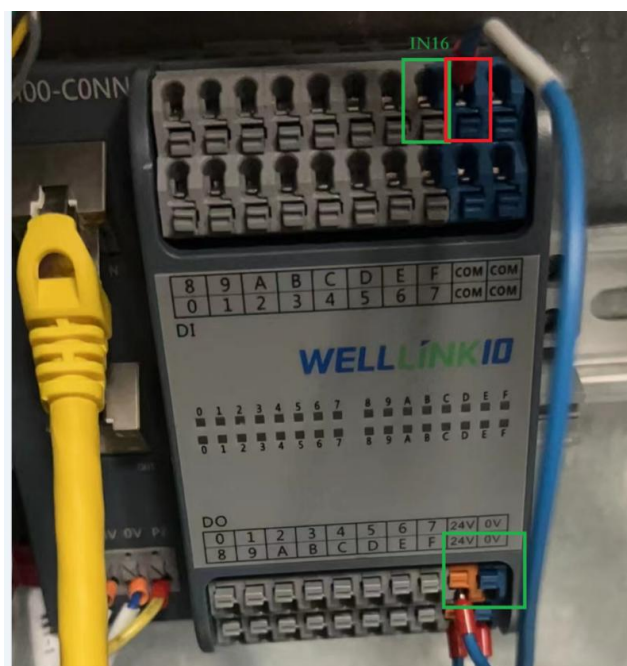


Figure 2-2 IO Wiring Diagram

2.1.1.2 [Status IO Configuration] Interface Configuration

Click [Arc Weld] - [Basic Config] - [Status IO Switch] to configure the relevant parameters.

The setting interface is shown in Figure 2-3:

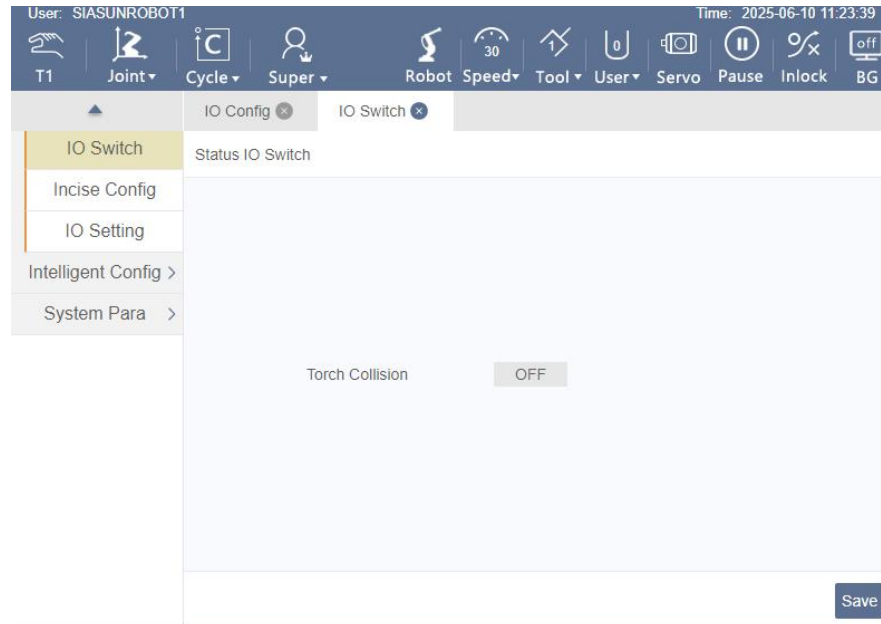


Figure 2-3 Status IO Switch Configuration Interface

2.1.1.3 [Status IO Configuration] Interface Configuration

This interface is a dynamic display interface. The display switch parameter items are determined by the enable status of the parameter items in [Status IO Configuration] interface. Only when the enable status is on can the corresponding parameter switch in the [Status IO Switch] interface be set. The interface correspondence example is shown in the following figure:

Example 1:

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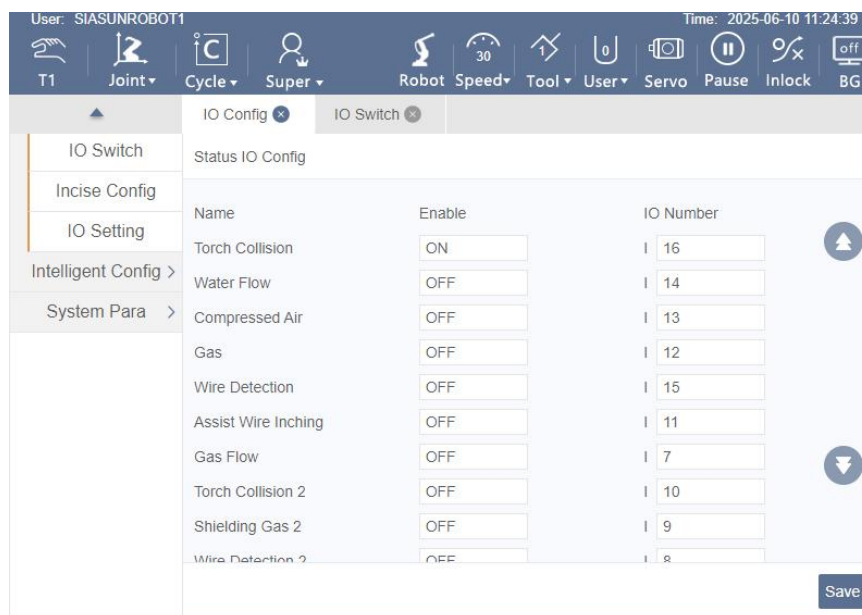


Figure 2-4 Status IO Configuration Interface

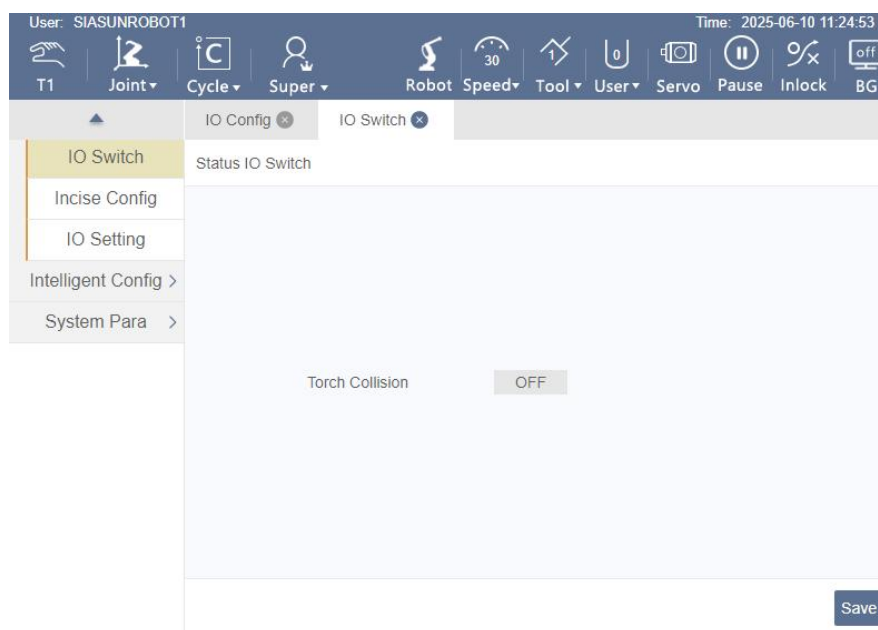


Figure 2-5 Status IO Switch Configuration Interface

Example 2:

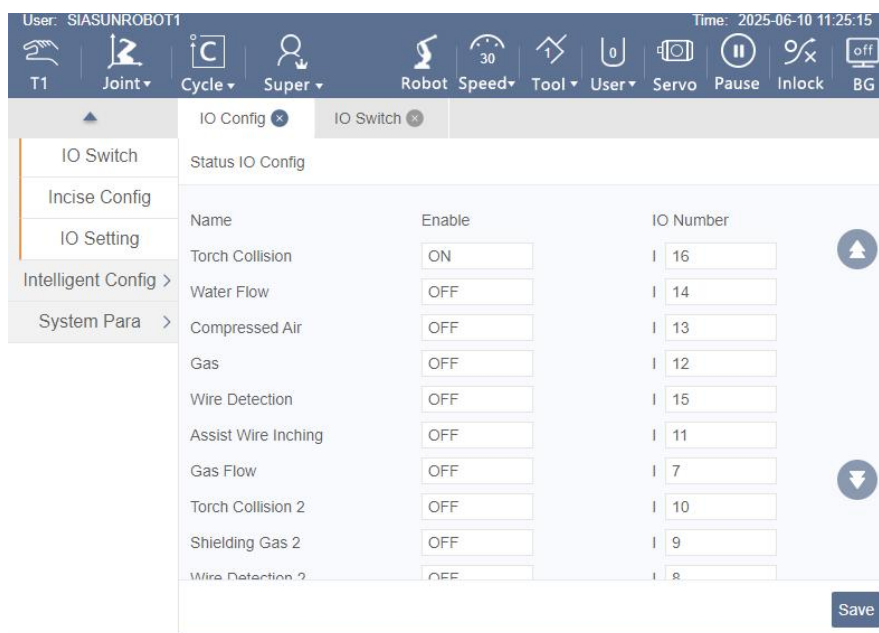


Figure 2-6 Status IO Configuration Interface

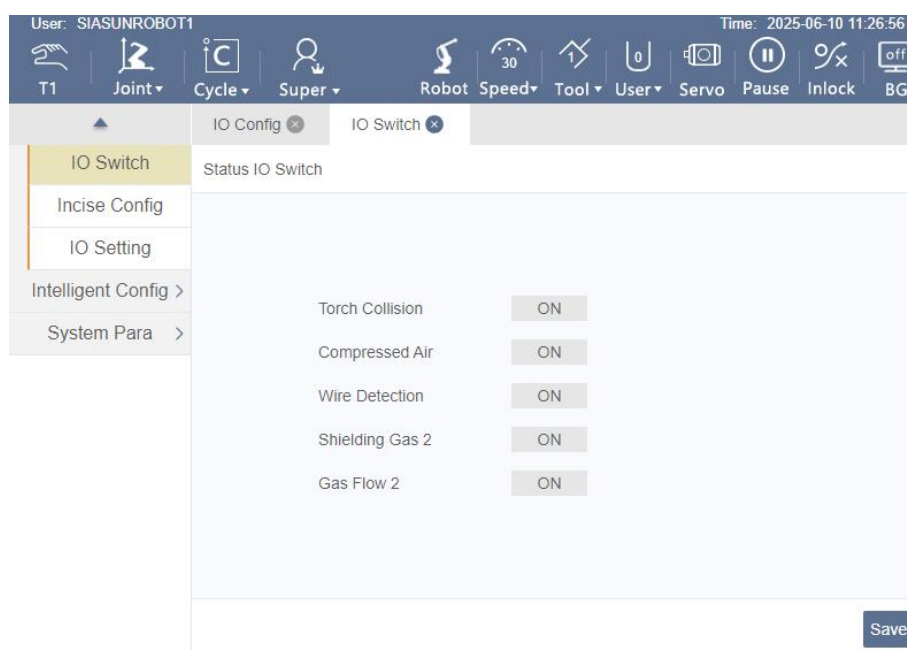


Figure 2-7 Status IO Switch Configuration Interface

Note: When triggering an abnormal status alarm, enter [Status IO Switch] for shielding. [Status IO Configuration] setting is prohibited.

2.2 Arc Welding Frame

The offset calculation for functions such as multi-pass welding and weave, as well as functions like offset/restart and starting point detection, are based on the weld frame. The general

method for establishing the weld frame is unified as follows: The basic weld frame is established by taking the welding reference plane under World or the positioner as the Z+direction of the weld frame, and the welding forward direction as the X+direction of the weld frame.

At present, for functions such as multi-pass welding, weave library, starting point detection, contact detection sensor (TDS), offset/restart, laser tracking, laser sensing, bevel cutting, intersection point calculation, and pose transformation, the weld frames are established in the above-mentioned way, and they also support use in multiple positioner systems.

2.2.1 Welding reference plane

Since the establishment of the current weld frame is based on the welding reference plane, it is necessary to determine the current welding reference plane through the reference plane specified instruction before each function is used. When the welding reference plane instruction is executed, the welding reference planes under the world and under the positioner (if any) will be determined simultaneously. Generally, the two are different and related to the position of the current positioner. For subsequent functions, a weld frame is established using the corresponding welding reference plane direction based on the corresponding system environment.

When the welding reference plane is not set, the initial default direction of the welding reference plane in the system is the Z+ direction under the world. For systems that require the use of positioners for linkage, the direction of the welding reference plane must be specified through the welding reference plane instruction before each function is executed. After being specified, the direction of the welding reference plane is bound to the positioner, and the welding reference plane changes along with the change of the positioner position, as shown in Figure 2-8, to ensure the normal execution of subsequent functions such as offset under interlock conditions.

The welding reference plane instruction currently supports four specified modes:

WREFP TYPE = 1(0,0,1)Z+direction under World.

WREFP TYPE = 2 (0,1,0) Y+ direction under World.

WREFP TYPE = 3(1,0,0)X+direction under World.

WREFP TYPE = 4 (x,y,z) Z- direction of welding torch (TCP) under current world

Calculate the corresponding downward direction of the positioner based on the current position of the positioner.

After the welding reference plane is specified, this direction remains valid until the next specification is made. After the robot is restarted, the reference plane direction returns to the

default value.

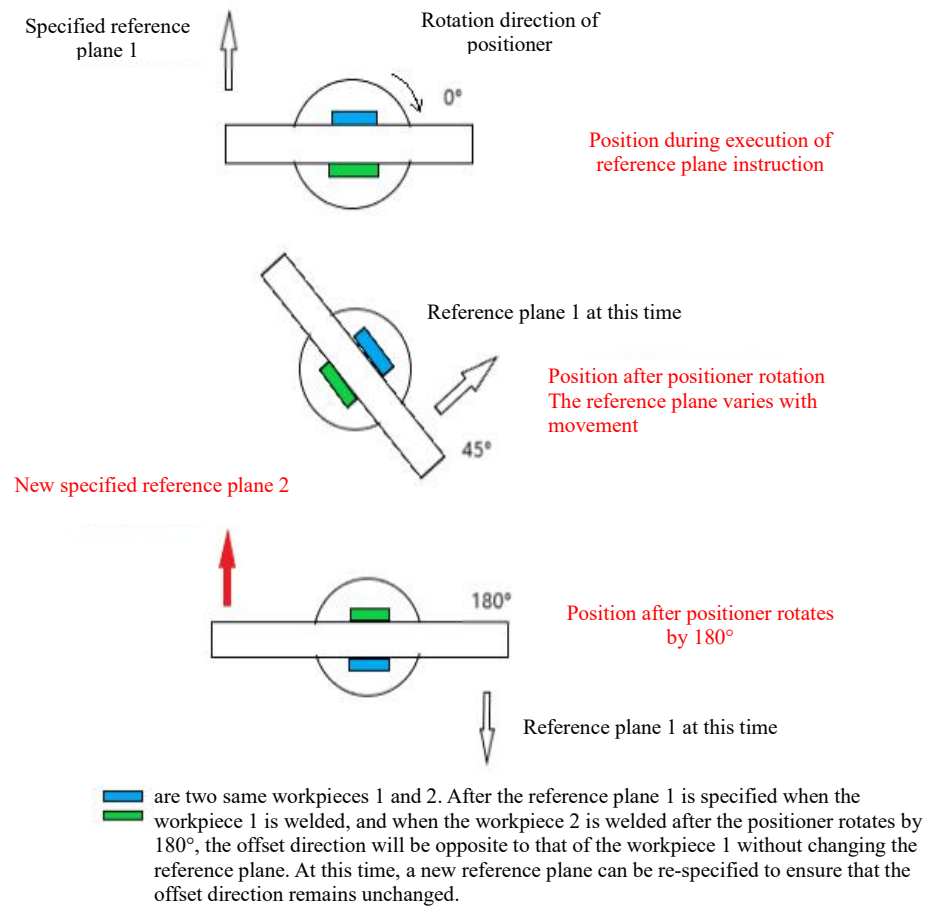


Figure 2-8 Introduction to reference plane

2.3 Weave File

It can be configured according to the actual situation. There are a total of 32 configuration files.

Click [Arc Weld] - [Weave Para] - [Weave File], as shown in Figure 2-9:

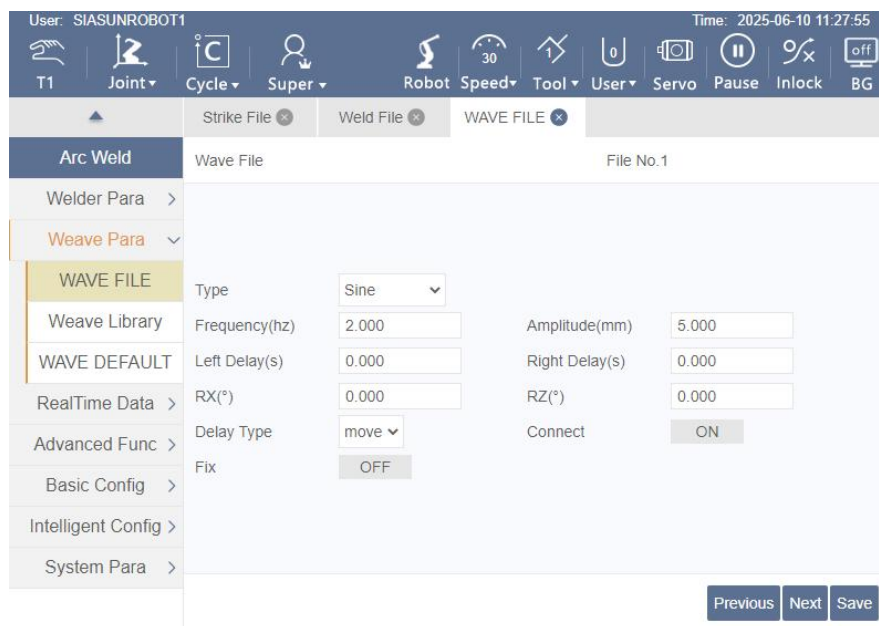


Figure 2-9 Weave File Configuration Interface

Table 2-1 Introduction Table of Weave File Parameters

Parameter	Parameter meaning	Parameter range	Other description
Weave type	Configure the wave mode of the current file	Custom: You can fill in the wave library number to select a custom wave library (1-100) Sine type: Weave library 101 Pendulum: Weave library 102 Triangular type: Weave library 103 L type: Weave library 104 Circular type: Weave library 105 Figure-8 type: Weave library 106	
Weave library number	Custom mode, select custom wave library	1-100	

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Frequency	Reciprocal of the time required to swing one cycle	0.1-99.99Hz	
Swing	Maximum distance at swing endpoint when swinging for one cycle	0.1-99.99mm	
Left residence time	When [Move] is selected, the robot swings to the endpoint position. The robot will move along the	0-100s	
Right residence time	forward direction of the swing frame at the current position until the end of the dwell time, and then continue with the remaining swing motion		
Inclined angle RX	The angle RX at which the swing path rotates along the forward direction X of the swing frame	-90°-90°	
Azimuth RZ	The angle RX at which the swing path rotates along the Z direction of the swing frame	-90°-90°	
Radius	Swing in X direction	0.1-99.99mm	Custom, Circular, and Figure-8 types include radius options
Residence	When [Move] is selected,	At present, there is only one	If both the left

delay type	the robot swings to the endpoint position. The robot will move along the forward direction of the swing frame at the current position until the end of the dwell time, and then continue with the motion	option [Move]	and right residence times are 0, select [Move], and do not stay at the endpoint position
Weave connection	When [ON] is selected, the weave action passes through the teaching point; when [OFF] is selected, the weave action does not necessarily pass through the teaching point.	[ON]/[OFF]	

2.4 Fixed Point Swinging Arc

2.4.1 Interface

Click [Arc Weld] - [Weave Para] - [Weave File], add [Fix] switch, set the switch to [OFF], then [Reference Point IP] and [Stop Signal] are concealed; set the switch to [ON], then the two parameters are displayed.

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The figure displays two screenshots of the SIASUN Robot configuration interface, specifically the 'Wave File' configuration for 'File No. 1'.

Top Screenshot:

- User: SIASUNROBOT1
- Time: 2025-06-10 11:28:29
- Navigation: T1, Joint, Cycle, Super, Robot, Speed, Tool, User, Servo, Pause, Inlock, BG
- File Selection: Strike File, Weld File, WAVE FILE (selected)
- Configuration Tab: Arc Weld, Welder Para, Weave Para (selected)
- WAVE FILE Parameters:
 - Type: Sine
 - Frequency(hz): 2.000
 - Amplitude(mm): 5.000
 - Left Delay(s): 0.000
 - Right Delay(s): 0.000
 - RX(°): 0.000
 - RZ(°): 0.000
 - Delay Type: move
 - Fix: OFF
- Buttons: Previous, Next, Save

Bottom Screenshot:

- User: SIASUNROBOT1
- Time: 2025-06-10 11:28:40
- Navigation: T1, Joint, Cycle, Super, Robot, Speed, Tool, User, Servo, Pause, Inlock, BG
- File Selection: Strike File, Weld File, WAVE FILE (selected)
- Configuration Tab: Arc Weld, Welder Para, Weave Para (selected)
- WAVE FILE Parameters:
 - Type: Sine
 - Frequency(hz): 2.000
 - Amplitude(mm): 5.000
 - Left Delay(s): 0.000
 - Right Delay(s): 0.000
 - RX(°): 0.000
 - RZ(°): 0.000
 - Delay Type: move
 - Fix: ON
 - Reference PR: 1
 - Stop Signal: 1
- Buttons: Previous, Next, Save

Figure 2- 10 Fixed Point Weave File Configuration Interface

Parameter description:

Reference point PR: Establish the weave frame between the fixed-point pendulum arc point and the reference point.

Stop signal: When the IO signal is triggered, the current weave action ends.

2.4.2 Job Example

Table 2-2 Table of Fixed-Point Pendulum Arc Job Example

Job	Description
NOP	
ARCL P1 VL = 200 ACC =100 CNT = 0	Safety point
ARCL P2 VL = 200 ACC =100 CNT = 0	Weave point
ARCON #1 #1	Strike the arc using Strike File 1 and weld with Weld File 1
WAVE ON #1	Call Weave File 1
ARCOFF #1	Quench the arc using Quench File
WAVE OFF	Stop weaving. For fixed-point weave stop, a signal to stop weave needs to be added.
ARCL P3 VL = 200 ACC =100 CNT = 0	Safety point
END	

2.4.3 Operation Instructions

- 1) Set the weave parameters according to the actual situation.

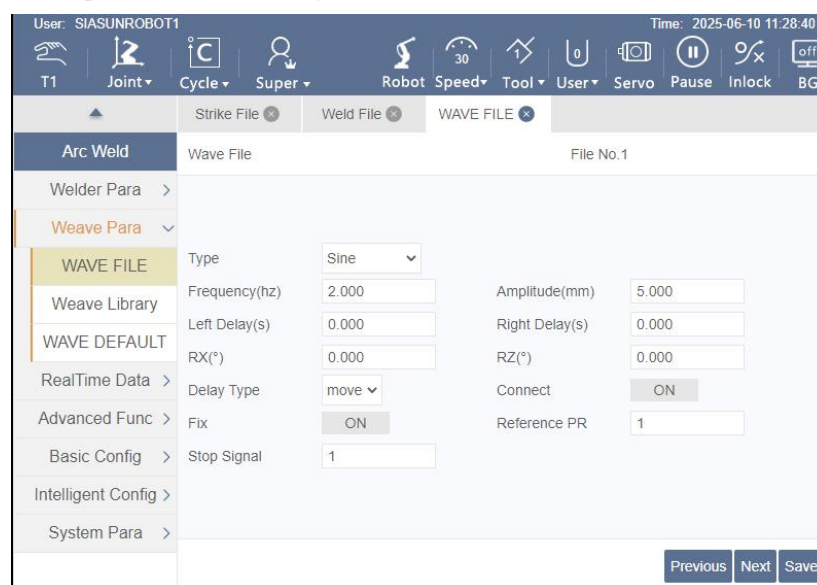


Figure 2- 11 Fixed Point Weave Parameter Configuration Interface

- 2) Teach the fixed-point weave operation (Note: It is essential to teach the reference point PR).

- 3) Perform fixed-point weave job.
- 4) If you want to stop weave, set IO corresponding to [Stop Signal] to ON.

2.5 Weave Library

The weave library function can cover all weave requirements. 100 user-editable weave libraries can be stored. Weave is supported in manual mode.

Note: Arc welding application should be configured. Before use, the tool frame and pose of the welding torch must be calibrated accurately.

2.5.1 Weave Reference

- 1) The weave reference can be specified through instructions. Currently, three specification modes are supported:

WAVEREF TYPE = 1 Welding reference plane.

WAVEREF TYPE = 2 Welding torch reference (welding torch TCP Z- direction).

WAVEREF TYPE = 3 Welding torch entry direction reference (the starting point points to the direction of the previous safety point).

Meanwhile, for types 1 and 3, the corresponding downward direction of the positioner is calculated based on the current position of the positioner to achieve the specification of the swing reference in the linked state of the positioner.

Welding reference plane (horizontal)

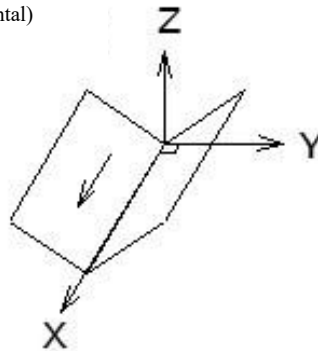


Figure 2- 12 Weave takes the welding reference plane as reference

Welding torch reference

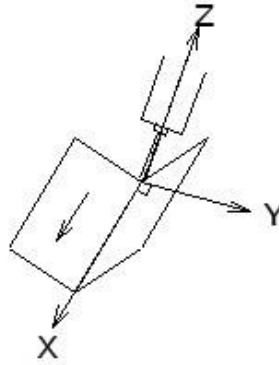


Figure 2-13 Weave takes the welding torch direction as reference

Welding torch entry direction

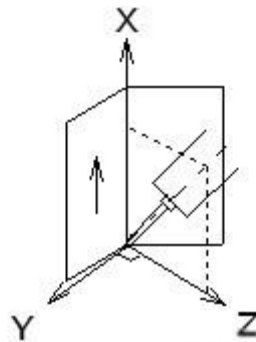


Figure 2-14 Weave takes the welding torch entry direction as reference

2) Establishment of weave frame:

The motion direction is the positive direction of the X-axis of the weave frame. The z-axis of the weave frame is determined based on the swing reference. The positive direction of the Y-axis of the weave frame is obtained according to the right-hand rule. Meanwhile, the weave frame can be rotated according to the rotation angles R_x , R_y , and R_z in the weave library parameters. The final weave action is executed based on the final weave frame. Currently, the system only supports rotating the weave reference plane in the R_x direction.

2.5.2 File Configuration

Click [Arc Weld] - [Weave Para] - [Weave Library] to enter the wave parameter configuration interface, as shown in Figure 2-15:

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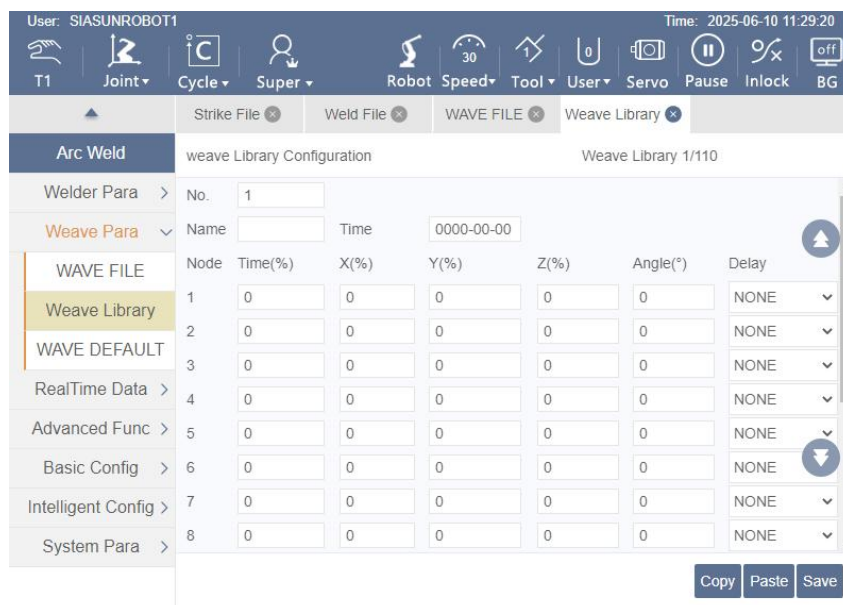


Figure 2- 15 Weave Library Configuration Interface

Table 2- 3 Introduction Table of Weave Library Parameters

Parameter	Parameter meaning	Parameter range	Other description
Number	Select weave library number	1-110	1 to 100 are custom weave libraries, and the parameters within the interface can be modified. 101-110 are the default weave libraries. The parameters cannot be modified, as shown below: Sine type: Weave library 101 Crescent type: Weave library 102 Triangular type: Weave library 103 L type: Weave library 104 Circular type: Weave library 105 Figure-8 type: Weave library 106
Name	Weave library name		
Time	Current time	Click Save to	

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		automatically obtain the system time	
Time (%)	Weave action time within the cycle	0-99 (0 indicates the start of the cycle, 100 indicates the end of the cycle, and 100 is automatically replenished by the system and cannot be manually configured)	Rule: The time within one cycle starting from node 1 cannot be set to 0, or the time of all nodes cannot be set to 0; otherwise, the unreasonable parameter alarm will be given, and the configuration will fail.
Front and rear X (%)	They are respectively the expected offsets in the X/Y/Z directions when reaching the current node. For example, filling in 50 indicates that the forward offset reaches 50% of the swing.	-100-100	The sum of the absolute values of the maximum and minimum values within the cycle must not exceed 100. If it exceeds 100, an alarm will be triggered during the motion.
Horizontal Y (%)			
Vertical Z (%)			
Angle (°)	Rotation angles of two nodes before and after along the X-axis of the weave frame	-10-10	Only the end pose during motion is changed without altering the motion path
Residence time	The robot swings to the endpoint position. The robot will move along the forward direction of the swing frame at the current position until the end of the dwell time, and	[None], [Left residence time], [right residence time]	Each cycle can contain a maximum of 1 left residence time and 1 right residence time. Exceeding this limit will trigger an alarm, resulting in save failure. The left and right residence times correspond to the default weave instructions and the left and right

	then continue with the motion		residence times in the weave file interface.
--	-------------------------------	--	--

2.5.3 Node input

Take the 90-degree V-type weave as an example (Example 1). Input the offset values (based on the swing percentage) of node 1, node 2 and node 3 relative to the teaching origin on the job interface. Node 1 is to make an offset of a*50% in the Y+ direction and an offset of a*50% in the Z+ direction at the first 25% time of 1/f ms (one cycle). Node 2 is to return to the origin position at the second 25% of time in 1/f ms (one cycle), and node 3 is to make an offset of a*50% in the y-direction and an offset of a*50% in the Z+ direction at the third 25% of time in 1/f ms (one cycle). The fourth 25% of the time of 1/f ms (one cycle) returns to the origin position. There is no need to fill it in on the interface here. The software defaults to returning to the origin.

Note: The input of each weave library will have default starting node and ending node (with offset value of 0), and no input is required.

The example of the system weave library part is as follows:

Table 2-4 No. 101 Triangle Residence Weave (Inward in Forward Direction)

Point	Time (%)	Front and rear X (%)	Horizontal Y (%)	Vertical Z (%)	Angle (°)
1	12	0	50	50	0
2	38	0	50	50	0
3	62	0	-50	50	0
4	88	0	-50	50	0

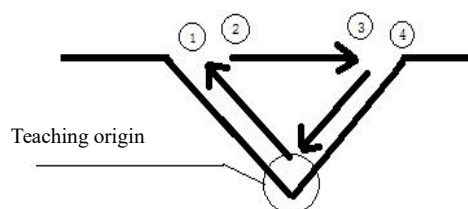


Figure 2- 16 No. 101 Triangle Residence Weave Diagram

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Table 2-5 No. 102 Right-angle Weave (Inward in Forward Direction)

Point	Time (%)	Front and rear X (%)	Horizontal Y (%)	Vertical Z (%)	Angle (°)
1	25	0	50	50	0
2	50	0	0	0	0
3	75	0	-50	50	0

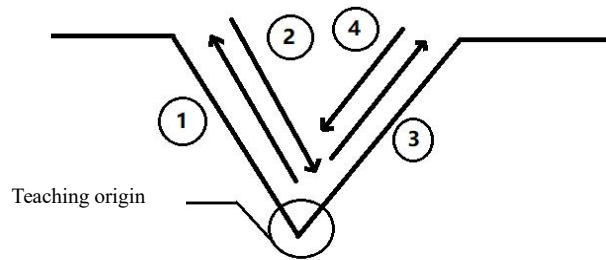


Figure 2-17 No. 102 Right-angle Weave Diagram

Table 2-6 No. 103 45-degree V-type Weave

Point	Time (%)	Front and rear X (%)	Horizontal Y (%)	Vertical Z (%)	Angle (°)
1	25	0	0	50	0
2	50	0	0	0	0
3	75	0	50	50	0

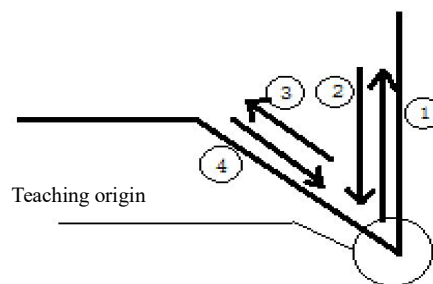


Figure 2-18 No. 103 45-degree V-type Weave Diagram

Table 2-7 No. 104 Linear Weave with Changing Forward Speed (Sometimes Fast, Sometimes

Slow)

Note: This mode does not support arc tracking

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Point	Time (%)	Front and rear X (%)	Horizontal Y (%)	Vertical Z (%)	Angle (°)
1	25	50	0	0	0
2	50	0	0	0	0
3	75	-50	0	0	0

Table 2-8 No. 105 Two-side Residence Horizontal Welding

Point	Time (%)	Front and rear X (%)	Horizontal Y (%)	Vertical Z (%)	Angle (°)
1	20	0	50	0	0
2	30	0	50	0	0
3	50	0	0	0	0
4	70	0	-50	0	0
5	80	0	-50	0	0

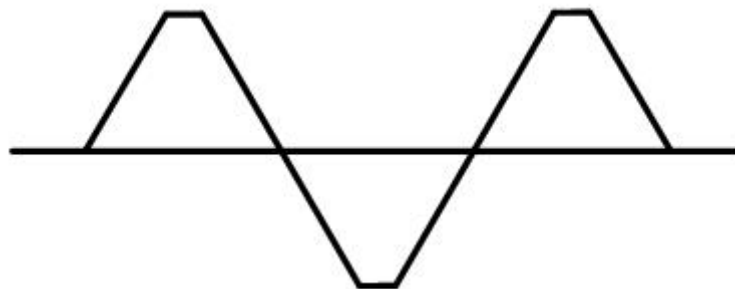


Figure 2-19 No. 105 Two-side Residence Horizontal Welding Diagram

Table 2-9 No. 106 Crescent Weave (Swing Angle 2°) adds angle variation on the basis of

horizontal weave

Point	Time (%)	Front and rear X (%)	Horizontal Y (%)	Vertical Z (%)	Angle (°)
1	25	0	50	0	2
2	50	0	0	0	0
3	75	0	-50	0	-2

Table 2-10 No. 107 Horizontal Weave

Point	Time (%)	Front and rear X (%)	Horizontal Y (%)	Vertical Z (%)	Angle (°)
-------	----------	----------------------	------------------	----------------	-----------

	(%)				
1	25	0	50	0	0
2	50	0	0	0	0
3	75	0	-50	0	0

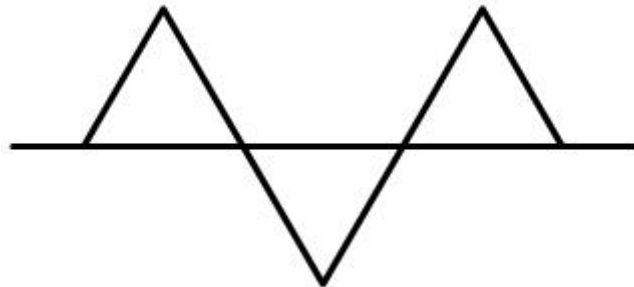


Figure 2-20 No. 107 Horizontal Weave Diagram

Note: The percentage of weave time cannot be filled in as 0%, and the percentage of weave amplitude cannot exceed 100%.

2.5.4 Instruction Description

WAVEREF	Function	Specify the corresponding weave reference plane according to the corresponding system environment.	
	Format	WAVEREF TYPE = <Parameter 1>	
	Description	Parameter 1	Weave reference type: 1: Welding reference plane. 2: Welding torch reference (welding torch TCP z- direction). 3: Welding torch entry direction reference (the starting point points to the direction of the previous safety point).
	Examples	WAVEREF TYPE = 1	
WAVE ON#	Function	Call the parameters in the weave file (1-32) corresponding to # to perform subsequent weave movements	
	Format	WAVE ON #<Parameter 1>	
	Description	Parameter 1	Weave File. Range: 1-32

	Examples	WAVE ON #1	
WAVE ON	Function	Weave instruction, customizing weave by setting parameter items.	
	Format	WAVE ON <Parameter 1> F=<Parameter 2> A=<Parameter 3> SEC=<Parameter 4> SEC=<Parameter 5>	
	Description	Parameter 1	There are six types of instructions in total. Sine: Sine weave Crescent: Crescent weave Triangle: Triangle weave L-Type: L-type weave Circle: Circular weave (no residence time) Type: Figure-8 weave (no residence time)
		Parameter 2	Weave frequency, range: 0.1-99.99Hz;
		Parameter 3	Swing, range: 0.1-99.99mm;
		Parameter 4	SEC: The first SEC is the left residence time, ranging from 0 to 100 seconds;
		Parameter 5	SEC: The second SEC is the right residence time, ranging from 0 to 100 seconds.
	Examples	WAVE ON Sine F=1.00 A=2.00 SEC=0 SEC=0	
WAVE OFF	Function	Weave stops in the weave library mode.	
	Format	WAVE OFF	
	Description	Parameter 1	This instruction has no parameter configuration.
	Examples	WAVE OFF	

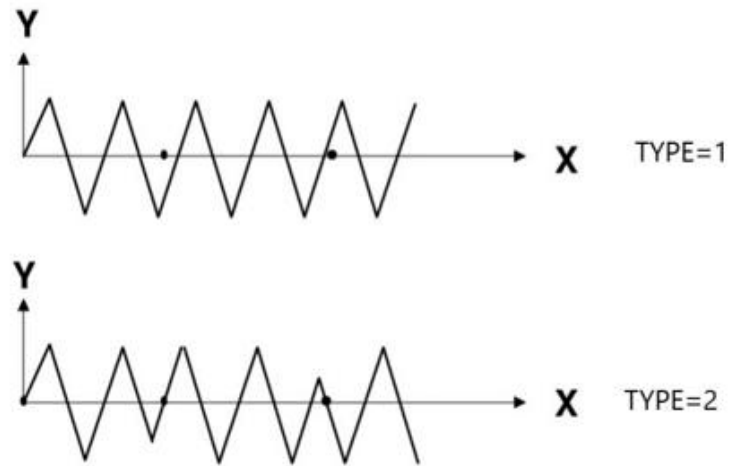


Figure 2-21 Example of Weave Instruction TYPE Parameters

Notes:

- 1) Weave is only effective on the welding path.
- 2) Pause during the weave process, and the robot returns to the center line; when making an emergency stop, the robot does not return to the center line; when resuming from a breakpoint, the robot first returns to the center line and then proceeds with the breakpoint recovery process.

2.5.5 Job Example

All the arc welding application instructions in the job are detailed in the arc welding instruction description in Appendix 7 of this document. The job example is as follows:

Job	Description
NOP	
ARCL P1 VJ = 10 ACC =100 CNT = 0	Safety point
ARCL P2 VL = 200 ACC =100 CNT = 0	Weave point
ARCON #1 #1	Strike the arc using Strike File 1 and weld with Weld File 1
WAVeref TYPE=2	Specify the weave reference as the welding torch reference (welding torch TCP Z-direction)
WAVE ON #1	Call weave reference file 1
ARCL P3 VL = 6 ACC =100 CNT = 0	Weave end point

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ARCOFF #1	Quench the arc using Quench File
WAVE OFF	Stop weaving
ARCL P4 VJ = 10 ACC =100 CNT = 0	Safety point
END	

NOP	
ARCL P1 VJ = 10 ACC =100 CNT = 0	Safety point
ARCL P2 VL = 200 ACC =100 CNT = 0	Weave point
ARCON #1 #1	Strike the arc using Strike File 1 and weld with Weld File 1
WAVeref TYPE=2	Specify the weave reference as the welding torch reference (welding torch TCP Z-direction)
WAVE ON Sine F=1.0 A=2.0 SEC=0.00 SEC=0.00	Weave according to the weave parameter item
ARCL P3 VL = 6 ACC =100 CNT = 0	Weave end point
ARCOFF #1	Quench the arc using Quench File
WAVE OFF	Stop weaving
ARCL P4 VJ = 10 ACC =100 CNT = 0	Safety point
END	

2.6 Arc Welding Real-time Display

Arc welding real-time display refers to the fact that the robot can obtain data in real time in both execution mode and teaching mode and display it on the teaching box interface, ensuring that users can accurately obtain the working information of the robot during the robot debugging process.

Click [Arc Weld] - [RealTime Data] - [RealTime Display] to enter the real-time display interface of arc welding, and you can view the real-time data of arc welding according to the displayed configuration items. It includes Welding Parameters, Arc Tracking, Multi-pass Welding, External Tools, Statistical Information - All Information, Statistical Information - Current Workpiece, Sensor Information - Current Offset, and Laser Tracking. If the RealTime Display is

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empty, the corresponding data can only be displayed after the corresponding function is executed.

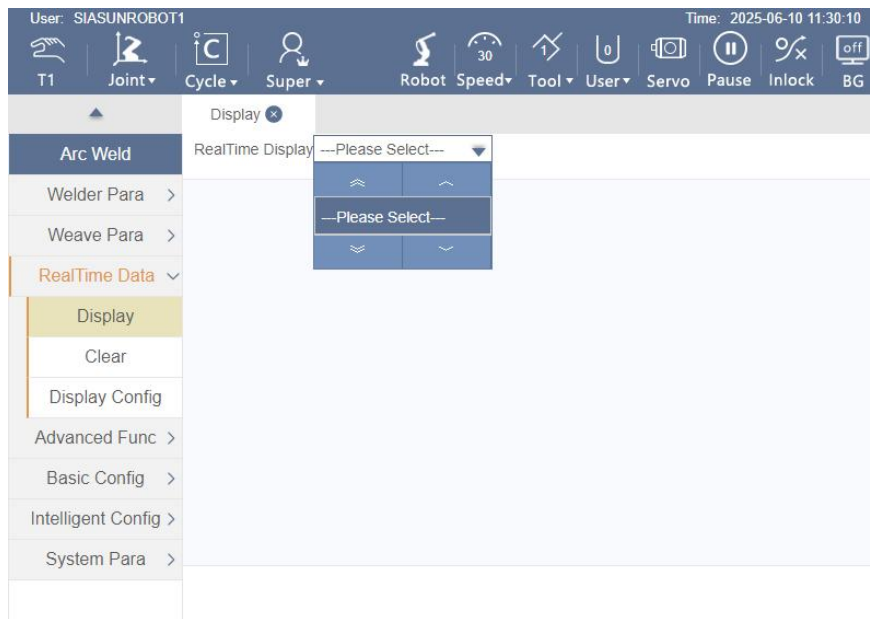


Figure 2-28 Arc Welding Real-time Display Interface

Click [Arc Weld] - [Arc Weld Real-Time] - [Clear]: Enter the real-time clearing interface to reset the corresponding data and re-record it. Under Normal User, only the number of workpieces is reset to zero. Under Advanced User or Super User, the rest of the items can be reset to zero.

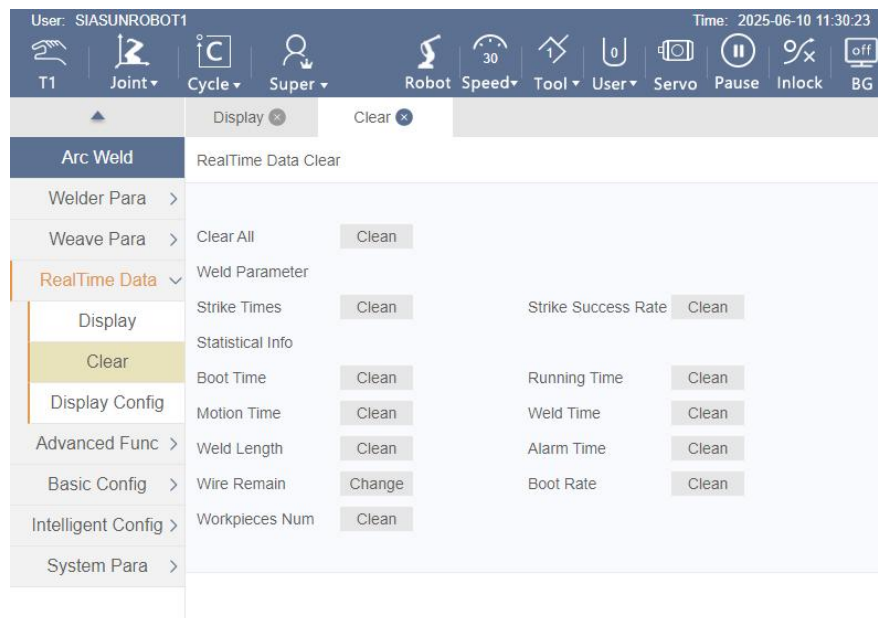


Figure 2-29 Arc Welding Real-time Display Interface

Click [Arc Weld] - [Arc Weld RealTime] - [Display] to enter the display configuration interface and select the content to be displayed in real time as needed.

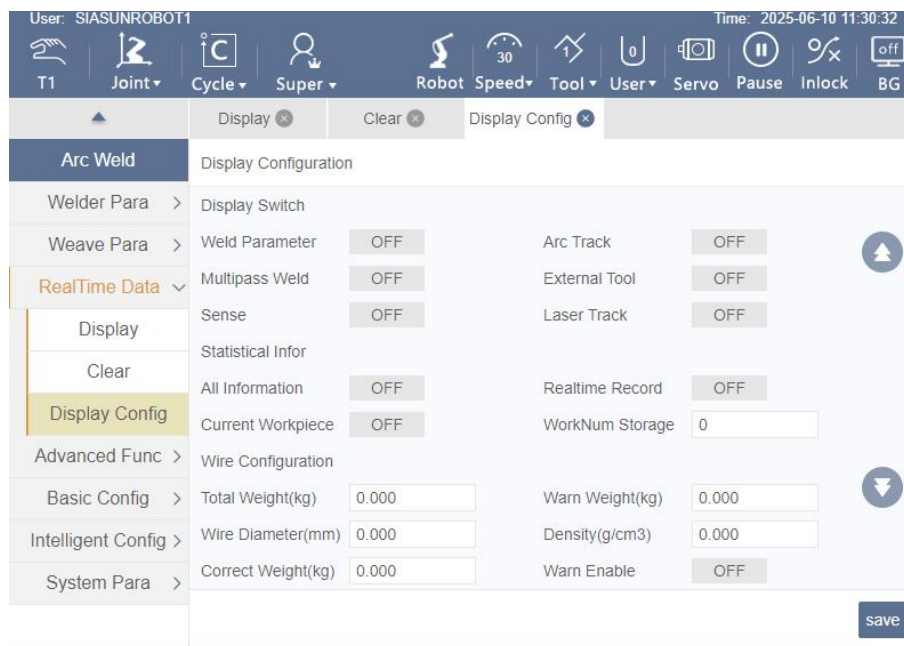


Figure 2-29 Display Configuration Interface

Real-time Recording: If it is ON, the startup time, running time and other data can still be recorded after shutdown. If it is OFF, the recording will resume after shutdown. It is OFF by default.

Welding Wire Configuration: To ensure the accuracy of the display of remaining welding wire quantity and welding wire usage information, the welding wire configuration parameters need to be accurately filled in according to the actual working conditions.

2.7 Weave Quench

2.7.1 File Configuration

Click [Arc Weld] - [Welder Para] - [Quench File], and set Swing Quench Enable to ON.

When using this function, simply turn on the weave quench enable switch for the quench file.

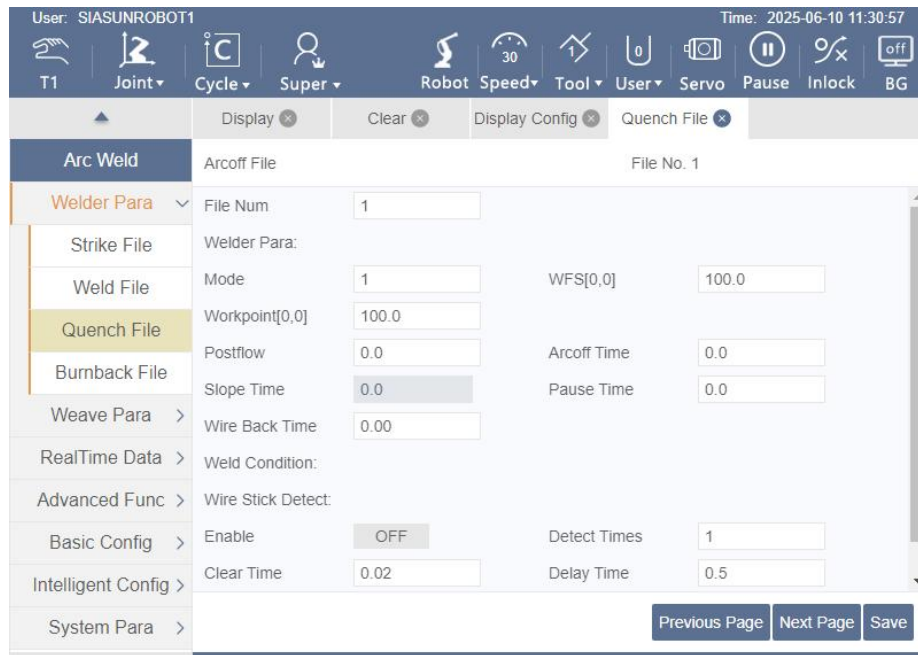


Figure 2- 30 Weave Quench Configuration

Notes:

Sequence of job instructions: WAVE OFF must be after ARCOFF; otherwise, there will be no weave quenching effect.

Between the arc striking and arc quenching instructions, there must be a motion instruction; otherwise, an alarm will be triggered.

2.8 Reservation

To use the reservation function, the users need to design their own special reservation boxes (multiple) corresponding to multiple workstations. Each console has a start button, and each start button corresponds to a job. Users can start the job corresponding to each button in the order they press the start button. There are other buttons on the reservation box, including pause, reset and emergency stop. There should be no enable (power on) button on the reservation box. Currently, this function is not supported.

2.8.1 Hardware Connection

The button module on a normal reservation box basically consists of two lines: one is connected to the user's io input board, and the other is connected to the 0v output. (For those with

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indicator lights, two $\pm 24V$ wires should be added and connected to 24v and 0v of the robot.) The button modules on the reservation box can be connected in parallel, and a single 0v line can be connected (the pause line is different and needs to be taken out separately).

1) Reservation start button:

Each start button represents a reserved job. The 0v line of the start button is connected in parallel with other buttons and connected to the user's io input board. The input and output of the corresponding job to the number in the reservation setting will be filled in accordingly.

2) Alarm reset and clear button:

It is nearly the same as the start button, with the only difference that you need to fill in the number of the connected user io input board in [io] - [Remote io] - [Input].

3) External pause button:

Except for the connection method of multiple pause buttons to be connected in series, the modification method is the same as that of the alarm reset button configuration.

Notes:

- a) **The external pause electrical component has normally closed interfaces. All the other interfaces are normally open.**
- b) **SR10C robot: For external pause on the remote input page, you must enter 12 in [io] - [Remote io] - [Input]. However, the flash arc robot does not have this requirement.**

Function	Input Number	Shield
Call Main Job:	4082	☐
Reset Alarm:	4083	☐
External Stop:	Unused	☒
External Start:	4085	☐
External Pause:	Unused	☒
External Servo Off:	Unused	☒
External Servo On:	4088	☐
Safety Gate Request	Unused	☒
Energy saving	Unused	☒

Save

Figure 2-49 Remote Input Configuration Interface

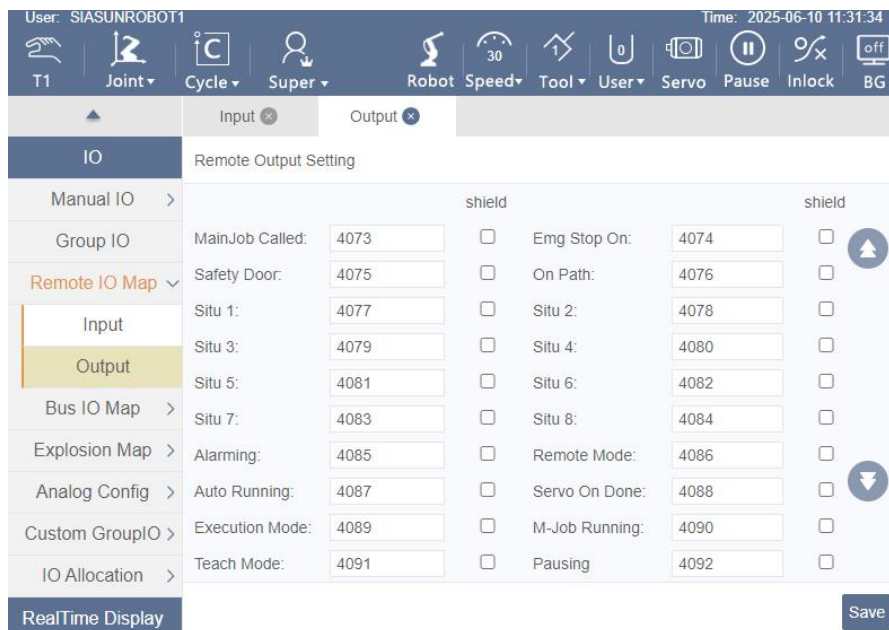


Figure 2-50 Remote Output Configuration Interface

4) Emergency stop button:

The emergency stop safety circuit is located at the P12 terminal block of the cabinet door interface board. Generally, interfaces 1 to 12 must form a circuit. If there is a safety door, please consult the electrical personnel or refer to Electrical Operation and Maintenance Manual for SIASUN Robot. Similar to the external pause button, the emergency stop button of the reservation box should be connected in series and then connected to ports 1 to 4.

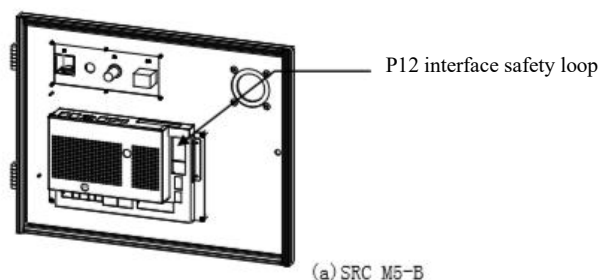


Figure 2-51 Position Diagram of Emergency Stop Safety Circuit

2.8.2 Robot Configuration

2.8.2.1 Reservation Job Switch

To use the reservation function, it is necessary to turn on Enable in the Execute Switch interface.

Click [System Setup] - [Execution Switch] to enter the execution switch interface and turn on the reservation switch. (Once it is turned on, the start button in the execution mode of the teaching box and the external start in remote io cannot be used. It can only be controlled through the start button of the reservation box.)

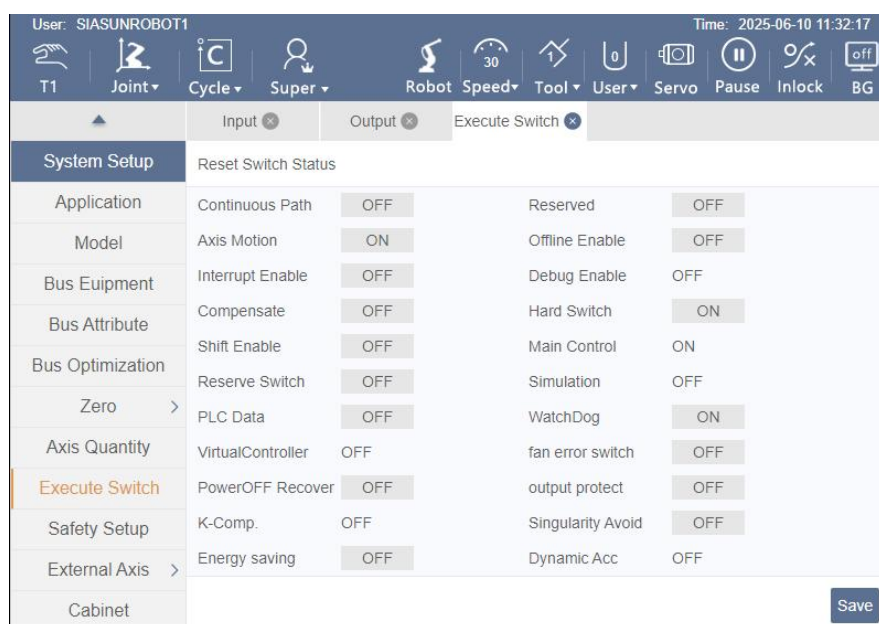


Figure 2- 52 Reservation Job Position Diagram

2.8.2.2 Reservation Job Name and Input/Output Configuration

Record the job name of each workstation and the input and output signals when starting on the reservation configuration interface. In this way, the job can be started by pressing the start button on the workstation.

Click [User Configuration] - [Reservation] - [Reservation Setting] to enter the reservation setting interface. Enter the job name, input and output signals for starting the job according to the job corresponding to the reservation box (fill in the user io input port for input and the user io output port for output). AllClear on the interface can completely clear the job names and input/output io numbers filled in.

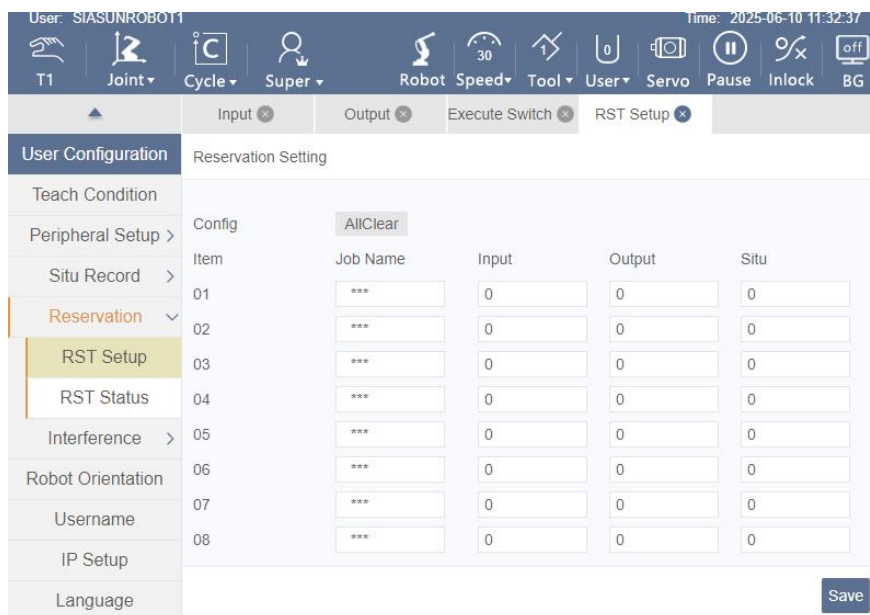


Figure 2- 53 Reservation Setting Interface

2.8.2.3 Job Reservation Status

You can check the current status of the robot reservation on the reservation status interface.

Click [User Configuration] - [Reservation] - [Reservation Status] to enter the reservation status interface. If you find that the job cannot be called out after using the start button, first enter this interface to check whether the job reservation status is normal. Or if the reserved job cannot proceed due to other reasons, and you want to restart the reserved job, you must also enter this interface to select and clear all existing job statuses before starting the job.

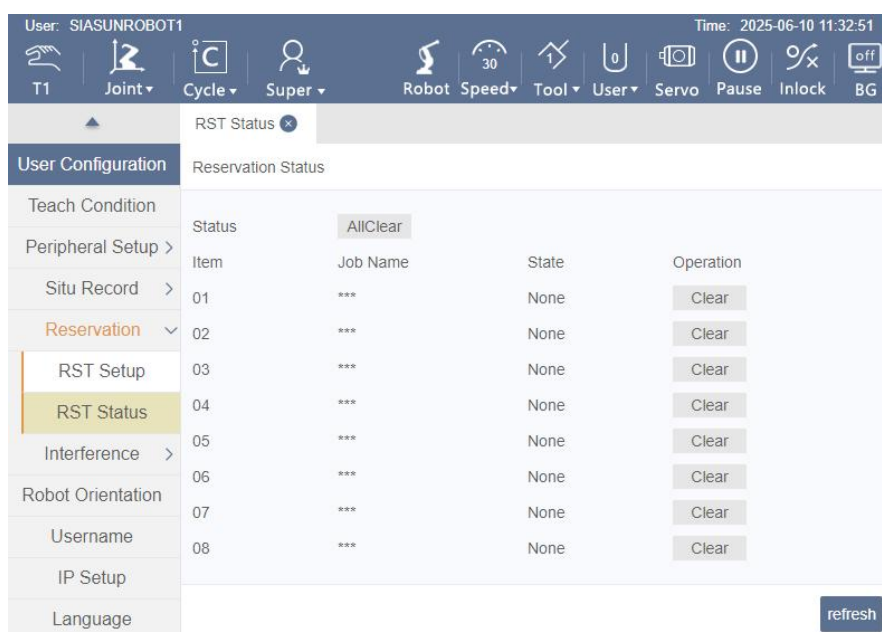


Figure 2- 54 Reservation Status Check Interface

2.8.2.4 Origin Setup

The reservation function requires each job to start from the origin. Therefore, when using the reservation function, it is necessary to set the origin and make the start and end of each job stop at the origin. So, teach the robot to a common safety point, turn it into a moving point and save it in the first and last instructions of all reserved jobs.

Click [User Configuration] - [Situ Record] - [Situ], click Situ Calibration and then click [Save]. Up to 8 situ values can be set.

User Configuration		Situ Setting and Calibration	
		Situ 1	Calibration
Teach Condition			
Peripheral Setup >			
Situ Record ▾			
Situ			
Range			
External Axis			
Reservation >			
Interference >			
Robot Orientation			
Username			
IP Setup			

Situ Calibration ☐ Current origin sequence number

J1:

J2:

J3: Direct setting

J4:

J5:

J6:

Figure 2- 55 Situ Record and Calibration Interface

2.8.3 Notes

- 1) After starting the job using the reservation function, the job will only be executed in a single cycle once and then stop.
- 2) The origin positions of the first and last instructions in each reserved job must be the same before it can be started.
- 3) After the reservation switch is turned on, the start button on the teaching box becomes

invalid.

- 4) For detailed information, please refer to the reservation function in General Operating Manual for SIASUN Robot.

2.9 IO Setting Interface

The IO setting interface is used to set the starting bit and length of the input and output IO of the welder.

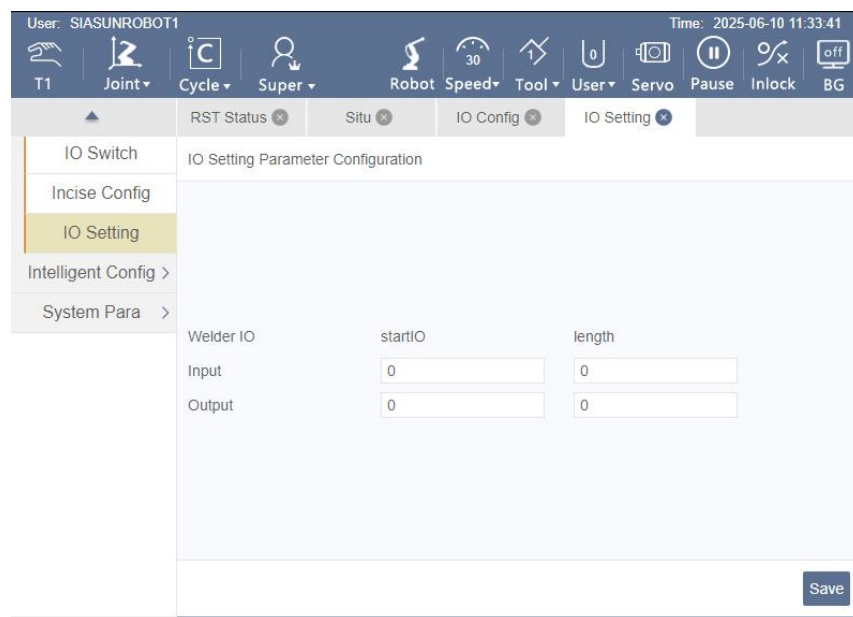


Figure 2-57 IO Setting Interface

1) Configuration requirements:

In this interface, the set parameters are required to meet the IO range of the arc welding configuration in the logic IO allocation. The logic IO allocation is shown in the following figure:

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User: SIASUNROBOT1 Time: 2025-06-10 11:34:29

Input IO Allocation Page:1

NO	App	Start	Length	Status	Annotation
1	None	1	0	UNASG	
2	None	1	0	UNASG	
3	None	1	0	UNASG	
4	None	1	0	UNASG	
5	None	1	0	UNASG	
6	None	1	0	UNASG	
7	None	1	0	UNASG	
8	None	1	0	UNASG	

Previous Page Next Page Save

User: SIASUNROBOT1 Time: 2025-06-10 11:34:38

Output IO Allocation Page:1

NO	App	Start	Length	Status	Annotation
1	None	1	0	UNASG	
2	None	1	0	UNASG	
3	None	1	0	UNASG	
4	None	1	0	UNASG	
5	None	1	0	UNASG	
6	None	1	0	UNASG	
7	None	1	0	UNASG	
8	None	1	0	UNASG	

Previous Page Next Page Save

Figure 2-58 Logical IO Allocation Interface

If the IO range in the IO configuration is modified, resulting in IO not within the range of logical IO, an alarm will be triggered when saving. Moreover, if the arc welding IO range in the logical IO allocation is modified, resulting in a conflict with that in the IO setting interface, the robot will give an alarm prompt.

The range of logic IO allocation can be larger than the range in the set IO.

2) Shielding setup:

If this function is not needed, all the data in the IO setting can be changed to 0, and

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accordingly, the alarm will be shielded.

3) User IO category

After the configuration in the IO settings is saved, the corresponding IO will be set as the IO of the arc welder. In the IO category, it can be quickly viewed.

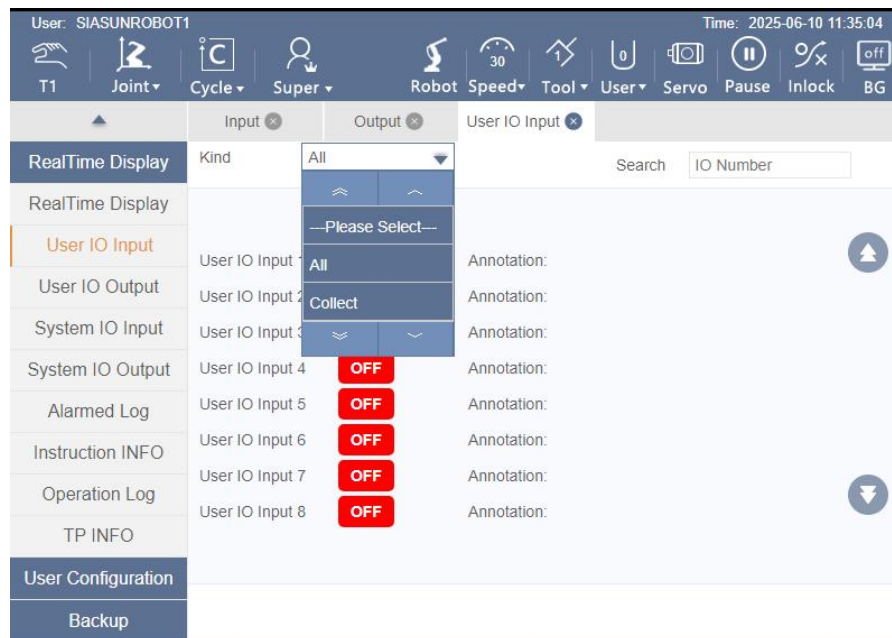


Figure 2-59 IO Category Interface

4) Kit System

After the welder is modified to a kit and restarted, the arc welding kit system, will automatically set the corresponding logical IO allocation and the parameters on the IO setting interface.

3 Appendix

3.1 Arc Welding Motion Instructions

ARCJ	Function	Movement between two points is executed in joint interpolation mode.	
	Format	ARCJ P/PR (Parameter 1) VJ = (Parameter 2) ACC = (Parameter 3) CNT = (Parameter 4) INTERLOCK= (Parameter 5)	
	Description	Parameter 1	P<Position variable number>, range: 1-12000 PR<Position variable number>, range: 1-5630

		Parameter 2	Meaning: Joint movement speed Unit: % Range: 1-100
		Parameter 3	Meaning: Joint movement acceleration Unit: % Range: 1-120
		Parameter 4	Meaning: Approach level Range: 0 and 1
		Parameter 5	Meaning: INTERLOCK function (for details, please refer to INTERLOCK Description in General Operating Manual for SIASUN Robot).
	Examples	ARCJ P3 VJ = 20 ACC = 100 CNT = 0 INTERLOCK = 1; // Move to point P3 at joint speed.	
ARCL	Function	Movement between two points is executed in linear interpolation mode.	
	Format	ARCL P/PR(Parameter 1) VL = (Parameter 2) ACC = (Parameter 3) CNT = (Parameter 4) ET/ COORD = (Parameter 5) INTERLOCK = (Parameter 6)	
	Description	Parameter 1	P<Position variable number>, range: 1-12000 PR<Position variable number>, range: 1-5630
		Parameter 2	Meaning: Linear movement speed Unit: mm/s Scope: Limited by the model of the robot
		Parameter 3	Meaning: The maximum acceleration at the end of the robot Unit: % Range: 1-120
		Parameter 4	Meaning: Approach level Range: 0-100
		Parameter 5	ET meaning: ET stands for external tool

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			number, that is, external TCP, range: 1-3; COORD meaning: Positioner system number Range: 1: Positioner System 1 2: Positioner System 2 3: Positioner System 3
		Parameter 6	Meaning: INTERLOCK function. For details, please refer to the interlock description. Range: 0 and 1
	Examples	ARCL P4 VL = 200 ACC =100 CNT = 0 COORD= 3 INTERLOCK=1; // In the user coordinate system, move at a right-angle speed to point P4.	
ARCC	Function	Movement among three points is executed in circular interpolation mode.	
	Format	ARCC P/PR(Parameter 1) VC = (Parameter 2) ACC = (Parameter 3) CNT = (Parameter 4) ET/ COORD = (Parameter 5) INTERLOCK= (Parameter 6)	
	Description	Parameter 1	P<Position variable number>, range: 1-12000 PR<Position variable number>, range: 1-5630
		Parameter 2	Meaning: Circular movement speed Unit: mm/s Scope: Limited by the model of the robot
		Parameter 3	Meaning: The maximum acceleration at the end of the robot Unit: % Range: 1-120
		Parameter 4	Meaning: Approach level Range: 0-100
		Parameter 5	ET meaning: ET stands for external tool number, that is, external TCP, range: 1-3; COORD meaning: Positioner system number Range: 1: Positioner System 1 2: Positioner

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			System 2 3: Positioner System 3
		Parameter 6	Meaning: INTERLOCK function. For details, please refer to the interlock description. Range: 0 and 1
	Examples	ARCL P4 VL = 200 ACC =100 CNT = 0 COORD= 3 INTERLOCK=1; // arc starting point ARCC P5 VC = 200 ACC =100 CNT = 0 COORD= 3 INTERLOCK=1; arc process point ARCL P4 VL = 200 ACC =100 CNT = 0 COORD= 3 INTERLOCK=1; // arc end point	
ARCC IRA	Function	By using the small circle instruction to teach the center point, the welding of the entire circle can be achieved.	
	Format	ARCCIRA P/PR Parameter 1 VC=Parameter 2 ACC=Parameter 3 CNT=Parameter 4 R=Parameter 5 OFFSET=Parameter 6 OVER=Parameter 7 UF=Parameter 8 FIFL=Parameter 9	
	Description	Parameter 1	Center point position, local/global position variable number.
		Parameter 2	Arc speed, with the maximum limit of 2000, in mm.
		Parameter 3	Percentage of maximum acceleration, unit: %, range: 1-120
		Parameter 4	Approach level. In this instruction, the default value is 0, and no other values are accepted
		Parameter 5	Radius of target circle, in mm, range: 0.1-100
		Parameter 6	Starting point offset (unit: mm): OFFSET>0 indicates that the starting point is outside the circle; OFFSET=0 indicates that the starting point is on the circle; OFFSET<0 indicates that the starting point is within the circle;

			(-100,100)
		Parameter 7	End point offset; over weld quantity, in mm; range: (-100,100); by using R, convert the unit of OVER into the angle; if the offset angle is greater than 60 degrees, an alarm should be triggered, that is, OVER (°) <60°;
		Parameter 8	User frame number, 1-128, used to confirm the circular plane;
		Parameter 9	Weld file number, range: 1-32;
	Examples	ARCCIRA P1 VC=100 ACC=100 CNT=0 R=100 FFSET=0 OVER=80 UF=1 FIFL=1	

3.2 Arc Welding Logic Instructions

3.2.1 Welding Instructions

ARCON	Function	Welding start.	
	Format	ARCON #<Parameter 1> #<Parameter 2>	
	Description	Parameter 1	Strike file number, range: 1-32.
		Parameter 2	Weld file number, range: 1-32.
	Examples	ARCON #1 #3 // Strike using the strike file number 1 and begin welding with the welding parameters in the weld file number 3.	
ARCOFF	Function	Welding end.	
	Format	ARCOFF #<Parameter 1>	
	Description	Parameter 1	Quench file number, range: 1-32.
	Examples	ARCOFF #1 // Quench using the quench file number 1.	
ARCON2	Function	Quick-change function of welding torch: When connecting the two welders, the welder 2 begins welding.	
	Format	ARCON2 #<Parameter 1> #<Parameter 2>	
	Description	Parameter 1	Strike file number, range: 1-32.
		Parameter 2	Weld file number, range: 1-32.

	Examples	ARCON2 #1 #1 // Strike using the strike file number 1 and begin welding with the welding parameters in the weld file number 1.	
ARCOFF2	Function	Quick-change function of welding torch: When connecting the two welders, the welder 2 ends welding.	
	Format	ARCOFF #<Parameter 1>	
	Description	Parameter 1	Quench file number, range: 1-32.
	Examples	ARCOFF2 #1 // Quench using the quench file number 1.	
PARA_S	Function	It is used when modifying welding parameters.	
	Format	PARA SWITCH #<Parameter 1>	
	Description	Parameter 1	Weld file number, range: 1-32.
	Examples	PARA SWITCH #1 // Perform welding using the parameters of weld file number 1.	
ARCON_UI	Function	The strike instruction can set the strike current and voltage through the instruction parameters.	
	Format	ARCON_UI #Parameter 1 #Parameter 2 I=Parameter 3 U=Parameter 4	
	Description	Parameter 1	Strike file number, range: 1-32.
		Parameter 2	Weld file number, range: 1-32.
		Parameter 3	Strike current value.
		Parameter 4	Strike voltage value.
	Examples	ARCON_UI #1 #1 I=200 U=20	
ARCOFF_UI	Function	The quench instruction can set the quench current and voltage as well as the quench time through the instruction parameters.	
	Format	ARCOFF_UI #Parameter 1 I=Parameter 2 U=Parameter 3 T=Parameter 4	
	Description	Parameter 1	Quench file number, range: 1-32.
		Parameter 2	Quench current.
		Parameter 3	Quench voltage.
		Parameter 4	Quench time.

	Examples	ARCOFF_UI #1 I=150 U=15 T=1.0	
PARA_S_UI	Function	The welding parameter instruction can set the welding current and voltage through the instruction parameters.	
	Format	PARA_S_UI #Parameter 1 I=Parameter 2 U=Parameter 3	
	Description	Parameter 1	Weld file number, range: 1-32.
		Parameter 2	Switch the welding current.
		Parameter 3	Switch the welding voltage.
	Examples	PARA_S_UI #1 I=250 U=24	
ARCON_UI2	Function	The strike instruction of the welder 2 can set the strike current and voltage through the instruction parameters.	
	Format	ARCON_UI2 #Parameter 1 #Parameter 2 I=Parameter 3 U=Parameter 4	
	Description	Parameter 1	Strike file number, range: 1-32.
		Parameter 2	Weld file number, range: 1-32.
		Parameter 3	Strike current value.
		Parameter 4	Strike voltage value.
	Examples	ARCON_UI2 #1 #1 I=200 U=20	
ARCOFF_UI2	Function	The quench instruction of the welder 2 can set the quench current and voltage as well as the quench time through the instruction parameters.	
	Format	ARCOFF_UI2 #Parameter 1 I=Parameter 2 U=Parameter 3 T=Parameter 4	
	Description	Parameter 1	Quench file number, range: 1-32.
		Parameter 2	Quench current.
		Parameter 3	Quench voltage.
		Parameter 4	Quench time.
	Examples	ARCOFF_UI2 #1 I=150 U=20 T=1.0	
PARA_S_UI2	Function	The welding parameter instruction of the welder 2 can set the welding current and voltage through the instruction parameters.	
	Format	PARA_S_UI2 #Parameter 1 I=Parameter 2 U=Parameter	

		3	
	Description	Parameter 1	Weld file number, range: 1-32.
		Parameter 2	Switch the welding current.
		Parameter 3	Switch the welding voltage.
	Examples	PARAMETER_S_UI2 #1 I=250 U=24	

3.2.2 Arc Welding Auxiliary Instructions

WELDER	Function	Switch the welder when using the quick-change function of the welding torch.	
	Format	WELDER TYPE=<Parameter 1>	
	Description	Parameter 1	1: Welder 1; 2: Welder 2;
	Examples	WELDER TYPE=1	
WALLDIR	Function	Wall direction setting instruction.	
	Format	WALLDIR TYPE=Parameter 1	
	Description	Parameter 1	Wall direction 1: Right wall; 2: Left wall
	Examples	WALLDIR TYPE=1	
FWD FWDWIRE1	Function	Automatic wire discharge instruction.	
	Format	FWDWIRE T=<Parameter 1>	
	Description	Parameter 1	Wire discharge time.
	Examples	FWDWIRE T=3	
FWD FWDWIRE2	Function	Automatic wire discharge instruction 2; usage: Dual-machine welding.	
	Format	FWDWIRE2 T=<Parameter 1>	
	Description	Parameter 1	Wire discharge time.
	Examples	FWDWIRE2 T=3	
FWD DFWDWIRE	Function	Multi-welder automatic wire discharge instruction.	
	Format	DFWDWIRE T=<Parameter 1> NO=<Parameter 2>	
	Description	Parameter 1	Wire discharge time
		Parameter 2	0: Multiple welders discharge wire simultaneously.

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			1: Only welder 1 discharges wire. 2: Only welder 2 discharges wire.
	Examples	DFWDWIRE T=3 NO=01	
WREFP	Function	Specify the corresponding welding reference plane according to the corresponding system environment.	
	Format	WREFP TYPE = <Parameter 1>	
	Description	Parameter 1	The Z direction of the welding reference plane is specified as follows: 1: z+direction under World. 2: y+direction under World. 3: x+direction under World. 4: Welding torch (TCP) z- direction.
	Examples	WREFP TYPE=1	
WAVEREFE	Function	Specify the corresponding weave reference plane according to the corresponding system environment.	
	Format	WAVEREF TYPE = <Parameter 1>	
	Description	Parameter 1	Weave reference type: 1: Welding reference plane. 2: Welding torch reference (welding torch TCP z- direction). 3: Welding torch entry direction reference (the starting point points to the direction of the previous safety point).
	Examples	WAVEREF TYPE = 1	
WAVE ON#	Function	Call the parameters in the weave file (1-32) corresponding to # to perform subsequent weave movements	
	Format	WAVE ON #<Parameter 1>	
	Description	Parameter 1	Weave File. Range: 1-32
	Examples	WAVE ON #1	

WAVE ON	Function	Weave instruction, customizing weave by setting parameter items.	
	Format	WAVE ON <Parameter 1> F=<Parameter 2> A=<Parameter 3> SEC=<Parameter 4> SEC=<Parameter 5>	
	Description	Parameter 1	There are six types of instructions in total. Sine: Sine weave Crescent: Crescent weave Triangle: Triangle weave L-Type:L-type weave Circle:Circular weave (no residence time) Type: Figure-8 weave (no residence time)
		Parameter 2	Weave frequency, range: 0.1-99.99Hz;
		Parameter 3	Swing, range: 0.1-99.99mm;
		Parameter 4	SEC: The first SEC is the left residence time, ranging from 0 to 100 seconds;
		Parameter 5	SEC: The second SEC is the right residence time, ranging from 0 to 100 seconds.
	Examples	WAVE ON Sine F=1.00 A=2.00 SEC=0 SEC=0	
WAVE OFF	Function	Weave stops in the weave library mode.	
	Format	WAVE OFF	
	Description	Parameter 1	This instruction has no parameter configuration.
	Examples	WAVE OFF	
IMWON	Function	The function of intermittent welding is enabled.	
	Format	IMWON TYPE=<Parameter 1> #<Parameter 2>. This instruction needs to be added after the starting position of	

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		the weld line (motion instructions such as ARCL, including ARCL_P ARCC SARCL).	
	Description	Parameter 1	File number, corresponding parameter interface; there are a total of eight files.
		Parameter 2	The function type of intermittent welding is 1. Other numbers are currently useless.
	Examples	IMWON TYPE=1 #1	
IMWOFF	Function	The intermittent welding function ends. It should be inserted after the last point of the weld line.	
	Format	IMW OFF	
	Description	Parameter 1	This instruction has no parameter configuration.
	Examples	IMW OFF	
LTWLT	Function	Quickly change the forward angle of the welding torch to the set angle within a short distance to meet the requirements of the welding process. Note: This instruction needs to be inserted between two motion instructions.	
	Format	LTWLT A=<Parameter 1> #<Parameter 2>	
	Description	Parameter 1	Target angle to which the forward angle of the welding torch needs to be changed.
		Parameter 2	File number that the instruction calls, corresponding to the parameter interface setting.
	Examples	LTWLT A=30 #1	
ARCON_S	Function	Weld by intermittent welding method.	
	Format	ARCON_S #Parameter 1 V1=Parameter 2 T1=Parameter 3 V2=Parameter 4 T2=Parameter 5	
	Description	Parameter 1	File number: File number for continuous spot welding function, range: 1-32.

		Parameter 2	Motion speed during welding (greater than or equal to 0, with 0 indicating welding in place).
		Parameter 3	Motion time during welding (greater than 0).
		Parameter 4	Motion speed during welding interrupt (greater than or equal to 0).
		Parameter 5	Motion time during welding interrupt (greater than 0).
	Examples	ARCON_S #1 V1=3 T1=2 V2=1 T2=1	
ARCOFF_S	Function	Achieve arc quenching of continuous spot welding.	
	Format	ARCOFF_S	
	Description	None	
	Examples	ARCOFF_S	
ARCON_S2	Function	The welder 2 welds by intermittent welding method.	
	Format	ARCON_S2 #Parameter 1 V1=Parameter 2 T1=Parameter 3 V2=Parameter 4 T2=Parameter 5	
	Description	Parameter 1	File number: File number for continuous spot welding function, range: 1-32.
		Parameter 2	Motion speed during welding (greater than or equal to 0, with 0 indicating welding in place).
		Parameter 3	Motion time during welding (greater than 0).
		Parameter 4	Motion speed during welding interrupt (greater than or equal to 0).
		Parameter 5	Motion time during welding interrupt (greater than 0).
	Examples	ARCON_S2 #1 V1=5 T1=1 V2=10 T2=1	
ARCOFF_S2	Function	The welder 2 achieves arc quenching of continuous spot welding.	

	Format	ARCOFF_S2	
	Description	None	
	Examples	ARCOFF_S2	
POSTRANS	Function	Perform pose transformation on the position PR variable.	
	Format	POSTRANS #<Parameter 1>	
	Description	Parameter 1	It is used to calculate the position of the point to be positioned after positioning when certain positions of the weld seam are not suitable for positioning. This instruction calculates the intersection point of each PR variable according to the setting parameters of the corresponding file.
	Example	POSTRANS #1	
POTCAL	Function	Calculate the intersection point of each PR variable according to the setting parameters of the corresponding file.	
	Format	POTCAL #<Parameter 1>	
	Description	Parameter 1	File number for intersection point calculation, corresponding to the parameter interface.
	Examples	POTCAL # 1	
FWD FWDGAS1	Function	Gas discharge instruction.	
	Format	FWDGAS T=Parameter 1	
	Description	Parameter 1	Gas discharge time.
	Examples	FWDGAS T=1.50	
FWD FWDGAS2	Function	Gas discharge instruction of welder 2.	
	Format	FWDGAS2 T=Parameter 1	
	Description	Parameter 1	Gas discharge time.
	Examples	IMWON TYPE=1 #1	
FWD	Function	Gas discharge instruction of double wire welding.	

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DFWDGAS	Format	DFWDGAS T=Parameter 1 NO=Parameter 2	
	Description	Parameter 1	Gas discharge time.
		Parameter 2	0: Simultaneous gas discharge 1: Gas discharge of welder 1 2: Gas discharge of welder 2
	Examples	DFWDGAS T=1 NO=0	
ARCSET ARC_P_SUB	Function	Calculate the difference between the components of the two position variables and store it in the R variable.	
	Format	ARC_P_SUB PR Parameter 1.Parameter 2 –PR Parameter 3. Parameter 4 =R Parameter 5	
	Description	Parameter 1	Position variable number, range: 1-5630.
		Parameter 2	A component for position variable.
		Parameter 3	Position variable number, range: 1-5630.
		Parameter 4	A component for position variable.
		Parameter 5	Float variable for difference storage.
	Examples	ARC_P_SUB PR2.1 –PR3.1 =R99	
ARCSET ARC_P_EXT T	Function	Assign values to the external axes of the position variable, and assign the external axis value of the position variable of Parameter 3 to the position variables corresponding to Parameter 1 to Parameter 2.	
	Format	ARC_P_EXT PR Parameter 1–PR Parameter 2 PR Parameter 3	
	Description	Parameter 1	Position variable number, range: 1-5630.
		Parameter 2	Position variable number, range: 1-5630.
		Parameter 3	Position variable number, range: 1-5630.
	Examples	ARC_P_EXT PR2–PR8 PR99 Assign the external axis	

		value of the position variable 99 to the external axis values of the position variables 2-8	
WLINE_INIT	Function	Set the total number of weld lines of anti-skip function.	
	Format	WLINE_INIT Count=Parameter 1	
	Description	Parameter 1	Total number of weld lines, compared with the actual number of weld lines.
	Examples	WLINE_INIT Count=5	
WLINE_CHECK	Function	Weld line detection of anti-skip function.	
	Format	WLINE_CHECK	
	Description	Set a comparison between the total number of weld lines and the actual number of weld lines. If the two are not equal, an alarm will be triggered.	
	Examples	WLINE_CHECK	

4 Alarm information

4.1 Alarm

Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6490	Data storage CF card: Checksum verification failed. Data has been restored to default values. Please take note!	-	-	Resolution: System error. Please contact technical support.
6491	Welding torch collision. Unclamp brakes 4/5/6.	-	-	Move the robot to a safe position.

4.2 Warnings

Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6500	Torch collision	1	Torch 1, signal source is external	Resolution: Remove the collision and check the IO status

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			IO	corresponding to the collision of welding gun 2.
		2	Torch 2, signal source is external IO	Resolution: Remove the collision and check the IO status corresponding to the collision of welding torch 2.
		3	Torch 1, signal source is welder feedback	Resolution: Remove the collision and check the welding torch collision signal status corresponding to welder 1.
		4	Torch 2, signal source is welder feedback	Resolution: Remove the collision and check the collision signal status of welding gun corresponding to welder 2.
6501	Welder error	-	-	Resolution: The welder is faulty, please reset the welder fault
6502	Arc loss	-	-	Resolution: Check whether there is no welding wire or blocked wire, and replace the welding wire before starting
6503	Strike error	-	-	Resolution: Check whether there is welding slag or oil in the arc starting position or the ground wire is not connected."
6504	Wire stick	-	-	Resolution: Welding wire and workpiece stick, please remove.
6505	Arcon and arcoff inst not match	-	-	Resolution: Check inst format.
6506	Welder status	-	-	Resolution: Check whether the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	disconnect			welder ready signal is set ON.
6509	Lack of wire	1	Lack of wire 1	Resolution: No welding wire, check the welding status IO missing wire 1 Check whether the signal is set ON.
		2	Lack of wire 2	Resolution: No welding wire, check the welding status IO missing wire 2 Check whether the signal is set ON
6512	Weld parameter set error	-	-	Resolution: The welding parameters sent by multiple machines are abnormal. Check the configuration.
6518	Welder is in strike state	-	-	Resolution: If the arc starts again under the current arcing state, the alarm is raised and the stack needs to be emptied.
6526	Welder is in quench state			Currently in arc extinction state
6535	Parameter Set error	-	-	Resolution: Multi-machine wire, gas instruction parameters are abnormal, please check the Settings.
6538	Compressed air error	-	-	Resolution: If there is no compressed air, check the compressed air signal status in the welding status monitoring.
6543	Cooling water error	-	-	Resolution: If the water flow is low, check the water flow signal

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				status in the welding status monitoring.
6544	Gas error	1	Gas 1 error	Resolution: Lack of protection gas, check the protection gas 1 signal status in welding status monitoring.
		2	Gas 2 error	Resolution: Lack of protection gas, check the protection gas 2 signal status in welding status monitoring.
6548	Main welder set error	-	-	Resolution: The host parameter Settings on multiple hosts are abnormal. Check the configuration.
6553	Get clock beat error, please reconfigure arc weld parameter	-	-	Resolution: System error. Please contact technical support.
6555	Please update parameter again	-	-	Resolution: Lincoln arclink welder parameter setting error, please reset.
6556	Please check current weld parameter	-	-	Resolution: Lincoln arclink welder 30762 alarm, please check.
6557	Welder is not online because welder is upgrading, please wait	-	-	Resolution: Lincoln arclink welder parameter setting error, please reset.
6558	Feeder is not online because welder is upgrading, please	-	-	Resolution: Lincoln arclink welder setup error, please check.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	wait			
6560	Welder not ready	-	-	Resolution: Multi-machine ready signal is not set, please check.
6561	Feeder not ready	-	-	Resolution: Multi-machine feeder feedback signal is not set, please check.
6566	Welder error, please check	-	-	Resolution: The arc of the welder starts in the system arc quenching state, please stop the arc quenching immediately.
6571	Welder type error:	1	Welder 1 type	Resolution: System welder 1 Obtain an error, contact technical personnel.
		2	Welder 2 type	Resolution: System welder 2 Obtain an error, contact technical personnel.
		3	Instruction type error	Resolution: The system welding machine number is incorrect, please contact technical personnel.
6572	Weld and quench parameter error	-	-	Resolution: The system arc extinguishing parameters are incorrectly obtained. Contact technical personnel.
6574	Strike sequence:	0	Parameter get error	Resolution: Incorrect system parameters are obtained. Contact technical support.
		1	Strike execute error	Resolution: The arc starting state machine is abnormal. Contact technical support

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		2	Strike point offset error	Resolution: Please check the arc offset parameter Settings.
		3	Strike parameter send error	Resolution: Please check the arcing parameter Settings
		4	Perflow set error	Resolution: Please check the front gas parameter Settings.
		5	Welder turn on error	Resolution: Please check the arcing signal Settings.
		6	Welder feed on error	Resolution: Please check the wire feed signal Settings.
		7	Strike detect error	Resolution: Please check the arcing detection Settings.
		8	Arc restart error	Resolution: Check arc restart parameter Settings.
		9	Strike point return error	Resolution: Check arc restart parameter Settings.
		10	Strike parameter send error 1	Resolution: Please check the arcing parameter Settings.
		11	Arc status error	Resolution: Please check the feedback arc success signal status of the welder.
		12	Machine status error	Resolution: Please check whether the robot has an alarm.
		13	Strike error	Resolution: Check the alarm log for other alarms.
		14	Variable parameters-Weld parameter send error	Resolution: Check the file corresponding to the parameter change.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		15	Strike detection-Welder feed off error	Resolution: Please check the feed off signal configuration.
		16	Strike detection-Welder turn off error	Resolution: Please check the signal configuration of welding mechanism.
		17	Strike detection-Wire inching error	Resolution: Please check the wire signal configuration.
		18	Strike detection-Welder turn on error	Resolution: Please check the welder open signal configuration.
		19	Strike detection-Welder feed on error	Resolution: Please check the feed open signal configuration.
		20	Strike detection-Welder turn off error 1	Resolution: Please check the signal configuration of welding mechanism 1.
		21	Strike detection-Welder feed off error 1	Resolution: Please check the feed off 1 signal configuration.
		22	Arc restart-Lift error	Resolution: Check arc restart parameters.
		23	Arc restart-Retrack error	Resolution: Check arc restart parameters.
		24	Arc restart-Move to target error	Resolution: Check alarm log and arc restart parameter configuration.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		25	Arc restart-Restart parameter send error	Resolution: Check the arc restart starting parameters.
		26	Arc restart-Welder turn on error	Resolution: Check arc restart parameters.
		27	Arc restart-Welder feed on error	Resolution: Check arc restart parameters.
		28	Arc restart-Return to starting point error	Resolution: Check arc restart parameters.
		29	Arc restart-Arc detection error	Resolution: Check arc restart parameters.
		30	Wire inching-Touch Sense on error	Resolution: Please check the open sensing signal status.
		31	Wire inching-feed forward on error	Resolution: Please check the feed signal status.
		32	Wire inching-feed forward off error	Resolution: Please check the stop wire feed signal status.
		33	Wire inching-Touch Sense off error	Resolution: Check the status of the off sensing signal.
		34	Wire inching-feed backward on error	Resolution: Please check the drawing signal status.
		35	Wire inching-feed backward off error	Resolution: Please check the stop drawing signal status
		36	Arc restart-Welder feed off error	Resolution: Please check the stop wire feed signal status.
		37	Arc restart-Welder	Resolution: Please check the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			turn off error	arc welding success signal status.
6575	Quench sequence:	0	Parameter get error	Resolution: Failed to obtain the parameters. Contact technical support.
		1	Quench execute error	Resolution: The arc extinguishing state machine is abnormal. Contact technical support.
		2	Weave quench execute error	Resolution: Check the alarm log to see if there are any other alarms.
		3	Transition parameter send error	Resolution: Check the arcing parameter Settings.
		4	Burn back parameter send error	Resolution: Please check the backburn parameter Settings.
		5	Welder feed off error	Resolution: Failed to stop wire feed, please check the stop wire feed signal.
		6	Welder turn off error	Resolution: Welder failed to shut down, please check the welder shutdown signal.
		7	Stick detection-Arc is not extinguished	Resolution: The arc state of the sticky wire detection is abnormal, please stop the arc immediately
		8	Stick detection-touch sense on error	Resolution: Please check to turn on the sensing signal.
		9	Stick	Resolution: Check the detection

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			detection-detect condition error	condition configuration.
		10	Stick detection-touch sense off error	Resolution: Please check to turn off the sensing signal.
		11	Stick detection-Welder turn on error	Resolution: Please check the welder open signal configuration.
		12	Stick detection-Welder turn off error	Resolution: Please check the signal configuration of welding mechanism.
		13	Stick detection-touch sense on error 1	Resolution: Please check the status of open sensor signal 1.
		14	Stick detection-detect condition error 1	Resolution: Check the status of detection condition 1
		15	Stick detection-touch sense off error 1	Resolution: Check the status of off sensing signal 1.
		16	Stick detection process error	Resolution: Please check the thread detection configuration.
		17	Welder reset error	Resolution: Please check the welder reset signal configuration.
		18	Arc is not extinguished when quench	Resolution: Please stop the arc immediately and contact technical personnel.
		19	Robot error	Resolution: Check alarm log to check robot status.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		20	Quench process error	Resolution: Please check if there is any other alarm information.
		21	Start wireback error	Resolution: Please check the draw back signal configuration.
		22	Stop wireback error	Resolution: Please check the stop drawing signal configuration.
		23	Gas off error	Resolution: Please check the gas off signal configuration.
6576	Process recovery cannot be performed because welder switch is changed	-	-	Resolution: Check the switch configuration of the welding machine.
6577	Welder all turned off	-	-	Resolution: Check the switch configuration of the welding machine.
6579	Welder switch error	-	-	Resolution: Check the switch configuration of the welding machine.
6580	No workpiece was touched during start point detection process	-	-	Resolution: Check the maximum detection distance setting in the start point configuration file.
6581	Start point offset is too large	-	-	Resolution: Check whether the welding wire has touched the workpiece when moving to the sensing starting point.
6582	Wire always touche workpiece during detection	-	-	Resolution: Check whether the welding wire has touched the workpiece before sensing.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6584	Seam frame establish error, please check	-	-	Resolution: Check whether the moving points before and after the arc command coincide.
6585	Detection deviation exceed maximum range, please check	-	-	Resolution: Check the offset limit Settings in the start point configuration file.
6587	Instruction not support this seam, please check	-	-	Resolution: Weld type is wrong, please check whether there is no initialization instruction.
6588	Laser start point search error	-	-	Resolution: Check the sensor Settings and parameter Settings.
6589	Please adjust start point detection parameter	-	-	-
6590	Detect instruction sequence error	-	-	Resolution: Check that the command editing logic in the job is correct.
6592	Point laser parameter error, please contact technical staff	-	-	Resolution: RSPL laser sensor parameters are abnormal. Check the configuration.
6593	TDS detect error	-	-	Resolution: TDS function is abnormal, please check whether there are other alarms.
6594	Touch sense get welder parameter error	-	-	Resolution: System error. Please contact technical support.
6595	Touch sense signal delay compensation data process error	-	-	Resolution: System error. Please contact technical support.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6596	Detection instruction teach is too close, please adjust	-	-	Resolution: 3D sensing point teaching position is too close, please check.
6597	TSDF detect error	-	-	Resolution: TSDF function is abnormal, please check whether there are other alarm information.
6598	Save location error	-	-	Resolution: System error. Please contact technical support.
6599	Please turn on point laser analog switch	-	-	Resolution: Point laser scan data is not turned on analog switch, please check.
6600	This function requires advanced point laser calibration	-	-	Resolution: Point laser scanning function is not calibrated analog, please check.
6601	Two points are too close	-	-	Resolution: Analog calibration is too close to IO calibration point, please adjust.
6602	Incomplete detection instruction	-	-	Resolution: The number of 2D or 3D sensing instructions is not complete, please check.
6603	Point laser search error	0	Parameter file number out of bounds	Resolution: Check the command parameter Settings.
		1	Establish weld fream error	Resolution: Check the teaching position of the assignment.
		2	No signal change detected	Resolution: Check the sensor I/O Settings
		3	Point laser not calibrated	Resolution: Checkpoint laser coordinate system setting

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				interface.
		4	Thick detect error	Resolution: Test plate thickness test related Settings.
		5	Simulate scan action error	Resolution: Detects the action library configuration.
		6	OFF-ON signal before detection is ON	Resolution: Please change the start detection location.
		7	ON-OFF signal before detection is OFF	Resolution: Please change the start detection location
		8	Switch detect action error	Resolution: Please check the action library configuration
		9	Detect value out of limiting lave	Resolution: Check the offset limit Settings in the PLSS file.
		10	Safety protection-RX too large	Resolution: Adjust the position of the teaching point.
		11	Safety protection-RY too large	Resolution: Adjust the position of the teaching point.
		12	Safety protection-Rz too large	Resolution: Adjust the position of the teaching point.
		13	Speed get error	Resolution: System error. Please contact technical support.
		14	Direction not set	Resolution: Check the action library Settings
		15	Parameter file	Resolution: Check the command

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			number out of bounds	parameter Settings.
		16	Safety protection-RX too large	Resolution: Adjust the position of the teaching point.
		17	Safety protection-RY too large	Resolution: Adjust the position of the teaching point.
		18	Safety protection-Rz too large	Resolution: Adjust the position of the teaching point.
6605	Axles information inconformity, please check	-	-	Resolution: The axis group information of the offset to be superimposed is inconsistent with the current axis group information. Please check.
6606	Point laser auto scan error	0	Instruct section cross	Resolution: Check the job instruction editing logic.
		1	No adapt templet instruct	Resolution: Check whether the job has no template instructions.
		2	Scan templet filenum out of bounds	Resolution: Check the command parameter Settings.
		3	Scan templet para get error	Resolution: System error. Contact technical support.
		4	Weldgun return pos calculate error	Resolution: System error. Contact technical support.
		5	Action to weldgun retpos error	Resolution: Check the rollback distance parameter Settings.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		6	Scanning motion error	Resolution: Check for additional alarm information.
		7	Scanning data get error	Resolution: Checkpoint laser sensor Settings.
		8	Return to weldgun retpos error	Resolution: Check the rollback distance parameter Settings.
		9	Weld area no lift error	Resolution: System error. Contact technical support.
		10	manual mode weldline	Resolution: Check the mode selection on the adaptive configuration page.
6607	Laser error	0	The current sensor is not configured with relevant parameters	Resolution: Check sensor Settings.
		1	The current sensor number exceeds	Resolution: Check sensor Settings.
		2	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.
		3	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.
		4	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		5	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.
		6	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.
		7	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.
		8	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.
		9	Sensor serial number exceeds the limit, please confirm	Resolution: Check sensor Settings.
6608	Line laser sensor initial error			Resolution: System error. Please contact technical support.
6609	Sensor configuration error	-	-	Resolution: The sensor type is incorrectly set. Check the configuration.
6610	Sensor feedback data is not good, storage error	-	-	Resolution: Adjust sensor Settings.
6611	Sensor detect err,	-	-	Resolution: The sensing process is

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	please check			abnormal, please check whether there is other alarm information.
6612	Setting offset inst, the coordinate type is not supported	-	-	Resolution: The coordinate system is abnormal when the sensor offset is set, please check.
6613	Search Time cannot be longer than total time	-	-	Resolution: Check the location search time Settings.
6614	Search Failure	-	-	Resolution: The location search failed. Check the sensor status.
6615	Positioning between calibration near and far point, please adjust parameter!	-	-	Resolution: Please adjust the positioning distance parameter Settings in the action library.
6620	Laser track no data, please check			Resolution: Check whether the sensor has scanned the job.
6621	Laser track position error is too large	-	-	Resolution: Adjust the minimum delivery distance in the tracing parameters.
6622	Laser track offline buffer write error	-	-	Resolution: System error. Please contact technical support.
6623	Forward sight distance buffer data is too little	-	-	Resolution: Adjust the sampling interval parameter in the tracing parameter.
6625	Laser track buffer is empty			Resolution: System error. Please contact technical support.
6628	Please select correct file number			Resolution: The tracing file number is abnormal. Please check.
6629	High-low current	-	-	Resolution: Arc tracing high and

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	learning error			low current learning failure, please try again.
6632	Collected current number is 0, please check configuration			
6633	Groove type error			Resolution: Adaptive The specified groove type is incorrect. Please check.
6634	Laser track attitude gradual transfer parameter error	-	-	Resolution: The laser tracking attitude gradient parameter is abnormal. Please adjust it.
6635	Instruction teach is too close, please adjust			Resolution: The distance between two points of laser tracking is too close, please adjust.
6636	Invalid laser search mode	-	-	Resolution: The laser search mode is abnormal. Check the configuration.
6639	Track cumulative deviation overrun			Resolution: Please check the tracking deviation limit Settings.
6640	Current difference learning success	-	-	Resolution: Current difference learning is complete, please continue.
6641	Weave parameter not suitable for track	-	-	Resolution: Please check the swing parameters.
6642	Track pause	-	-	Resolution: Laser tracking process triggers pause, please check.
6643	Non track point exceed maximum	-	-	Resolution: If no data quantity exceeds the forward-looking distance, check the sensor and

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				workpiece.
6644	Laser track forward sight distance set error	-	-	Resolution: Please check the forward looking distance Settings in the tracking parameters.
6645	Sensor communication disconnected	-	-	Resolution: Check the sensor circuit.
6646	Arc track current data is incorrect	-	-	Resolution: System error. Please contact technical support.
6674	Communication disconnect			Resolution: The communication of migamk welding machine is abnormal, please check the communication line.
6675	Welder disconnect	-	-	Resolution: Welding machine communication is disconnected, please check the welding machine and robot communication, there may be interference and other problems.
6678	Welder disconnect	-	-	Resolution: please check the communication configuration interface, there may be interference and other problems.
6679	Sensor disconnect	-	-	Resolution: The connection of the Yinglai laser is abnormal. Check the sensor status.
6680	Communication registration exception	0	TCP registration times exceeded	Resolution: System error. Contact technical support.
		1	UDP registration	Resolution: System error. Contact

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			times exceeded	technical support.
		2	TCP abnormal registration information	Resolution: System error. Contact technical support.
		3	UDP abnormal registration information	Resolution: System error. Contact technical support.
6681	Incorrect response between robot and welder	-	-	Resolution: Please check the Lincoln arclink welder Settings.
6682	The communication of arc welding peripherals is abnormal	-	-	Resolution: Please check the arc welding peripheral communication lines.
6683	Logical IO allocation is out of scope	-	-	Resolution: Check the arc welding configuration screen for logical IO assignment.
6691	Strike number in cycle interval is greater than 20 times	-	-	Resolution: Check that a maximum of 20 welding wires are supported in the cycle.
6692	Realtime point number record is greater than maximum, please adjust	-	-	Resolution: Check that a single welding line supports a maximum of 1000 teaching points.
6693	Path record error	-	-	Resolution: Please check the teaching point of multi-pass welding and recommend to teach again

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6694	Exercise type get error	-	-	Resolution: Please check the current movement type.
6695	Multi-pass weld parameter initialization error, please restart	-	-	Resolution: System error. Please contact technical support.
6696	Seam parameter not existed	-	-	Resolution: Multipass welding template does not exist, please check the job editing.
6697	Multi template is inconsistent with Settings, Please check	-	-	Resolution: Check the number of tracks inserted into the database.
6698	The combined cycle interval does not support the specified pass	-	-	Resolution: Please check the type of multipass welding cycle in the job.
6699	No MLON command before switching of multi-pass welding template	-	-	Resolution: Please check the job editing instructions.
6700	The start offset point was incorrectly calculated	-	-	Resolution: Please check the multi-pass welding starting offset setting and show the teaching point again.
6701	The end offset point was incorrectly calculated	-	-	Resolution: Please check the multi-pass welding end offset setting and show the teaching point again.
6702	The two point	-	-	Resolution: Point calculation

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	distance calculate error			error, please check the teaching point.
6703	The center of the planned arc is incorrectly calculated	-	-	Resolution: The center of the circle is wrong. Please check the teaching point.
6704	The specified pass teaching point was not obtained	-	-	Resolution: The specified teaching point of the weld is abnormal. Please check the teaching point.
6705	The intelligent welding function does not directly start	0	Multi-pass welding The path jump is prohibited within the cycle interval	Resolution: Please manually move the robot outside the loop.
		1	No interlocking forward movement is allowed in the welding interval	Resolution: Close the chain or select Pause to start.
		2	No moving cursor is allowed to start directly in the welding zone	Resolution: Move back to the cursor or select pause to start.
6706	Multi-pass welding does not support manual mode interlocking execution	-	-	Resolution: Manual mode is not supported for multi-pass welding, please note.
6708	Reproduction track is empty	-	-	Resolution: Check the multi-pass root path parameter Settings.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6709	Reproduction track instruction number error	-	-	Resolution: Please check the job editing.
6710	Scan instructions error	-	-	Resolution: Please check the job editing.
6711	Seam scan failure	-	-	Resolution: Check the scan path and sensor status.
6712	-	-	-	-
6713	Multi-pass welding search point abnormal	-	-	Resolution: Please check the job editing.
6714	Calculate the offset coordinate exception	-	-	Resolution: Overlay channel offset coordinate system calculation is abnormal, please check the point position.
6740	Offline move error	-	-	Resolution: Motion planning problem location alarm, check alarm log, contact technician.
6741	Plan library 1.0 is supported for arc,current plan library is 2.0. Please check	-	-	Resolution: Change the planned library in user configuration to 1.0
6742	Instruction target point get error, please check	-	-	Resolution: The command target point is not obtained. The point planning is incorrect. Contact technical personnel.
6743	Too small distance between two points	-	-	Resolution: The distance between the two points of the intermittent

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				welding function is too small, please adjust it.
6744	Current instruction only supports offline motion	-	-	Resolution: Please check the current motion mode.
6745	Weave library parameter not entered	-	-	Resolution: Swing parameters are not set, please check the command parameters and swing library configuration interface.
6747	Instruction position error	-	-	Resolution: The current instruction is not in the intermittent welding interval, please check.
6748	Fixed point weave error	-	-	Resolution: If the point replanning process is abnormal, show the teaching point again and contact technical personnel.
6749	Point re-plannings error	-	-	Resolution: Point planning error, please check whether there are other alarm information.
6750	Position variable operation error	-	-	Resolution: System error. Please contact technical support.
6751	Angle of WAVE and MOVE is too small, not between 10 and 70 degrees	-	-	Resolution: The oscillation coordinate system establishes an anomaly and changes the oscillation reference.
6752	Not find code of move	-	-	Resolution: Please check the job editing.
6753	Arc planning error of	0	Two points over	Resolution: Please check the job

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	arc welding		180 degrees	editing point.
		1	Two points are too close	Resolution: Please check the job editing point.
		2	Two points are too close	Resolution: Please check the job editing point.
		3	Two points are too close	Resolution: Please check the job editing point.
		4	Two points are too close	Resolution: Please check the job editing point.
		5	Two points are too close	Resolution: Please check the job editing point.
		6	Two points are too close	Resolution: Please check the job editing point.
		7	Two points are too close	Resolution: Please check the job editing point.
		8	Two points are too close	Resolution: Please check the job editing point.
6754	The arc section of arc welding is entered incorrectly	-	-	Resolution: Please check the job editing.
6755	Description Failed to obtain the middle point of the arc	-	-	Resolution: Please check the teaching point.
6756	The starting point of the arc trajectory has no linear moving point	-	-	Resolution: Please check the job editing.
6757	The job that include arce inst is illegal	-	-	Resolution: Please check the job editing.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6760	Arc restart read CF card error. It needs to be reconfigured	-	-	Resolution: An error occurs when data is written to the CF card. Contact technical support.
6761	Welder parameter read CF card error. It needs to be reconfigured	-	-	Resolution: An error occurs when data is written to the CF card. Contact technical support.
6764	Data pointer is empty	-	-	Resolution: The pointer data is empty. Contact technical personnel.
6765	Invalid data	-	-	Resolution: The pointer data is empty. Contact technical personnel.
6766	IMW instruction function error	-	-	Resolution: The intermittent welding function is abnormal, please check the editing of the job.
6768	File operation: File open error	-	-	Resolution: If reading or writing data fails, contact technical personnel.
6769	File operation: Folder create error	-	-	Resolution: If reading or writing data fails, contact technical personnel.
6770	File operation: CF data read error	-	-	Resolution: If reading or writing data fails, contact technical personnel.
6772	CF card: Device initialization error	-	-	Resolution: Data synchronization fails. Contact technical support.
6773	CF card: synchronization	-	-	Resolution: Data synchronization fails. Contact technical support.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	completed, with more than 3 consecutive failures			
6774	Calculate intersection type error	-	-	Resolution: Please check the type selection parameter configuration of the intersection calculation.
6775	Point cannot be calculated	-	-	Resolution: Please check the teaching position.
6776	Frame establish error	-	-	Resolution: Please check the teaching position.
6778	Wrong seam skip position	-	-	Resolution: Please re-operate the welding wire jump.
6779	Automatic positioning error	-	-	Resolution: System error. Please contact technical support.
6780	File operation: CF data write error	-	-	Resolution: If reading or writing data fails, contact technical personnel.
6781	File operation: CF load error	-	-	Resolution: If reading or writing data fails, contact technical personnel.
6782	File operation: File delete error	-	-	Resolution: If reading or writing data fails, contact technical personnel.
6783	Laser automatic calibration error:	0	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		1	Data error	Resolution: Please check the calibration parameters and sensor configuration.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		2	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		3	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		4	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		5	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		6	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		7	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		8	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		9	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		10	Data error	Resolution: Please check the calibration parameters and sensor configuration.
		11	Exceed safety protection limit	Resolution: Please check the calibration parameters and sensor

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			value, please contact technical staff	configuration.
		12	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		13	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		14	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		15	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		16	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		17	Exceed safety	Resolution: Please check the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			protection limit value, please contact technical staff	calibration parameters and sensor configuration.
		18	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		19	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		20	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
		21	Exceed safety protection limit value, please contact technical staff	Resolution: Please check the calibration parameters and sensor configuration.
6784	Automatic laser calibration error, please manually move robot to a safe point	-	-	Resolution: Move to a safe position and try again.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6785	Job edit error	-	-	Resolution: Adaptive weld information error, please check the job editing.
6786	Seam data error	-	-	Resolution: Failed to obtain the adaptive weld information. Check the configuration.
6787	Self-adaption database parameter error	-	-	Resolution: Check the adaptive database parameter Settings.
6788	Self-adaption mode error	-	-	Resolution: Check the adaptive mode Settings.
6789	Groove Analysis Data Error	-	-	Resolution: The groove analysis data is abnormal. Check the groove analysis parameter Settings.
6790	Folder open error	-	-	Resolution: System error. Please contact technical support.
6791	Universal welder data error	0	Group IO out of bounds	Resolution: Check the I/O configuration page of the user-defined group.
		1	Group IO no setting	Resolution: Check the I/O configuration page of the user-defined group.
		2	Group IO set out of bounds	Resolution: Check the I/O configuration page of the user-defined group.
		3	Data Trans filenum out of bounds	Resolution: Check the data conversion configuration page Settings.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		4	Data Trans denominator is 0	Resolution: Check the data conversion configuration page Settings.
		5	NULL pointer	Resolution: The pointer data is empty. Contact technical personnel.
		6	Group IO out of bounds	Resolution: Check the I/O configuration page of the user-defined group.
		7	Data Trans filenum out of bounds	Resolution: Resolution: Check the data conversion configuration page Settings.
		8	Group IO no setting	Resolution: Check the I/O configuration page of the user-defined group.
		9	Group IO set out of bounds	Resolution: Check the I/O configuration page of the user-defined group.
		10	Data Trans filenum out of bounds	Resolution: Check the data conversion configuration page Settings.
		11	NULL pointer	Resolution: The pointer data is empty. Contact technical personnel.
		12	Group IO out of bounds	Resolution: Check the I/O configuration page of the user-defined group.
		13	Data Trans filenum out of	Resolution: Check the data conversion configuration page

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			bounds	Settings.
		14	Group IO no setting	Resolution: Check the I/O configuration page of the user-defined group.
		15	Group IO set out of bounds	Resolution: Check the I/O configuration page of the user-defined group.
		16	Data Trans denominator is 0	Resolution: Check the data conversion configuration page Settings.
6792	Data processing error	-	-	Resolution: If the trigonometric function data is out of bounds or divided by zero, contact technical personnel.
6793	Ring queue is full, please reset	-	-	Resolution: If the data store cache is abnormal, contact technical support.
6794	Data exceed valid value	-	-	Resolution: If the data is abnormal, contact technical support.
6795	Instruction position variable number out of bounds	-	-	Resolution: The laser sensing position variable is out of bounds, please check the Settings.
6796	Instruction parameter out of bounds	-	-	Resolution: The number of oscillating libraries exceeds the limit. Contact technical support.
6797	Arc weld data is empty, please check	-	-	Resolution: The pointer data is empty. Contact technical personnel.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6798	Breakpoint restore err	0	Scan process pause	Resolution: Press pause during scanning. Please cancel to continue.
		1	Welding line jump function invalid	Resolution: Welding wire jump switch is not open, please open.
		2	Pause process no done	Resolution: The pause process is not complete. Please delay the start.
6799	Insufficient welding seams, number of:	-	-	Resolution: The number of arc welds is less than the initial set total, please check the workpiece.
6800	Multiple welding seams, number of:	-	-	Resolution: The number of arc welds exceeds the initial set total, please check the workpiece.
6801	Megmeet DEX welding machine data error	0	Group IO out of bounds	Resolution: Magmet DEX set machine, check group IO configuration.
		1	Group IO not configured	Resolution: Magmet DEX set machine, check group IO configuration.
		2	Group IO configuration out of bounds	Resolution: Magmet DEX set machine, check group IO configuration.
		3	Data conversion file number out of bounds	Resolution: Magmet DEX set, check data conversion files.
		4	The denominator of the data conversion is zero	Resolution: Magmet DEX set, check data conversion files.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		5	NULL pointer	Resolution: System error. Contact technical support.
		6	Group IO out of bounds	Resolution: Magmet DEX set machine, check group IO configuration.
		7	Data conversion file number out of bounds	Resolution: Magmet DEX set, check data conversion files.
		8	Group IO not configured	Resolution: Magmet DEX set machine, check group IO configuration.
		9	Group IO configuration out of bounds	Resolution: Magmet DEX set machine, check group IO configuration.
		10	The denominator of the data conversion is zero	Resolution: Magmet DEX set, check data conversion files.
		11	NULL pointer	Resolution: System error. Contact technical support.
		12	Group IO out of bounds	Resolution: Magmet DEX set machine, check group IO configuration.
		13	Data conversion file number out of bounds	Resolution: Magmet DEX set, check data conversion files.
		14	Group IO not configured	Resolution: Magmet DEX set machine, check group IO configuration.
		15	Group IO	Resolution: Magmet DEX set

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			configuration out of bounds	machine, check group IO configuration.
		16	The denominator of the data conversion is zero	Resolution: Magmet DEX set, check data conversion files.
6802	Megmeet ARTSEN welding machine data error	0	Group IO out of bounds	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		1	Group IO not configured	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		2	Group IO configuration out of bounds	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		3	Data conversion file number out of bounds	Resolution: Magmet ARTSEN set, check data conversion files.
		4	The denominator of the data conversion is zero	Resolution: Magmet ARTSEN set, check data conversion files.
		5	NULL pointer	Resolution: System error. Contact technical support.
		6	Group IO out of bounds	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		7	Data conversion file number out of bounds	Resolution: Magmet ARTSEN set, check data conversion files.
		8	Group IO not	Resolution: Magmet ARTSEN set

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			configured	machine, check group IO configuration.
		9	Group IO configuration out of bounds	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		10	The denominator of the data conversion is zero	Resolution: Magmet ARTSEN set, check data conversion files.
		11	NULL pointer	Resolution: System error. Contact technical support.
		12	Group IO out of bounds	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		13	Data conversion file number out of bounds	Resolution: Magmet ARTSEN set, check data conversion files.
		14	Group IO not configured	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		15	Group IO configuration out of bounds	Resolution: Magmet ARTSEN set machine, check group IO configuration.
		16	The denominator of the data conversion is zero	Resolution: Magmet ARTSEN set, check data conversion files.
6803	Aotai MIG welding machine data error	0	Group IO out of bounds	Resolution: Aotai Mig set machine, check group IO configuration.
		1	Group IO not	Resolution: Aotai Mig set

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			configured	machine, check group IO configuration.
		2	Group IO configuration out of bounds	Resolution: Aotai Mig set machine, check group IO configuration.
		3	Data conversion file number out of bounds	Resolution: Aotai Mig set, check data conversion files.
		4	The denominator of the data conversion is zero	Resolution: Aotai Mig set, check data conversion files.
		5	NULL pointer	Resolution: System error. Contact technical support.
		6	Group IO out of bounds	Resolution: Aotai Mig set machine, check group IO configuration.
		7	Data conversion file number out of bounds	Resolution: Aotai Mig set, check data conversion files.
		8	Group IO not configured	Resolution: Aotai Mig set machine, check group IO configuration.
		9	Group IO configuration out of bounds	Resolution: Aotai Mig set machine, check group IO configuration.
		10	The denominator of the data conversion is zero	Resolution: Aotai Mig set, check data conversion files.
		11	NULL pointer	Resolution: System error. Contact

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				technical support.
		12	Group IO out of bounds	Resolution: Aotai Mig set machine, check group IO configuration.
		13	Data conversion file number out of bounds	Resolution: Aotai Mig set, check data conversion files.
		14	Group IO not configured	Resolution: Aotai Mig set machine, check group IO configuration.
		15	Group IO configuration out of bounds	Resolution: Aotai Mig set machine, check group IO configuration.
		16	The denominator of the data conversion is zero	Resolution: Aotai Mig set, check data conversion files.
6805	Power off data size out of bounds	-	-	Resolution: System error. Please contact technical support.
6806	Power off data verification failed	-	-	Resolution: System error. Please contact technical support.
6830	Gas flow error	0	Gas flow 1 error	Resolution: Gas flow 1 is insufficient. Check the welding monitoring gas flow 1 Set IO.
		1	Gas flow 2 error	Resolution: Gas flow 2 is insufficient. Check the welding monitoring gas flow 2 Set IO.
6831	Fly weld process error	-	-	Resolution: Abnormal flight welding process, check whether there are other alarm information.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6832	Stitch weld error	0	No ARCOFF_S Instruction	Resolution: Please check the job editing.
		1	Get Weld Path Error, Please Check Move Instruction	Resolution: Please check the motion position.
		2	Curve Distance Calculate Error	Resolution: Please check the motion position.
		3	Stitchweld and Self-adapt Incompatible	Resolution: Please check the job editing.
		4	-	Resolution: Please contact technical support.
		5	Teach Distance Too Short	Resolution: Please check the job editing.
		6	Pause Time mismatch Weld Time	Resolution: Please check the continuous spot welding instruction Settings and arc recovery parameters interface.
6833	Assist wire error	-	-	Resolution: Plug the wire, please check the welding monitor auxiliary wire feed setting IO.
6834	Working mode not existed, please change	-	-	Resolution: Check whether the signal of the sending welder mode corresponds to the current sending mode of the welder protocol.
6835	Arc restart error:	1	Welding interval did not find endpoint 1	Resolution: Check that the teaching assignment has an arc command.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		2	Endpoint calculation error 2	Resolution: Point calculation error, please change the teaching point and contact the technician.
6836	Continuous spot welding instruction error	-	-	Resolution: Please check the job editing.

4.3 Prompt

Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
6853	Save complete	-	-	Resolution: Point laser or line laser data is successfully stored in P variable without processing
6855	Parameter mapping initialization error, please check	-	-	Resolution: Welding parameter mapping failed. Check the configuration.
6856	Network search in progress, please wait	-	-	Resolution: arclink network welding machine IP search begins.
6857	Search complete, please select	-	-	Resolution: The IP search of the welder in the arclink network is complete.
6860	Arc weld parameter backup failed	0	File1: Intelligent welding parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		1	File2: Multi-pass welding template configuration	Resolution: Please reinsert the USB flash drive or restart the robot.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
		2	File3:Arc welding parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		3	File4:Configure welding status monitoring I/O parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		4	File5:Configure contact sensing parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		5	File6:Configure arc tracking parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		6	File7:Attitude change parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		7	File8:Set breakpoint recovery parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		8	File9:Sensor action library parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		9	File10:Double wire welding parameters configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		10	File11:Configure	Resolution: Please reinsert the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			arc restart parameters	USB flash drive or restart the robot.
		11	File12:Laser automatic calibration parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		12	File13:Configure common welding parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		13	File14:Welding common parameters - Switch configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		14	File15:Welding common parameter - arcing parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		15	File16:Welding common parameters - arc initiation and arc leakage detection parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		16	File17:Welding common	Resolution: Please reinsert the USB flash drive or restart the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			parameter - flight arcing parameter configuration	robot.
		17	File18:Welding common parameter - arc quenching parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		18	File19:Welding common parameter - Sticky wire detection parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		19	File20:Welding common parameters - flight arc and swing arc parameters configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		20	File21:Parameter configuration of each welding machine	Resolution: Please reinsert the USB flash drive or restart the robot.
		21	File22:CAN and ARCLINK welder configurations	Resolution: Please reinsert the USB flash drive or restart the robot.
		22	File23:Adaptive	Resolution: Please reinsert the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			parameter configuration	USB flash drive or restart the robot.
		23	File24:Adaptive database parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		24	File25:Contact sensing function select parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		25	File26:Point laser sensing parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		26	File27:Groove analysis File parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		27	File28:Point laser function parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		28	File29:Arc welding production monitoring parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		29	File30:Continuous spot welding parameter	Resolution: Please reinsert the USB flash drive or restart the robot.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			configuration	
		30	File31:Line laser tracking parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		31	File32:Line laser sensor parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		32	File33:Parameter mapping Parameter setting	Resolution: Please reinsert the USB flash drive or restart the robot.
		33	File34:Configure start point detection parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		34	File35:Parameter Settings are displayed in real time	Resolution: Please reinsert the USB flash drive or restart the robot.
6861	Arc weld parameter recovery failed	0	File1:Intelligent welding parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		1	File2:Multi-pass welding template configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		2	File3:Arc welding parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		3	File4:Configure welding status	Resolution: Please reinsert the USB flash drive or restart the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			monitoring I/O parameters	robot.
		4	File5:Configure contact sensing parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		5	File6:Configure arc tracking parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		6	File7:Attitude change parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		7	File8:Set breakpoint recovery parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		8	File9:Sensor action library parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		9	File10:Double wire welding parameters configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		10	File11:Configure arc restart parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		11	File12:Laser automatic calibration	Resolution: Please reinsert the USB flash drive or restart the robo

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			parameter configuration	
		12	File13:Configure common welding parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		13	File14:Welding common parameters - Switch configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		14	File15:Welding common parameter - arcing parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		15	File16:Welding common parameters - arc initiation and arc leakage detection parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		16	File17:Welding common parameter - flight arcing parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		17	File18:Welding common	Resolution: Please reinsert the USB flash drive or restart the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			parameter - arc quenching parameter configuration	robot.
		18	File19:Welding common parameter - Sticky wire detection parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		19	File20:Welding common parameters - flight arc and swing arc parameters configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		20	File21:Parameter configuration of each welding machine	Resolution: Please reinsert the USB flash drive or restart the robot.
		21	File22:CAN and ARCLINK welder configurations	Resolution: Please reinsert the USB flash drive or restart the robot.
		22	File23:Adaptive parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		23	File24:Adaptive database parameter	Resolution: Please reinsert the USB flash drive or restart the robot.

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			configuration	
		24	File25:Contact sensing function select parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		25	File26:Point laser sensing parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		26	File27:Groove analysis File parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		27	File28:Point laser function parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		28	File29:Arc welding production monitoring parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		29	File30:Continuous spot welding parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		30	File31:Line laser tracking parameter configuration	Resolution: Please reinsert the USB flash drive or restart the robot.
		31	File32:Line laser	Resolution: Please reinsert the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			sensor parameter configuration	USB flash drive or restart the robot.
		32	File33:Parameter mapping Parameter setting	Resolution: Please reinsert the USB flash drive or restart the robot.
		33	File34:Configure start point detection parameters	Resolution: Please reinsert the USB flash drive or restart the robot.
		34	File35:Parameter Settings are displayed in real time	Resolution: Please reinsert the USB flash drive or restart the robot.
6862	Invalid track type, please reset			
6864	Initial point detection configuration changed, please restart			
6865	Invalid welder type, please reset	-	-	Resolution: Funis welding machine type is abnormal, please check
6872	Backup data permission is different from current permission	-	-	Resolution: The discovery permission of data restoration is inconsistent with that of backup. Reconfigure intelligent welding and restore data.
6873	Laser automatic calibration completed	-	-	Resolution: Laser automatic calibration complete, no

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				processing
6874	Group IO number is not configured,and the output value is default value	-	-	Resolution: WM/ Funis unit IO is not set, please check the group IO configuration.
6875	The corresponding group io configuration is out of range	-	-	Resolution: The I/O Settings of the EWM/ Funis unit are out of bounds. Check the group I/O configuration.
6876	The starting address is 0,which is invalid.Please configure it	-	-	Resolution: The start address of the EWM/ Funis machine is 0. Check the Settings.
6877	The starting address does not match the logical IO range,please check	-	-	Resolution: The start address of the EWM/ Funis machine is not in the logical I/O range. Check the Settings.
6878	Sending silk	-	-	Resolution: Arc welding shortcut key trigger welding gun wire prompt, no need to deal with.
6879	Wire feeding completed	-	-	Resolution: The arc welding shortcut key triggers the welding gun to stop the wire prompt, no processing is required.
6880	Returning silk	-	-	Resolution: Arc welding shortcut key trigger welding gun wire drawing prompt, no need to deal with.
6881	Wire withdrawal	-	-	Resolution: The arc welding

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	completed			shortcut key triggers the welding gun to stop the wire drawing prompt, no processing is required.
6882	Testing gas	-	-	Resolution: The arc welding shortcut key triggers the welding gun to vent the gas, no need to deal with it.
6883	Gas test completed	-	-	Resolution: The arc welding shortcut key triggers the welding gun to stop the gas prompt, no processing is required.
6884	Error cleared	-	-	Resolution: Arc welding shortcut key trigger alarm reset prompt, no need to deal with.
6885	Shortcut key calibration accuracy execution	-	-	Resolution: Arc welding shortcut key trigger line laser accuracy test tips, no processing required.
6886	Shortcut key for weld type switching execution	-	-	Resolution: Arc welding shortcut key trigger line laser recognition weld switching prompt, no processing required.
6887	Shortcut key starts reading execution	-	-	Resolution: The arc welding shortcut key triggers the line laser to start reading the data prompt without processing.
6888	Shortcut key to stop reading and execute	-	-	Resolution: Arc welding shortcut key trigger line laser end reading data prompt, no processing.
6889	Test Arcweld	0	Test welding start	Resolution: shortcut key test

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				welding function prompt, no processing.
		1	Test welding end	Resolution: shortcut key test welding function prompt, no processing.
6911	High-low current learning success	-	-	Resolution: Arc tracking high and low current learning success, no need to deal with.
6913	Adaptive function not support current groove type	-	-	Resolution: The line laser sensor does not support the groove type to obtain data. Check.
6914	Laser search success	-	-	Resolution: Line laser search successful, no action required.
6915	Track and posture gradual functions are invalid in teach mode	-	-	Resolution: Line laser tracking attitude gradient does not support teaching mode, please check.
6920	Current seam parameter is empty	-	-	Resolution: Multipass welding database is empty, please check.
6921	Seam name is empty	-	-	Resolution: Multi-pass welding database name is empty, please check.
6922	Execution pass number is greater than template pass	-	-	Resolution: Multipass welding execution layer number is greater than the maximum 20, please check.
6924	Database already exist	-	-	Resolution: Multi-pass welding database already exists, please reset.
6925	Name length cannot	-	-	Resolution: Multi-pass welding

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	be greater than 8			database name length more than 8, please check.
6926	Database library is full	-	-	Resolution: The number of multi-pass welding database exceeds 500, please check.
6927	Pass number exceed maximum	-	-	Resolution: Multi pass welding cycle up to 20, please check.
6929	Please set database first	-	-	Resolution: Multi-pass welding database weld Settings are not set a name, please check.
6930	Cover pass forward and backward motion only support teach mode	-	-	Resolution: Please note that multi-pass welding manual forward and forward motion only supports the execution of teaching points.
6931	Scan start	-	-	Resolution: line laser 0 layer scanning starts
6932	Scan end	-	-	Resolution: Line laser layer 0 scanning is complete.
6933	Please set correct position variable number	-	-	Resolution: Laser detection accuracy function, select the wrong position variable number, please check.
6935	Scan Skip successful	-	-	Resolution: Point laser automatic scanning succeeds, no action is required.
6936	The template specified is inconsistent, accordin	-	-	Resolution: If the specified pass name is inconsistent with the current execution, execute as the

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
	g to teaching track			current teaching trajectory offset, please note.
6950	Welder reconnected	-	-	Resolution: Welder and robot communication re-established tips.
6952	Torch collision turned off	1	Torch 1, signal source is external IO	Resolution: Function closed prompt, you can turn on welding status monitoring welding torch 1 collision switch.
		2	Torch 2, signal source is external IO	Resolution: Function closed prompt, you can turn on welding status monitoring welding torch 2 collision switch.
		3	Torch 1, signal source is welder feedback	Resolution: The function is off prompt, you can turn on the welding torch 1 collision switch in welding parameters.
		4	Torch 2, signal source is welder feedback	Resolution: The function is off prompt, you can turn on the welding torch 2 collision switch in welding parameters.
6953	Arc loss detection switch turned off	-	-	Resolution: The leakage arc detection function has been turned off. You can turn on the leakage arc detection switch in the arc starting parameter.
6956	Weld switch turned off	-	-	Resolution: Welding allows to close without arc, Shortcut keys can be used to switch the working

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				mode to normal mode.
6957	Strike detection switch turned off	-	-	Resolution: The arcing detection function is disabled. You can turn on the arcing detection switch in the arcing parameters.
6959	Arc restart switch is off			Resolution: The arc restart function has been turned off, and the arc restart switch in the arc starting parameter can be turned on.
6961	Compressed air switch turned off	-	-	Resolution: Check the welding status monitoring configuration and turn on the compressed air switch.
6962	Water flow switch turned off	-	-	Resolution: Check the welding status monitoring configuration and turn on the water flow switch.
6963	Gas switch turned off	0	-	
		1	Gas switch 1	Resolution: Check the welding status monitoring configuration and turn protection gas switch 1 on.
		2	Gas switch 2	Resolution: Check the welding status monitoring configuration and turn protection gas switch 2 on.
6964	Wire detection switch turned off	0	-	
		1	Wire detection switch 1	Resolution: Check the welding status monitoring configuration

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				and turn on the missing wire detection switch 1.
		2	Wire detection switch 2	Resolution: Check the welding status monitoring configuration and turn on the missing wire detection switch 2.
6965	Assist wire inching switch turned off	-	-	Resolution: Check the welding status monitoring configuration and turn on the auxiliary wire feed switch.
6966	Wire is little, please change wire	-	-	Resolution: Real-time display prompt Missing wire prompt, please check the use of welding wire and real-time display configuration.
6967	Air flow switch is closed	0	-	
		1	Gas flow switch 1	Resolution: Check the welding status monitoring configuration and turn on the gas flow switch 1.
		2	Gas flow switch 2	Resolution: Check the welding status monitoring configuration and turn on the gas flow switch 2.
6968	Welder warn 154	-	-	Resolution: The welding machine has an alarm, please reset the welding machine alarm.
6969	Parameter send error during welding	-	-	Resolution: Failed to send welding parameters. Check the welding file configuration.
6970	Clear arc weld stack	-	-	Resolution: The stack has been

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				cleared and the job can be executed in the current location.
6971	Touch sense bit feedback is 1, please check	-	-	Resolution: When the contact sensor is turned on, the welder will feedback the contact sensor setting. Please check the wiring.
6972	The offset function is turned on, and the point position can be modified	-	-	Resolution: Offset Hold is turned on, please note.
6973	Stitch weld prompt	0	Pause Time is 0	Resolution: The arc recovery time is 0. Check the configuration.
6975	The data in the tool coordinate is not zero, please confirm	-	-	Resolution: Please note that the coordinates are not set to 0 after the laser automatic calibration fails.
6976	Groove data overrun	-	-	Resolution: Automatically obtain groove data out of bounds. Check the configuration.
6977	UDP send error	-	-	Resolution: Lincoln arclink communication abnormal, please check.
6978	Adaptive correction data overrun	-	-	Resolution: V-weld adaptive speed adjustment range 0.5-20, please check.
6979	Offline data length error	-	-	Resolution: The length of the data collection function exceeds the limit. Contact technical personnel.
6980	Sensing detect switch	0	Tds switch status	Resolution: Check the sensor

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
			off	function switch configuration.
		1	Search switch status off	Resolution: Check the sensor function switch configuration.
		2	Detect switch status off	Resolution: Check the sensor function switch configuration.
		3	Tsdf switch status off	Resolution: Check the sensor function switch configuration.
		4	Circle switch status off	Resolution: Check the sensor function switch configuration.
		5	Positioning reference switch status off	Resolution: Check the sensor function switch configuration.
6981	Sense inst para err	-	-	Resolution: Sensing parameters are out of bounds. Contact technical support.
6982	Inst type err	-	-	Resolution: The motion command type of the sensor is abnormal. Please check.
6983	Invalid sensor type	0	GET Inst	Resolution: The sensing type was invalid when the offset was obtained. Please check.
		1	GET Inst	Resolution: The sensing type was invalid when the offset was obtained. Please check.
		2	SET Inst	Resolution: The sensing type is invalid when setting the offset, please check.
		3	SET Inst	Resolution: The sensing type is invalid when setting the offset,

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Alarm Code	Alarm information	Subcode	Subcode information	Troubleshooting
				please check.
6984	Get the offset, the coordinate type is not supported	-	-	Resolution: The obtained offset coordinate type is not supported. Please check.
6985	Successful point update after sensing	-	-	Resolution: The sensing point is updated to the teaching point, please note.
6986	The corresponding security door request is paused	-	-	Resolution: Pause in response to security gate request, please be advised.
6987	Inspection of the number of welds	0	Same	Resolution: The weld is normal and no treatment is required.
		1	Little	Resolution: Abnormal weld detection, please check.
		2	More	Resolution: Abnormal weld detection, please check.
6988	Drive parameter write error	-	-	Resolution: Failed to initialize the driver bus optimization parameters. Please check the configuration.
6989	Flexible collision is set, please restart	-	-	Resolution: The flexible collision has been set. Please restart.

Revision history of Manual

Version	Date	Changes
01	2025.6	Initial version writing